Thermodynamic Dynamics of Observer-Memory Entanglement

Memory, Ego, and Self-Referential Calibration in Persistent Far-from-Equilibrium Systems

Complete Collected Volume

Papers I, II, and III

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Abstract

This volume develops conditional models of persistent agency under thermodynamic, computational, and calibration constraints.

The three papers are not presented as an irreversible proof of necessary observer architecture. Paper I gives a restricted Markovian baseline and a conditional memory advantage when usable system–environment correlations are supplied. Paper II studies a declared archive and coordinate-projection model in which compression can reduce computational load. Paper III analyzes a specified self-referential calibration implementation and its costs under explicit likelihood, drift, convexity, and hardware assumptions.

Together, the papers define a running architecture of memory, reference-frame selection, and second-order calibration. The architecture is a model template and test bed, not a proof that all persistent agents require these layers or that the list is necessary, sufficient, minimal, or complete.

Keywords: non-Markovian dynamics, open quantum systems, memory kernel, reference-frame selection, bounded rationality, Fisher information, information geometry, self-referential calibration, Lyapunov stability, implementation cost

Contents

Abstract **iii**

1 Non-Markovian Memory and Temporal Accumulation: A Conditional Thermodynamic Analysis **1**

1.1 Introduction . . . 2
1.1.1 Context: The Problem of Persistence . . . 2
1.1.2 Position within the Series . . . 2
1.1.3 Relation to HAFF Paper F . . . 3
1.1.4 Scope and Disclaimers . . . 3
1.2 Mathematical Preliminaries . . . 4
1.2.1 Open Quantum Systems: The Markovian Baseline . . . 4
1.2.2 Beyond Markov: The Nakajima–Zwanzig Equation . . . 4
1.2.3 Thermodynamic Framework . . . 5
1.2.4 The Survival Functional . . . 6
1.3 A Restricted Markovian Baseline . . . 7
1.3.1 Spohn's Inequality . . . 7
1.3.2 The Restricted Autonomous Ceiling . . . 8
1.4 Initial-Correlation Accounting . . . 8
1.4.1 System–Environment Mutual Information . . . 8
1.4.2 The Information–Thermodynamic Identity . . . 9
1.4.3 The Survival Identity . . . 10
1.4.4 Three Regimes of Survival . . . 11
1.4.5 The Correlation Battery . . . 11
1.4.6 Connection to Non-Markovianity Measures . . . 13
1.4.7 Mechanism: The Surfer Analogy . . . 13
1.5 Operational Temporal Relations . . . 14
1.5.1 Thresholded Memory-Influence Relation . . . 14
1.5.2 A Fisher-Accretion Conjecture . . . 14
1.5.3 The Bridge to HAFF . . . 15
1.6 Worked Example: The Quantum Predictive Agent . . . 15
1.6.1 Model Setup . . . 16
1.6.2 Exact Decoherence Function . . . 16
1.6.3 Quantitative Evaluation . . . 16
1.7 The Cost of Memory . . . 19
1.7.1 The Landauer Debt . . . 19
1.7.2 Raw Archive Overload Scenario . . . 20
1.7.3 Conditional Motivation for Compression . . . 21
1.8 Numerical Demonstration . . . 21

1.8.1 Model . . . 21
1.8.2 Results . . . 22
1.8.3 Scope of This Demonstration . . . 23
1.9 Discussion . . . 23
1.9.1 Summary of Results . . . 23
1.9.2 What This Paper Does and Does Not Show . . . 24

2 Spontaneous Symmetry Breaking of Reference Frames as a Computational Cost Minimization Strategy 25
2.1 Introduction . . . 26
2.1.1 Context: The Problem of Overload . . . 26
2.1.2 Position within the Series . . . 26
2.1.3 Relation to Q-RAIF . . . 27
2.1.4 Relation to HAFF Paper G . . . 27
2.1.5 Scope and Disclaimers . . . 27
2.2 Mathematical Preliminaries . . . 28
2.2.1 Inherited Framework from Paper I . . . 28
2.2.2 The Agent's Internal Algebra . . . 29
2.2.3 Rate-Distortion Theory . . . 30
2.2.4 Bounded Rationality . . . 30
2.2.5 Fiber Bundle Formalism . . . 30
2.2.6 Standing Assumptions . . . 31
2.3 Raw-Channel Load Model . . . 31
2.3.1 The Information Processing Inequality for Bounded Agents . . . 32
2.3.2 Processing Collapse . . . 33
2.4 Conditional Frame-Selection Model . . . 33
2.4.1 Reference Frame as Gauge Fixing . . . 33
2.4.2 A Coordinate-Compression Model . . . 33
2.4.3 Conditional Symmetry-Breaking Interpretation . . . 35
2.4.4 The Four Bias Terms . . . 35
2.5 Emergent Structure: The Architecture of Ego . . . 36
2.5.1 The Ego as a Fiber Bundle Section . . . 37
2.5.2 The Effective Survival Functional . . . 37
2.5.3 The Computational Speedup . . . 38
2.5.4 Uniform-Channel Information Trade-off . . . 38
2.6 Worked Example: Qubit in a Two-Channel Bath . . . 39
2.6.1 Model Setup . . . 39
2.6.2 Raw Full-Sweep Budget Limit . . . 40
2.6.3 Frame Selection: Choosing σ_z . . . 40
2.6.4 The Broken Phase: Effective Processing . . . 41
2.6.5 Quantitative Evaluation . . . 41
2.6.6 The Pointer-State Connection . . . 42
2.7 The Cost of Ego . . . 43
2.7.1 The Rigidity Trap . . . 43
2.7.2 Stylized Drift Model . . . 43
2.7.3 The Prediction Error Divergence . . . 44
2.7.4 Visibility Crossover and Conditional Gradient Reversal . . . 44
2.7.5 The Origin of Paper III . . . 45

2.8 Numerical Demonstration . . . 45
2.8.1 Model . . . 45
2.8.2 Results . . . 46
2.8.3 Scope of This Demonstration . . . 47
2.9 Discussion . . . 49
2.9.1 Summary of Results . . . 49
2.9.2 Testable Predictions . . . 49
2.9.3 What This Paper Does and Does Not Show . . . 49

3 Fisher Information Geometry and Conditional Implementation Costs of Self-Referential Calibration 51
3.1 Introduction . . . 52
3.1.1 Context: The Delusion Trap . . . 52
3.1.2 Position within Papers I–III . . . 52
3.1.3 The Information-Geometric Insight . . . 53
3.1.4 Relation to Architectural Incompleteness . . . 53
3.1.5 Scope and Disclaimers . . . 53
3.2 Mathematical Preliminaries . . . 54
3.2.1 Inherited Framework from Papers I and II . . . 54
3.2.2 Fisher Information Metric . . . 55
3.2.3 Information Geometry . . . 56
3.2.4 Thermodynamic Length . . . 57
3.2.5 Second-Order Cybernetics . . . 57
3.2.6 Standing Assumptions . . . 58
3.3 The Drift Detection Problem . . . 58
3.3.1 Identifiability in an Even Residual Model . . . 58
3.3.2 The Agent's Statistical Manifold . . . 58
3.3.3 Self-Referential Fisher Information . . . 59
3.4 The Self-Referential Bound . . . 60
3.4.1 The Bayesian Framework . . . 60
3.4.2 The Self-Referential Cramér–Rao Bound . . . 61
3.4.3 The Rigidity-Sensitivity Trade-off . . . 61
3.5 The Calibration Loop . . . 62
3.5.1 The Natural Gradient Update Law . . . 62
3.5.2 Lyapunov Stability of the Loop . . . 62
3.5.3 Convergence Rate under Persistent Excitation . . . 63
3.5.4 Four-Part Model Template . . . 64
3.6 Thermodynamic Cost . . . 64
3.6.1 The Three Cost Components . . . 64
3.6.2 Conditional Implementation Bound . . . 65
3.6.3 Withdrawal of the Universal Persistence Budget . . . 65
3.7 Worked Example: Qubit in a Drifting Two-Channel Bath . . . 66
3.7.1 Model Setup . . . 66
3.7.2 Fisher Information under Drift . . . 66
3.7.3 Loop Dynamics: Self-Calibration in Action . . . 67
3.7.4 Thermodynamic Cost Evaluation . . . 67
3.8 Numerical Demonstration . . . 67
3.8.1 Direct Fisher-Signal Test . . . 67

3.8.2 Heuristic Calibration-Control Model 68
3.8.3 Results 69
3.8.4 Scope of This Demonstration 70
3.9 Discussion 71
3.9.1 Summary of Results 71
3.9.2 Conditional Interface Sequence 71
3.9.3 What This Paper Does and Does Not Show 71

Chapter 1

Non-Markovian Memory and Temporal Accumulation: A Conditional Thermodynamic Analysis

Paper I — "The Seed"

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Abstract

We investigate a conditional thermodynamic role for memory in open quantum systems interacting with stochastic environments. Working within stochastic and information thermodynamics, we define a *survival functional* $\mathcal{S} := \Delta F - W$ measuring the difference between non-equilibrium free energy gained and work invested by an agent.

For a fixed-Hamiltonian autonomous Davies/GKSL baseline with no measurement or feedback, we prove a restricted Markovian ceiling: $\mathcal{S} \leq 0$. This is not a universal ceiling for driven or feedback-controlled Markovian agents. We then derive an accounting identity for finite-dimensional autonomous evolution, with the interaction-energy term retained or a declared weak-coupling approximation made, expressing the survival functional through changes in system–environment mutual information and bath displacement. In models where prior interaction has supplied usable correlations, their consumption during non-Markovian backflow intervals can yield $\mathcal{S} > 0$, bounded by the initial correlation budget.

The result is therefore a conditional memory advantage, not a proof that memory or non-Markovianity is necessary or sufficient for all persistent systems. Fisher accretion and the comparison with the accessibility ordering of the Holographic Alaya-Field Framework (HAFF) are presented as heuristics. A worked spin-boson example with Lorentz–Drude spectral density illustrates how the accounting can operate in one supplied model.

Finally, using entropy rate and predictive information from computational mechanics, we study the implementation cost of a specified irreversible archive-reset model. Universal Landauer costs for merely storing or maintaining memory, and the resulting universal "Memory Catastrophe," are withdrawn; only the declared archive implementation is charged.

Keywords: non-Markovian dynamics, open quantum systems, Nakajima–Zwanzig equation, memory kernel, thermodynamic arrow of time, information backflow, entropy production, stochastic thermodynamics

1.1 Introduction

1.1.1 Context: The Problem of Persistence

A quantum system coupled to a thermal environment generically relaxes toward equilibrium. This is the content of the *zeroth crisis*: absent special structure, every open subsystem is eventually erased by thermal noise [10].

Yet the physical world contains persistent far-from-equilibrium structures—from molecular machines to living organisms—that maintain themselves against the entropic tide for timescales vastly exceeding their intrinsic relaxation times. What structural feature of their dynamics makes this possible?

The standard answer invokes free-energy input: a persistent system continuously imports low-entropy energy and exports high-entropy waste [29]. This paper asks the narrower accounting question of when pre-existing system–environment correlations can fund a later interval. It does not claim that memory distinguishes every persistent system from every non-persistent one.

1.1.2 Position within the Series

This paper is the first of three constituting the **T-DOME** (Thermodynamic Dynamics of Observer-Memory Entanglement) framework, the third pillar of a three-paper program.

Framework	**Question**		**Result**	**Status**
HAFF [41, 42]	How does geometry emerge?	Ocean	Conditional algebra/dictionary models	Revised
Q-RAIF [46, 47]	What algebra must an observer have?	Fish	Clifford representation as model input	Revised
T-DOME I (this work)	What can inter-epoch correlations fund?	Seed	ELV-type accounting identity; restricted ceiling	**This paper**
T-DOME II	How can a supplied archive be budgeted?	Ego	Conditional feature selection	Companion
T-DOME III	When can residual statistics track drift?	Loop	Conditional Fisher calibration	Companion

The three T-DOME papers form a conditional modelling sequence, not an irreversible theorem chain:

1. **Paper I (The Seed):** The accounting identity asks when correlations built in earlier epochs can fund a later interval. A deliberately uncompressed archive is then introduced only as motivation for a finite-budget problem.

2. **Paper II (The Ego):** Given that archive model and a supplied component representation, Paper II studies conditional feature selection under a processing budget.

3. **Paper III (The Loop):** Given a selected frame and an explicit residual likelihood, Paper III studies conditions under which a calibration controller can track drift.

The present paper addresses only the first link in this chain.

1.1.3 Relation to HAFF Paper F

Revised HAFF Paper F [44] treats record growth and algebra inclusion as protocol-relative orderings and does not derive a universal temporal partial order. Section 1.5 therefore uses it only as motivation: a memory-influence relation and a pointer-record ordering may be compared after both are supplied, but neither is proved to embed in the other.

1.1.4 Scope and Disclaimers

1. This work does *not* claim that memory or non-Markovianity is necessary for persistence. Continuously driven Markovian dissipative structures can persist. The narrower claim is that pre-existing inter-epoch correlations can fund a positive interval of the survival accounting functional.

2. The sign of that functional is governed by changes in initial correlation and bath-displacement resources, not by Markovianity as such. No strict comparison of Markovian and non-Markovian suprema is proved.

3. We do *not* derive the specific form of the memory kernel from first principles. The kernel is treated as a structural feature of the system-environment coupling.

4. The term "agent" is used in the control-theoretic sense (a subsystem that acts on its environment to maintain a target state) and carries no implication of consciousness, intention, or subjective experience.

5. A broader structural analogy with classical philosophical concepts of temporal persistence exists but is outside the scope of this paper.

Result-status ledger. The information–thermodynamic identity is a general finite-dimensional unitary accounting identity (GvN/THM/OK-cond, with the interaction-energy term retained or a declared weak-coupling approximation). The original time-dependent-control Markovian Ceiling is retracted; a narrower autonomous Davies-semigroup theorem is proved below (FD/I/THM/OK-cond). Fisher accretion and the HAFF bridge are heuristics. Universal Landauer costs for storing or maintaining memory and the resulting "Memory Catastrophe" are retracted; only a specified irreversible archive-reset model remains.

1.2 Mathematical Preliminaries

1.2.1 Open Quantum Systems: The Markovian Baseline

Consider a bipartite Hilbert space $\mathcal{H} = \mathcal{H}_R \otimes \mathcal{H}_E$, where R denotes the "agent" (reduced system) and E the environment. The total Hamiltonian is

$$H = H_R \otimes \mathbb{1}_E + \mathbb{1}_R \otimes H_E + \lambda\, H_{\text{int}}, \tag{1.1}$$

where λ parametrizes the coupling strength.

Under the Born–Markov and secular approximations, the reduced dynamics of $\rho_R(t) = \mathrm{Tr}_E[\rho(t)]$ is governed by the Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) master equation [21, 16]:

$$\dot{\rho}_R(t) = -i[H_{\text{eff}},\, \rho_R(t)] + \sum_k \gamma_k \left(L_k \rho_R(t) L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k,\, \rho_R(t)\} \right), \tag{1.2}$$

with $\gamma_k \geq 0$ and Lindblad operators $\{L_k\}$.

Remark 1.1 (Markovian = Memoryless)**.** *The GKSL equation is* time-local*: $\dot{\rho}_R(t)$ depends only on $\rho_R(t)$, never on $\rho_R(s)$ for $s < t$. Physically, this corresponds to an environment with vanishing correlation time ($\tau_E \to 0$): the bath "forgets" its interaction with the system instantaneously. The semigroup property $\Lambda(t+s) = \Lambda(t)\Lambda(s)$ ensures complete positivity at all times but precludes any information backflow from environment to system [26].*

1.2.2 Beyond Markov: The Nakajima–Zwanzig Equation

When the environmental correlation time τ_E is non-negligible, the Born–Markov approximation fails. The exact reduced dynamics is captured by the Nakajima–Zwanzig (NZ) integro-differential equation [23, 38]:

$$\dot{\rho}_R(t) = -i[H_{\text{eff}},\, \rho_R(t)] + \int_0^t ds\; \mathcal{K}(t,s)\, \rho_R(s), \tag{1.3}$$

where $\mathcal{K}(t,s)$ is the **memory kernel**—a superoperator encoding the influence of the system's entire history on its present dynamics.

Definition 1.2 (Memory Kernel)**.** *The memory kernel $\mathcal{K} : [0,\infty)^2 \to \mathcal{L}(\mathcal{B}(\mathcal{H}_R))$ is the superoperator satisfying* (1.3)*. It encodes two types of information:*

1. ***Environmental structure:*** *the spectral density, correlation functions, and non-equilibrium features of the bath;*
2. ***Temporal reach:*** *the effective support $\tau_{\text{mem}} := \inf\{\tau : \|\mathcal{K}(t,s)\| < \epsilon \;\forall\; t - s > \tau\}$, the "memory depth."*

The Markovian limit corresponds to $\mathcal{K}(t,s) \to \mathcal{K}_0\, \delta(t-s)$, recovering the GKSL generator.

Remark 1.3 (Information Backflow)**.** *Non-Markovian dynamics admits* information backflow*: the distinguishability of two initial states, as measured by trace distance $D(\rho_1(t), \rho_2(t))$, can temporarily increase [9]. This is the operational signature of memory—the environment returns previously absorbed information to the system.*

1.2.3 Thermodynamic Framework

We adopt the framework of stochastic thermodynamics for open quantum systems [15]. The following conventions are fixed throughout.

Definition 1.4 (Thermodynamic Setup)**.**

1. ***Hamiltonian decomposition.*** *The system Hamiltonian is* $H_S(t) = H_R + H_{\mathrm{ctrl}}(t)$*, where* H_R *is the* fixed *bare Hamiltonian and* $H_{\mathrm{ctrl}}(t)$ *is the agent's time-dependent control protocol. The bath Hamiltonian* H_E *and coupling* H_{int} *are as in* (1.1)*.*

2. ***Reference state.*** *The thermal equilibrium state of the bare Hamiltonian is*

$$\rho_{\mathrm{eq}} := \frac{e^{-\beta H_R}}{Z_R}, \qquad Z_R := \mathrm{tr}\big(e^{-\beta H_R}\big), \qquad \beta := (k_B T)^{-1}. \tag{1.4}$$

   Since H_R *is time-independent,* ρ_{eq} *is a well-defined, fixed reference throughout the protocol.*

3. ***Non-equilibrium free energy.*** *For any state* ρ *of the reduced system,*

$$F(\rho) := \mathrm{tr}(\rho\, H_R) + \beta^{-1}\,\mathrm{tr}(\rho \ln \rho) = \langle H_R \rangle_\rho - \beta^{-1} S(\rho), \tag{1.5}$$

   where $S(\rho) = -\,\mathrm{tr}(\rho \ln \rho)$ *is the von Neumann entropy. The equilibrium value is* $F_{\mathrm{eq}} = -\beta^{-1} \ln Z_R$*.*

4. ***Free energy–relative entropy identity.***

$$D_{\mathrm{KL}}(\rho \| \rho_{\mathrm{eq}}) = \beta\big(F(\rho) - F_{\mathrm{eq}}\big) \geq 0. \tag{1.6}$$

   Thus D_{KL} *measures the free-energy surplus in units of* $k_B T$*.*

5. ***Work.*** *The inclusive work performed on the total system by the control protocol over* $[0, \tau]$ *is*

$$W[0,\tau] := \int_0^\tau \mathrm{tr}\left(\rho(t)\, \frac{\partial H_{\mathrm{ctrl}}}{\partial t}\right) dt. \tag{1.7}$$

6. ***Entropy-production functional.*** *The* generalised entropy production *over* $[0, \tau]$ *is*

$$\Sigma[0,\tau] := \beta\big(W[0,\tau] - \Delta F\big), \tag{1.8}$$

   where $\Delta F = F(\rho(\tau)) - F(\rho(0))$*. For uncorrelated (product) initial states,* $\Sigma \geq 0$ *recovers the standard second-law bound. For initially correlated states,* Σ *can be* negative*, reflecting the consumption of pre-existing correlations (see Remark 1.25).*

Remark 1.5 (Why H_R is fixed)**.** *The bare Hamiltonian* H_R *defines the system's energy scale and hence the reference state* ρ_{eq}*. The agent acts on the world through* $H_{\mathrm{ctrl}}(t)$*, which may be time-dependent. This separation ensures that* ρ_{eq} *is well-defined and time-independent for the accounting identity. It does* not *make a driven generator satisfy detailed balance with respect to that fixed state. The ceiling result below therefore uses autonomous* $H_{\mathrm{ctrl}} = 0$ *dynamics.*

Definition 1.6 (Standing Assumptions)**.** *The following minimal assumptions are in force throughout Sections 1.4–1.6 unless stated otherwise. Every main result (Lemma 1.19, Theorem 1.21, Corollary 1.24) relies* only *on items* (A1)–(A5) *below.*

(A1) ***Finite-dimensional bipartite system.*** *$\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_E$, with total Hamiltonian* (1.1) *and global unitary evolution $U(t) = \mathcal{T}\exp\Big(-i\int_0^t H(s)\,ds\Big)$.*

(A2) ***Weak coupling.*** *The system–environment interaction satisfies $\lambda \ll 1$ in* (1.1)*, so that $\Delta\langle H_{\rm int}\rangle = O(\lambda)$ [10]. The first-law bookkeeping is $W = \Delta\langle H_R\rangle + \Delta\langle H_{\rm ctrl}\rangle + \Delta\langle H_E\rangle + \Delta\langle H_{\rm int}\rangle$. Dropping the last term is the declared weak-coupling approximation $\Delta\langle H_{\rm int}\rangle = O(\lambda)$; it is not an energy-conservation equation with a zero right-hand side.*

(A3) ***Fixed environmental reference.*** *$\rho_E^{\rm th} := e^{-\beta H_E}/Z_E$ is a fixed* bookkeeping *Gibbs state at inverse temperature β. The* actual *initial bath state $\rho_E(0)$ need not coincide with $\rho_E^{\rm th}$; when $\rho_E(0) \neq \rho_E^{\rm th}$, the quantity $D_{\rm KL}(\rho_E(t)\|\rho_E^{\rm th})$ tracks the nonequilibrium free energy stored in the bath relative to this reference. The bath Hamiltonian H_E is time-independent.*

(A4) ***Arbitrary initial state.*** *The total initial state $\rho_{SE}(0)$ is* not *required to be a product state. In particular, initial system–environment correlations $I(S{:}E;\,0) > 0$ and initial bath displacement $D_{\rm KL}(\rho_E(0)\|\rho_E^{\rm th}) > 0$ are both permitted.*

(A5) ***Regularity.*** *All quantum states appearing in the thermodynamic identities are assumed to have full rank (or are restricted to their support), so that all relative entropies $D_{\rm KL}(\rho\|\sigma)$ are finite.*

Remark 1.7 (Bookkeeping conventions)**.** *The heat absorbed by the environment is $Q := \Delta\langle H_E\rangle = \mathrm{Tr}[\rho_E(\tau)H_E] - \mathrm{Tr}[\rho_E(0)H_E]$ (matching Esposito* et al. *[15]). We define $\Sigma := \beta(W - \Delta F)$ as a* generalised entropy-balance functional*; for correlated initial conditions Σ need not be nonnegative (see Remark 1.25).*

1.2.4 The Survival Functional

We now define the central quantity of this paper.

Definition 1.8 (Survival Functional)**.** *For a reduced system R evolving under dynamics Λ over $[0,\tau]$, the* ***survival functional*** *is*

$$\mathcal{S}[\Lambda,\tau] := \Delta F - W[0,\tau] = \big[F(\rho(\tau)) - F(\rho(0))\big] - W[0,\tau]. \tag{1.9}$$

Equivalently, using (1.8)*,*

$$\beta\,\mathcal{S}[\Lambda,\tau] = -\Sigma[0,\tau]. \tag{1.10}$$

Note on nomenclature. We retain the term "survival functional" to emphasize the biological interpretation of persistence far from equilibrium; mathematically, $\mathcal{S}$ is strictly a *generalized entropy-balance functional* derived from the first and second laws.

Remark 1.9 (Interpretation)**.** *The survival functional has a transparent physical meaning:*

- *$\mathcal{S} > 0$: the system gained more free energy than was invested by the external protocol—a* thermodynamic profit*. The agent has extracted usable work from environmental correlations.*
- *$\mathcal{S} = 0$: the agent breaks even (reversible limit, $\Sigma = 0$).*

- $\mathcal{S} < 0$*: the agent paid more than it gained (the generic irreversible case).*

Under the standard second law ($\Sigma \geq 0$), $\mathcal{S} \leq 0$ always. As Section 1.4 shows, a positive interval requires consumption of a resource present at that interval's initial time, such as system–environment correlation or bath displacement. Whether a memory kernel preserves access to that resource is a model-specific dynamical question.

Remark 1.10 (Connection to Information Thermodynamics)**.** *In the Sagawa–Ueda framework [27, 28], a system under feedback control satisfies the generalized second law*

$$\Sigma \geq -I_{\text{feedback}}, \tag{1.11}$$

where $I_{\text{feedback}} \geq 0$ is the mutual information gained through measurement of the system. This permits $\Sigma < 0$ (and hence $\mathcal{S} > 0$) at the expense of information. Pre-existing correlations can enter the entropy balance in a role analogous to measurement information. A memory kernel is not automatically a measurement or feedback controller; a concrete model must show that it preserves and later exposes those correlations.

1.3 A Restricted Markovian Baseline

The former "Markovian Ceiling" allowed a time-dependent control Hamiltonian while using a fixed bare Gibbs state as the detailed-balance reference. A driven generator need not preserve that state, so the asserted balance law did not follow from Spohn's inequality. That theorem, its open-loop control class, and its temporal-blindness corollary are retracted. The autonomous result below is the narrower statement that survives.

1.3.1 Spohn's Inequality

Throughout this section we assume that the GKSL generator $\mathcal{L}$ is a *thermal Lindbladian*: it is obtained from the weak-coupling (Davies) limit of a system coupled to a single thermal bath at inverse temperature β, and satisfies **quantum detailed balance** (the KMS condition) [35, 10]. Under this assumption, the unique stationary state is the Gibbs state $\rho_{\text{ss}} = \rho_{\text{eq}}$ of (1.4), and the generator is self-adjoint with respect to the KMS inner product. This ensures that the entropy production rate below is well-defined and non-negative.

Definition 1.11 (Autonomous Davies baseline)**.** *The autonomous baseline has $H_{\text{ctrl}} = 0$, fixed system Hamiltonian H_R, and a time-independent Davies/GKSL generator $\mathcal{L}$ satisfying quantum detailed balance with respect to the faithful Gibbs state $\rho_{\text{eq}} = e^{-\beta H_R}/Z_R$. No measurement, feedback, or initial system–bath resource is included in this reduced semigroup model.*

Lemma 1.12 (Spohn [35])**.** *For any GKSL dynamical semigroup $\Lambda_t = e^{\mathcal{L}t}$ satisfying quantum detailed balance with unique invariant state ρ_{eq}, the entropy production rate*

$$\sigma(t) := -\operatorname{tr}\big(\mathcal{L}[\rho(t)]\,(\ln \rho(t) - \ln \rho_{\text{eq}})\big) \tag{1.12}$$

satisfies $\sigma(t) \geq 0$, equivalently $dD_{\text{KL}}(\rho(t)\|\rho_{\text{eq}})/dt \leq 0$.

Proof. This follows from the contractivity of CPTP maps under quantum relative entropy [35, 10]: $D_{\text{KL}}(\Lambda_t\rho\|\Lambda_t\rho_{\text{eq}}) \leq D_{\text{KL}}(\rho\|\rho_{\text{eq}})$ for all $t \geq 0$. Differentiating at $t = 0$ yields $\sigma(t) \geq 0$. □

1.3.2 The Restricted Autonomous Ceiling

Theorem 1.13 (Restricted Autonomous Ceiling)**.** *Under Definition 1.11, the survival functional obeys*

$$\mathcal{S}[0,\tau] = F(\rho(\tau)) - F(\rho(0)) \leq 0. \tag{1.13}$$

Equality requires no decrease of relative entropy on the interval (for a primitive semigroup, equilibrium is the stationary equality case).

Proof. Here $W = 0$. By (1.6) and Lemma 1.12, $\beta\Delta F = \Delta D_{\mathrm{KL}} \leq 0$, hence $\mathcal{S} = \Delta F \leq 0$. □

Remark 1.14 (Driven protocols require a different theorem)**.** *For time-dependent $H_S(t)$ one may define the instantaneous Gibbs state of the full driven Hamiltonian and assume a generator satisfying instantaneous detailed balance, together with inclusive work $\int \mathrm{tr}\Big(\rho\,\dot{H}_S\Big)dt$. Those hypotheses describe a different driven-thermodynamics result and are not proved here. Open-loop alone does not repair the missing relation between generator and reference state; feedback protocols require their own information balance.*

Remark 1.15 (Memorylessness and temporal correlations)**.** *Under the Born–Markov approximation, the bath correlation function is replaced by its white-noise limit $C(t,s) \to C_0\,\delta(t-s)$, and the GKSL dissipator depends only on the spectral density $J(\omega)$ evaluated at the system's Bohr frequencies. The reduced semigroup therefore omits explicit inter-epoch bath correlations. This modelling loss does not imply a universal spectral gap formula, a universal maintenance-work bound, or impossibility of continuously driven Markovian persistence.*

Remark 1.16 (Dissipative vs. Self-Nourishing Structures)**.** *The restricted theorem is a free-energy monotonicity statement for one autonomous detailed-balance class. It does not say that Markovian systems cannot persist under continuous driving, that non-Markovianity is necessary for persistence, or that a memory kernel alone supplies usable feedback. It is a controlled baseline for comparisons at fixed initial resources.*

1.4 Initial-Correlation Accounting

We now separate a dynamics-agnostic accounting identity from the model-dependent question of memory. Quantum mutual information $I(S{:}E)$ can be a consumable initial resource. A non-Markovian model may preserve such a resource across epochs, but non-Markovianity alone does not fix the sign of $\mathcal{S}$ and is neither necessary nor sufficient for a positive interval.

1.4.1 System–Environment Mutual Information

We work with the total system–environment state $\rho_{SE}(t)$, evolving unitarily under the total Hamiltonian (1.1). The quantum mutual information

$$I(S{:}E;\,t) := S(\rho_S(t)) + S(\rho_E(t)) - S(\rho_{SE}(t)) = D_{\mathrm{KL}}\big(\rho_{SE}(t)\,\big\|\,\rho_S(t)\otimes\rho_E(t)\big) \geq 0 \tag{1.14}$$

quantifies the total correlations (classical and quantum) between the system S and the environment E at time t.

Remark 1.17 (Role of Initial Correlations)**.** *Under the Born approximation, the initial state is taken as a product* $\rho_{SE}(0) = \rho_S(0) \otimes \rho_E^{\text{th}}$*, so* $I(S{:}E;\,0) = 0$*. For a system already interacting with its environment, a restart state* may *be correlated. If a specified memory mechanism preserves those correlations and a later interval consumes them, they contribute to that interval's resource balance. This is a conditional mechanism, not a generic property of every persistent system or memory kernel.*

1.4.2 The Information–Thermodynamic Identity

The following identity is the central technical tool of this section. It holds for **any** initial state—product or correlated—and relies only on unitarity and the definitions of mutual information and relative entropy. It is the Esposito–Lindenberg–Van den Broeck entropy-production decomposition [15], written so that arbitrary initial-correlation terms are retained explicitly.

Remark 1.18 (Relative-entropy chain rule)**.** *We repeatedly use the identity*

$$D_{\text{KL}}\big(\rho_{SE} \,\big\|\, \rho_S \otimes \sigma_E\big) = I(S{:}E)_{\rho_{SE}} + D_{\text{KL}}(\rho_E \| \sigma_E), \tag{1.15}$$

valid for arbitrary *(possibly correlated)* ρ_{SE} *and any full-rank reference state* σ_E*.*[1] *Importantly, this is a* pure algebraic identity *and does not assume product initial conditions.*

Lemma 1.19 (Information–Thermodynamic Identity)**.** *Let* $\rho_{SE}(t)$ *evolve unitarily under the total Hamiltonian. Then, for* ***any*** *initial state* $\rho_{SE}(0)$ *(product or correlated):*

$$\Delta I(S{:}E) + \Delta D_{\text{KL}}\big(\rho_E \,\big\|\, \rho_E^{\text{th}}\big) = \Delta S_S + \beta\, \Delta\langle H_E\rangle, \tag{1.16}$$

where $\Delta S_S = S(\rho_S(\tau)) - S(\rho_S(0))$ *is the change in the system's von Neumann entropy and* $\Delta\langle H_E\rangle = \text{Tr}[\rho_E(\tau)\, H_E] - \text{Tr}[\rho_E(0)\, H_E]$ *is the energy absorbed by the environment.*

Proof. Applying the chain rule (1.15) (Remark 1.18) with $\sigma_E = \rho_E^{\text{th}}$:

$$D_{\text{KL}}\big(\rho_{SE}(t) \,\big\|\, \rho_S(t) \otimes \rho_E^{\text{th}}\big) = I(S{:}E;\, t) + D_{\text{KL}}\big(\rho_E(t) \,\big\|\, \rho_E^{\text{th}}\big). \tag{1.17}$$

Expanding the left side using $\ln \rho_E^{\text{th}} = -\beta H_E - \ln Z_E$:

$$D_{\text{KL}}\big(\rho_{SE}(t) \,\big\|\, \rho_S(t) \otimes \rho_E^{\text{th}}\big) = -S(\rho_{SE}(t)) + S(\rho_S(t)) + \beta\langle H_E\rangle_t + \ln Z_E. \tag{1.18}$$

Since the total evolution is unitary, $S(\rho_{SE}(t)) = S(\rho_{SE}(0))$ for all t. Taking the difference between times τ and 0 cancels both $S(\rho_{SE})$ and $\ln Z_E$, yielding

$$\Delta\big[I(S{:}E) + D_{\text{KL}}(\rho_E \| \rho_E^{\text{th}})\big] = \Delta S_S + \beta\, \Delta\langle H_E\rangle. \tag{1.19}$$

□

Remark 1.20 (No assumption on the initial state)**.** *The proof of Lemma 1.19 uses* only *unitarity (*$\Delta S(\rho_{SE}) = 0$*) and the algebraic structure of the KL divergence. No assumption is made about the initial state* $\rho_{SE}(0)$*, the coupling strength, or the character (Markovian or non-Markovian) of the reduced dynamics. For a product thermal initial state, autonomous driving, and negligible endpoint interaction energy, the associated entropy balance reduces to the Esposito decomposition [15]:* $\Sigma = I(S{:}E;\,\tau) + D_{\text{KL}}(\rho_E(\tau) \| \rho_E^{\text{th}}) + O(\lambda)$*.*

[1] This follows from the definition of quantum relative entropy and $\ln(\rho_S \otimes \sigma_E) = \ln \rho_S \otimes \mathbb{1}_E + \mathbb{1}_S \otimes \ln \sigma_E$. See, e.g., M. M. Wilde, *Quantum Information Theory*, 2nd ed., Cambridge University Press (2017), Sec. 11; and M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press (2000), Ch. 11.

1.4.3 The Survival Identity

We now connect the information–thermodynamic identity (1.16) to the survival functional $\mathcal{S}$ defined in Section 1.2.4.

Theorem 1.21 (Survival Balance: General Form)**.** *Under Assumptions* (A1)–(A5) *of Definition 1.6, let $\rho_{SE}(t)$ evolve unitarily from an* arbitrary *(possibly correlated) initial state $\rho_{SE}(0)$. The survival functional satisfies*

$$\boxed{\beta\,\mathcal{S}[\Lambda,\tau] = -\Delta I(S{:}E) - \Delta D_{\mathrm{KL}}\big(\rho_E \,\big\|\, \rho_E^{\mathrm{th}}\big) - \beta\,\Delta\langle H_{\mathrm{ctrl}}\rangle - \beta\,\Delta\langle H_{\mathrm{int}}\rangle,} \tag{1.20}$$

where $\Delta\langle H_{\mathrm{ctrl}}\rangle = \mathrm{Tr}[\rho_S(\tau)\,H_{\mathrm{ctrl}}(\tau)] - \mathrm{Tr}[\rho_S(0)\,H_{\mathrm{ctrl}}(0)]$ is the change in the control-field energy.

For ***autonomous evolution*** *with negligible interaction-energy change at the endpoints ($H_{\mathrm{ctrl}} = 0$ and $\Delta\langle H_{\mathrm{int}}\rangle = O(\lambda)$), the weak-coupling form is*

$$\beta\,\mathcal{S}[\Lambda,\tau] = -\Delta I(S{:}E) - \Delta D_{\mathrm{KL}}\big(\rho_E \,\big\|\, \rho_E^{\mathrm{th}}\big) + O(\lambda). \tag{1.21}$$

Proof. The proof uses three ingredients: the definition of $\mathcal{S}$, the first law, and Lemma 1.19.
Step 1 (First law in weak coupling). Since $H_{\mathrm{ctrl}}(t)$ is the only explicitly time-dependent component of H, inclusive work satisfies the exact first law $W = \Delta\langle H\rangle = \Delta\langle H_R\rangle + \Delta\langle H_{\mathrm{ctrl}}\rangle + \Delta\langle H_E\rangle + \Delta\langle H_{\mathrm{int}}\rangle$.
Step 2 (Connecting Σ to the identity). From Definition 1.8 and (1.8), using $\Delta F = \Delta\langle H_R\rangle - \beta^{-1}\Delta S_S$:

$$\begin{aligned}\Sigma &= \beta(W - \Delta F) = \beta\big(W - \Delta\langle H_R\rangle\big) + \Delta S_S \\ &= \beta\big(\Delta\langle H_{\mathrm{ctrl}}\rangle + \Delta\langle H_E\rangle + \Delta\langle H_{\mathrm{int}}\rangle\big) + \Delta S_S \\ &= \big(\Delta S_S + \beta\,\Delta\langle H_E\rangle\big) + \beta\,\Delta\langle H_{\mathrm{ctrl}}\rangle + \beta\,\Delta\langle H_{\mathrm{int}}\rangle.\end{aligned} \tag{1.22}$$

By Lemma 1.19, the parenthesized term equals $\Delta I(S{:}E) + \Delta D_{\mathrm{KL}}(\rho_E \| \rho_E^{\mathrm{th}})$. Hence

$$\Sigma = \Delta I(S{:}E) + \Delta D_{\mathrm{KL}}\big(\rho_E \,\big\|\, \rho_E^{\mathrm{th}}\big) + \beta\,\Delta\langle H_{\mathrm{ctrl}}\rangle + \beta\,\Delta\langle H_{\mathrm{int}}\rangle. \tag{1.23}$$

Step 3 (Survival functional). $\beta\,\mathcal{S} = -\Sigma$ by (1.10), yielding (1.20). For $H_{\mathrm{ctrl}} = 0$ and weak endpoint coupling, the last term is $O(\lambda)$, yielding (1.21). □

Remark 1.22 (Nature of the result)**.** *Equation* (1.20) *is an exact* accounting identity*, not an inequality or optimality bound. It establishes that the sign is governed by the declared initial resources and endpoint bookkeeping: changes in correlation, bath displacement, control energy, and interaction energy. The identity itself is dynamics-agnostic and does not make non-Markovianity a resource. A memory kernel matters only when a model shows that it preserves correlations across epochs and later permits their consumption.*

Remark 1.23 (Scope of the theorem)**.** *Theorem 1.21 holds for* any *initial state $\rho_{SE}(0)$—product or correlated. The proof requires only Assumptions* (A1)–(A5) *of Definition 1.6 and the definitions of $\mathcal{S}$, $I(S{:}E)$, and D_{KL}. No assumption about the reduced dynamics (Markovian, non-Markovian, or otherwise) is needed. An interacting system may carry correlations at the chosen interval boundary. When present and subsequently consumed, those correlations are one thermodynamic resource; persistence does not require that every model use this resource.*

1.4.4 Three Regimes of Survival

We specialize to the autonomous, endpoint-decoupled weak-coupling convention ($H_{\text{ctrl}} = 0$ and the $O(\lambda)$ interaction term suppressed). This isolates the initial-resource accounting.

Corollary 1.24 (Three Regimes)**.** *Under autonomous evolution, identity* (1.21) *identifies three regimes:*

1. ***Product initial state*** *($I(S{:}E;\,0) = 0$, $D_{\mathrm{KL}}(\rho_E(0)\|\rho_E^{\mathrm{th}}) = 0$): Both ΔI and ΔD_{KL} are increases from zero to non-negative final values, so*

$$\beta\,\mathcal{S} = -\big(I(S{:}E;\,\tau) + D_{\mathrm{KL}}(\rho_E(\tau)\|\rho_E^{\mathrm{th}})\big) \le 0.$$

   This product-state unitary balance is dynamics-agnostic and is compatible with, but logically distinct from, the restricted Davies result (Theorem 1.13).

2. ***Correlated initial state*** *($I(S{:}E;\,0) > 0$): If the dynamics* consumes *pre-existing correlations ($\Delta I < 0$, i.e., $I(S{:}E;\,\tau) < I(S{:}E;\,0)$), the first term contributes* positively *to $\mathcal{S}$. Provided*

$$|\Delta I(S{:}E)| > \Delta D_{\mathrm{KL}}\big(\rho_E \,\big\|\, \rho_E^{\mathrm{th}}\big), \tag{1.24}$$

   the survival functional is strictly positive: $\mathcal{S} > 0$. The agent has converted pre-existing correlations into usable free energy.

3. ***Upper bound:*** *Since $I(S{:}E;\,\tau) \ge 0$ and $D_{\mathrm{KL}}(\rho_E(\tau)\|\rho_E^{\mathrm{th}}) \ge 0$, the maximum survival gain is bounded by*

$$\beta\,\mathcal{S} \le I(S{:}E;\,0) + D_{\mathrm{KL}}\big(\rho_E(0) \,\big\|\, \rho_E^{\mathrm{th}}\big). \tag{1.25}$$

   The thermodynamic profit cannot exceed the total initial "resource budget"—the pre-existing correlations plus the initial displacement of the bath from equilibrium.

1.4.5 The Correlation Battery

The three regimes of Corollary 1.24 raise a natural question: *where do the initial correlations $I(S{:}E;\,0) > 0$ come from?*

Remark 1.25 (The Correlation Battery)**.** *The answer is: from* ***prior interaction epochs****. A persistent agent does not begin its existence in a product state. Over any interaction interval, unitary evolution generically builds system–environment correlations ($\Delta I > 0$), at a thermodynamic cost ($\mathcal{S} < 0$ during this phase by Corollary 1.24(i)). A non-Markovian dilation can preserve correlations across the chosen epoch boundary, but whether they are later consumed with $\mathcal{S} > 0$ must be checked from the full balance.*

The process is analogous to a ***battery****:*

- ***Charging phase*** *(correlation building, $\Delta I > 0$): the agent "pays" free energy to build system–environment correlations. $\mathcal{S} < 0$.*
- ***Discharging phase*** *(correlation consumption, $\Delta I < 0$): the agent extracts free energy from the stored correlations. $\mathcal{S} > 0$.*

The repeated-product Born approximation removes this resource by construction at each reset. A reduced semigroup law by itself does not state that every microscopic dilation is uncorrelated. Conversely, a memory kernel only represents retained history; it does not guarantee that the history is usable as thermodynamic work.

Crucially, global thermodynamics remains respected. For any full cycle starting from an uncorrelated thermal state ($I(S{:}E;\,0)=0$, $D_{\mathrm{KL}}(\rho_E(0)\|\rho_E^{\mathrm{th}})=0$), the total survival functional satisfies

$$\beta\,\mathcal{S}[0,t] = -\Sigma[0,t] \;\le\; 0 \qquad \textit{(second law).} \tag{1.26}$$

The local positivity $\mathcal{S}[t^*,t]>0$ during the discharging phase is strictly funded by the free energy dissipated during the earlier charging phase (see Proposition 1.26 for the formal decomposition).

Proposition 1.26 (Full-cycle closure)**.** *Under the conditions of Theorem* 1.21 *with autonomous evolution ($H_{\mathrm{ctrl}}=0$) and negligible interaction-energy changes at the endpoints of each displayed balance, partition $[0,\tau]$ at any intermediate time t^* into a charging phase $[0,t^*]$ and a discharging phase $[t^*,\tau]$.*

(i) ***Charging*** *(product initial state, $I(S{:}E;\,0)=0$). By Corollary* 1.24(i)*,*

$$\beta\,\mathcal{S}[0,t^*] = -I(S{:}E;\,t^*) - D_{\mathrm{KL}}\big(\rho_E(t^*)\,\big\|\,\rho_E^{\mathrm{th}}\big) \;\le\; 0. \tag{1.27}$$

(ii) ***Discharging*** *(correlated initial state at t^*). Applying* (1.21) *to $[t^*,\tau]$:*

$$\beta\,\mathcal{S}[t^*,\tau] = -\big(I(S{:}E;\,\tau) - I(S{:}E;\,t^*)\big) - \big(D_{\mathrm{KL}}(\rho_E(\tau)\|\rho_E^{\mathrm{th}}) - D_{\mathrm{KL}}(\rho_E(t^*)\|\rho_E^{\mathrm{th}})\big), \tag{1.28}$$

which is positive whenever the decrease in correlations dominates the change in bath displacement (Corollary 1.24(ii)*).*

(iii) ***Full cycle.*** *Since $\mathcal{S}$ is additive over concatenated intervals, $\beta\,\mathcal{S}[0,\tau] = \beta\,\mathcal{S}[0,t^*] + \beta\,\mathcal{S}[t^*,\tau]$. Equivalently, applying* (1.21) *directly to $[0,\tau]$ with $I(S{:}E;\,0)=0$:*

$$\beta\,\mathcal{S}[0,\tau] = -I(S{:}E;\,\tau) - D_{\mathrm{KL}}\big(\rho_E(\tau)\,\big\|\,\rho_E^{\mathrm{th}}\big) \;\le\; 0. \tag{1.29}$$

The net thermodynamic profit over the full cycle is non-positive—the "interest" paid during charging meets or exceeds the "dividend" collected during discharging. But the local *positivity of $\mathcal{S}$ during discharge* (1.28) *is what enables the agent to survive through intervals that would kill a memoryless system.*

Proof. The survival functional is additive over concatenated intervals:

$$\mathcal{S}[0,\tau] = \underbrace{(\Delta F[0,t^*] - W[0,t^*])}_{\mathcal{S}[0,t^*]} + \underbrace{(\Delta F[t^*,\tau] - W[t^*,\tau])}_{\mathcal{S}[t^*,\tau]},$$

since both ΔF and W decompose additively. Items (i) and (iii) then follow from Theorem 1.21 (autonomous case) applied to $[0,t^*]$ and $[0,\tau]$ respectively, each starting from a product state. Item (ii) follows from Theorem 1.21 applied to $[t^*,\tau]$ with correlated initial state $\rho_{SE}(t^*)$. Inequality (1.29) holds because $I(S{:}E;\,\tau)\ge 0$ and $D_{\mathrm{KL}}(\rho_E(\tau)\|\rho_E^{\mathrm{th}})\ge 0$. □

1.4.6 Connection to Non-Markovianity Measures

Remark 1.27 (The Born Approximation Destroys the Resource)**.** *Under the Born (product-state) approximation, every infinitesimal time step begins from* $\rho_{SE} \approx \rho_S \otimes \rho_E^{\mathrm{th}}$*, enforcing* $I(S{:}E) = 0$ *at all times. Corollary 1.24(i) then guarantees* $\mathcal{S} \leq 0$ *for* every *finite interval. The Born approximation does not merely simplify the dynamics—it* eliminates the thermodynamic resource *(system–environment correlations) that would otherwise be available.*

Remark 1.28 (Connection to BLP Non-Markovianity)**.** *The Breuer–Laine–Piilo (BLP) measure of non-Markovianity [9] is defined via the temporary increase of trace distance between pairs of initial states:* $\mathcal{N}_{\mathrm{BLP}} := \max_{\rho_{1,2}} \int_{\dot{D}>0} \frac{d}{dt} D(\rho_1(t), \rho_2(t))\, dt$*. BLP backflow and correlation consumption can coincide in particular models, but neither condition implies the other in general [26]. The BLP measure therefore diagnoses one reduced-dynamics notion of non-Markovianity; it is not a witness for the sign of* $\mathcal{S}$ *without the accompanying energy and correlation balance.*

Remark 1.29 (Consistency with the Sagawa–Ueda Framework)**.** *In the Sagawa–Ueda framework [27, 28], measurement-based feedback permits* $\Sigma \geq -I_{\mathrm{feedback}}$*, where* I_{feedback} *is the mutual information gained through measurement. Pre-existing correlations are an analogous accounting resource, but a memory kernel is not itself a measurement record and the two protocols must not be identified. The total system (agent + bath) still satisfies* $\Sigma_{\mathrm{total}} \geq 0$*; the apparent "profit" for the agent is paid for by the correlations consumed from the system–environment entanglement. The bound* (1.25) *is the non-Markovian analogue of the Sagawa–Ueda bound* $\beta\,\mathcal{S} \leq I_{\mathrm{feedback}}$*.*

1.4.7 Mechanism: The Surfer Analogy

The physical mechanism admits an intuitive picture.

- **The Markovian Agent (The Stone):** A stone thrown into the ocean sinks. It interacts with the water only at the instant of contact, dissipates its kinetic energy, and thermalizes ($\mathcal{S} \leq 0$). Each collision builds system–environment correlations that are immediately discarded (Born approximation), so $I(S{:}E) = 0$ at all times. The wave structure is invisible to it.
- **The Non-Markovian Agent (The Surfer):** A surfer carries *memory* of past wave patterns—encoded in the correlations $I(S{:}E;\, t_0) > 0$ built up over previous interactions (the "charging phase" of Remark 1.25). During backflow intervals ($\Delta I < 0$), the surfer *spends* these stored correlations to extract free energy from the wave itself. The surfer remains far from equilibrium not by fighting the environment, but by converting temporal correlations into thermodynamic profit.

Remark 1.30 (Thermodynamic Rectification)**.** *The "surfing" mechanism is* ***thermodynamic rectification****: the memory kernel* $\mathcal{K}(t, s)$ *functions as a temporal filter that enables the system to accumulate correlations during one phase and consume them during another. Formally, the kernel enables access to the resource* $I(S{:}E;\, 0)$ *accumulated during previous interaction epochs—converting the environment's temporal correlations into the system's structural persistence via the* ΔI *term in Theorem 1.21.*

Remark 1.31 (Memory as Implicit Maxwell's Demon)**.** *The memory kernel functions as an* implicit Maxwell's demon*. A Markovian system interacts with each environmental fluctuation exactly once, at the moment of contact; the Born approximation resets $I(S{:}E) = 0$ after each step. A non-Markovian system retains a trace of past fluctuations (via $\mathcal{K}(t, s)$ with $s < t$) and can exploit correlations between past and present environmental states. This is not a violation of the second law but an instance of the Sagawa–Ueda generalization: the demon's cost is paid in the currency of memory maintenance (Landauer erasure), a point we quantify in Section 1.7. The total budget for "demonic profit" is capped by the bound* (1.25)*.*

1.5 Operational Temporal Relations

The accounting identity does not derive a temporal order. This section records two heuristic structures that can be studied after a reduced memory model is specified: an operational influence relation between times and a comparison with pointer-record growth. Neither is a universal partial order, and no HAFF theorem is inherited.

1.5.1 Thresholded Memory-Influence Relation

A non-Markovian memory kernel $\mathcal{K}(t, s)$ defines a causal link between a past state at s and the present dynamics at t. We define a thresholded relation based on the effective response.

Definition 1.32 (Operational memory-influence relation)**.** *For a declared reduced model, let $\mathcal{R}_{t\leftarrow s}$ denote the linear response of the predicted state or read-out at time t to a perturbation applied at time $s < t$. At tolerance $\epsilon > 0$, define*

$$s \rightsquigarrow_K t \quad \Longleftrightarrow \quad \|\mathcal{R}_{t\leftarrow s}\| > \epsilon. \tag{1.30}$$

The response norm, perturbation class, and tolerance are part of the protocol. This relation need not be transitive or antisymmetric, so it is not asserted to be a partial order.

An ideal delta-memory reduced model has no off-diagonal response in this description. A finite-memory model can have $s \rightsquigarrow_K t$ within its response window. This statement is about the chosen reduced model, not about the existence of time or a universal thermodynamic arrow.

1.5.2 A Fisher-Accretion Conjecture

One may ask whether a specified survival optimizer tends to retain parameter information in the forward protocol direction. The following statement is a conjecture; the survival identity alone does not create statistical irreversibility.

Conjecture 1.33 (Fisher Information Accretion)**.** *Let $\mathcal{I}_F(\theta;\, \rho(t))$ denote the Fisher information contained in the system state $\rho(t)$ regarding a parameter θ encoded in the environment at time $s < t$. For an agent whose dynamics maximize the survival functional* (1.9)*, the time-averaged Fisher information satisfies*

$$\overline{\frac{d}{dt}\mathcal{I}_F(\theta;\, \rho(t))} \geq 0, \tag{1.31}$$

where the overbar denotes a time average over scales larger than the bath correlation time τ_B.

Remark 1.34 (Status of the conjecture)**.** *The survival balance supplies no optimization domain, controller class, or monotonicity theorem for Fisher information. Mutual information and Fisher information are distinct functionals, and information backflow can make either non-monotone. Establishing the displayed inequality requires a concrete controlled process and objective; it is not used as an input to later theorems.*

1.5.3 The Bridge to HAFF

Revised HAFF Paper F [44] distinguishes an algebra-inclusion preorder from a protocol-relative redundancy diagnostic and does not prove generic monotone record expansion. The following comparison is therefore a heuristic interface, not a theorem.

Conjecture 1.35 (Conditional pointer-record comparison)**.** *Suppose a concrete branching/record model supplies (i) a pointer coarse-graining* Φ*, (ii) a fixed fragmentation, and (iii) a monotone record channel for a specified observation window. One may then test whether an influence* $s \rightsquigarrow_K t$ *is accompanied by nested record algebras,*

$$\mathcal{R}_s \subseteq \mathcal{R}_t. \tag{1.32}$$

The memory-influence relation alone does not imply this inclusion; backflow, record erasure, or a changed fragmentation can violate it.

Remark 1.36 (Scope of the Correspondence)**.** *Conjecture 1.35 states a test that can succeed in a branching record model. No general monotonicity, embedding, or duality between T-DOME and HAFF is established.*

Remark 1.37 (Dynamical and Algebraic Time)**.** *The two papers use distinct, protocol-dependent notions:*

	Paper F (HAFF)	***Paper I (T-DOME)***
Nature	*Algebraic*	*Dynamical*
Mechanism	*Record/algebra diagnostics*	*Reduced-model response*
Formalism	*Preorders/diagnostics*	*Relation* $\rightsquigarrow_K$ *on times*
Asymmetry source	*Declared record protocol*	*Declared kernel/response model*
Domain	*Classical (pointer) sector*	*Full quantum dynamics*

An explicit morphism between them remains open.

Remark 1.38 (The Seed and the Tree)**.** *The seed/tree language is a programme metaphor. It supplies neither a time variable nor a theorem connecting memory dynamics to geometry.*

1.6 Worked Example: The Quantum Predictive Agent

To illustrate free-energy monotonicity and a correlation-funded backflow interval *within one explicit model*, we employ an open quantum system model: the spin-boson model with Lorentz–Drude spectral density, which admits an exact analytic solution for the decoherence dynamics [10].

1.6.1 Model Setup

The total Hamiltonian is $H = H_S + H_B + H_I$. The agent is a two-level system with energy gap ω_0: $H_S = \frac{\omega_0}{2}\sigma_z$. The environment is a bosonic bath: $H_B = \sum_k \omega_k b_k^\dagger b_k$. The interaction is of the pure-dephasing form $H_I = \sigma_z \otimes \sum_k (g_k\, b_k + g_k^*\, b_k^\dagger)$.

The spectral density $J(\omega) = \sum_k |g_k|^2 \delta(\omega - \omega_k)$ characterizes the environment. We choose the Lorentz–Drude form:

$$J(\omega) = \frac{2\lambda\,\gamma\,\omega}{\omega^2 + \gamma^2}, \tag{1.33}$$

where λ is the reorganization energy and γ is the bath memory rate (inverse correlation time $\tau_B = 1/\gamma$). We place the system in the low-temperature regime $\beta\omega_0 \gg 1$ (i.e., $k_B T \ll \omega_0$). The bath correlation function in the $T \to 0$ limit is $C(t) = \lambda\gamma\, e^{-\gamma|t|}$, so the parameter γ directly controls the bath memory depth. For $\beta\omega_0 \geq 10$ the finite-temperature corrections to all quantities below are of order $O(e^{-\beta\omega_0}) \lesssim 5 \times 10^{-5}$ and are neglected throughout.[2]

1.6.2 Exact Decoherence Function

For the pure-dephasing spin-boson model in the $T \to 0$ limit, the off-diagonal element of the reduced density matrix $\rho_{01}(t) = \rho_{01}(0)\, p(t)$ is governed by the **decoherence function** [10]:

$$p(t) = e^{-\gamma t/2}\left[\cos(\Omega t) + \frac{\gamma}{2\Omega}\,\sin(\Omega t)\right], \tag{1.34}$$

where $\Omega := \frac{1}{2}\sqrt{4\lambda\gamma - \gamma^2}$. This solution is exact for the Lorentz–Drude spectral density.

Remark 1.39 (Non-Markovian Regime)**.** *The character of the dynamics is controlled by the discriminant $\Delta := 4\lambda\gamma - \gamma^2 = \gamma(4\lambda - \gamma)$:*

- *$\gamma > 4\lambda$ ($\Delta < 0$): Ω is imaginary, $p(t)$ decays monotonically. The dynamics is Markovian (no backflow).*
- *$\gamma = 4\lambda$ ($\Delta = 0$): Critical damping. $p(t) = (1 + \gamma t/2)\, e^{-\gamma t/2}$.*
- *$\gamma < 4\lambda$ ($\Delta > 0$): Ω is real and positive. $p(t)$ oscillates with envelope $e^{-\gamma t/2}$. The dynamics is* non-Markovian*: intervals where $|p(t)|$ increases correspond to information backflow [9].*

The non-Markovian regime $\gamma < 4\lambda$ is thus the regime of structured, long-memory baths.

1.6.3 Quantitative Evaluation

We now evaluate the survival functional explicitly. For the pure-dephasing model, populations are conserved ($p_0(t) = p_0(0)$, $p_1(t) = p_1(0)$), and the non-equilibrium free energy depends only on the coherence:

$$F(\rho(t)) - F(\rho_{\text{eq}}) = \beta^{-1}\, D_{\text{KL}}(\rho(t) \| \rho_{\text{eq}}). \tag{1.35}$$

[2] All plots and numerical values use the standard $T \to 0$ analytic expression for the decoherence function (1.34) (see Breuer and Petruccione [10], Sec. 12.3, for the Lorentz–Drude pure-dephasing solution), which provides an accurate proxy in the low-temperature regime; β is a well-defined bookkeeping parameter and β^{-1} a finite energy scale.

For a qubit with initial state $\rho(0) = \frac{1}{2}(\mathbb{1} + \vec{r}\cdot\vec{\sigma})$ and $r_z = 0$ (maximal coherence in the x–y plane), the relative entropy reduces to $D_{\mathrm{KL}}(\rho(t)\|\rho_{\mathrm{eq}}) \approx |p(t)|^2\,|\rho_{01}(0)|^2$ to leading order in the coherence (see, e.g., [10]). Since there is no external driving ($H_{\mathrm{ctrl}} = 0$, $W = 0$), the survival functional is simply

$$\beta\,\mathcal{S}(t) = D_{\mathrm{KL}}(\rho(t)\|\rho_{\mathrm{eq}}) - D_{\mathrm{KL}}(\rho(0)\|\rho_{\mathrm{eq}}) \propto |p(t)|^2 - 1. \tag{1.36}$$

The proportionality in (1.36) is specific to the **pure-dephasing model** with the chosen maximally coherent initial state ($r_z = 0$) and measurement in the pointer basis (σ_z). Under these conditions, the exact solution [10] ensures that population terms vanish from the free energy ($\Delta\langle H_S\rangle = 0$), leaving only the coherence contribution: $\beta\,\mathcal{S} = -\Delta S_S$ depends only on the coherence trajectory $|p(t)|$. The proxy $|p(t)|^2$ thus rigorously captures the sign and monotone behaviour of $\beta\,\mathcal{S}$; the exact numerical prefactor depends on the initial state and on β, but the qualitative conclusion—$\mathcal{S} > 0$ during backflow intervals—is robust and does not depend on the proxy normalization.

For a Markovian evolution, $|p(t)|$ decreases monotonically, so $|p(t)|^2 - 1 \le 0$ for all t: $\mathcal{S} \le 0$ always in this autonomous model (consistent with Theorem 1.13). For non-Markovian evolution with $\gamma < 4\lambda$, the oscillations in $p(t)$ produce intervals where $|p(t)|$ *increases after a previous decrease*, i.e., the system *re-coheres*.

Concrete parameters. Set $\omega_0 = 1$ (energy units), $\lambda = 1$, $\gamma = 0.5$ (deep non-Markovian regime: $\gamma/4\lambda = 0.125 \ll 1$). Then:

$$\Omega = \tfrac{1}{2}\sqrt{4\cdot 1\cdot 0.5 - 0.25} = \tfrac{1}{2}\sqrt{1.75} \approx 0.661. \tag{1.37}$$

The decoherence function (1.34) first reaches $p(t^*) = 0$ at $t^* = [\pi/2 + \arctan\big(\gamma/(2\Omega)\big)]/\Omega \approx 1.93/\Omega \approx 2.92$ (in units of ω_0^{-1}), where the system has fully decohered. Subsequently, the environment *returns* coherence: $|p(t)|$ increases, reaching a local maximum $|p(t_1)| \approx 0.31$ at $t_1 \approx 4.75/\omega_0$.

Over the backflow interval $[t^*, t_1]$, for the pure-dephasing qubit with the chosen initial state ($r_z = 0$, maximal coherence) and in the autonomous setting ($H_{\mathrm{ctrl}} = 0$, $W = 0$), the survival proxy (1.36) gives

$$\beta\,\mathcal{S}[t^*, t_1] \propto |p(t_1)|^2 - |p(t^*)|^2 \approx 0.093 - 0 = 0.093 > 0. \tag{1.38}$$

Equivalently, $\mathcal{S} \approx 0.093\,\beta^{-1}$ in the bookkeeping units set by β. The agent has gained a dimensionless survival advantage $\beta\,\mathcal{S} \approx +0.093$ *with zero work input* (autonomous evolution, $H_{\mathrm{ctrl}} = 0$), solely by exploiting the non-Markovian backflow. Figure 1.1 illustrates the contrast between Markovian and non-Markovian evolution.

Consistency with Theorem 1.21 and Proposition 1.26. Since this is autonomous evolution ($H_{\mathrm{ctrl}} = 0$), identity (1.21) applies exactly: $\beta\,\mathcal{S} = -\Delta I(S{:}E) - \Delta D_{\mathrm{KL}}(\rho_E\|\rho_E^{\mathrm{th}})$. The example realizes the *correlation battery* of Remark 1.25, with the charge–discharge decomposition of Proposition 1.26:

- **Charging** ($[0, t^*]$, eq. (1.27)): the system decoheres, building correlations $I(S{:}E;\,t^*) > 0$ at the cost of $\mathcal{S} < 0$.
- **Discharging** ($[t^*, t_1]$, eq. (1.28)): the correlations are consumed ($\Delta I < 0$ over this interval), returning $\beta\,\mathcal{S} \approx +0.093 > 0$.

The bound (1.25) is satisfied: $\beta\,\mathcal{S}[t^*, t_1] = 0.093 \le I(S{:}E;\,t^*)$. Full-cycle closure (1.29) is confirmed: $\beta\,\mathcal{S}[0, t_1] < 0$.

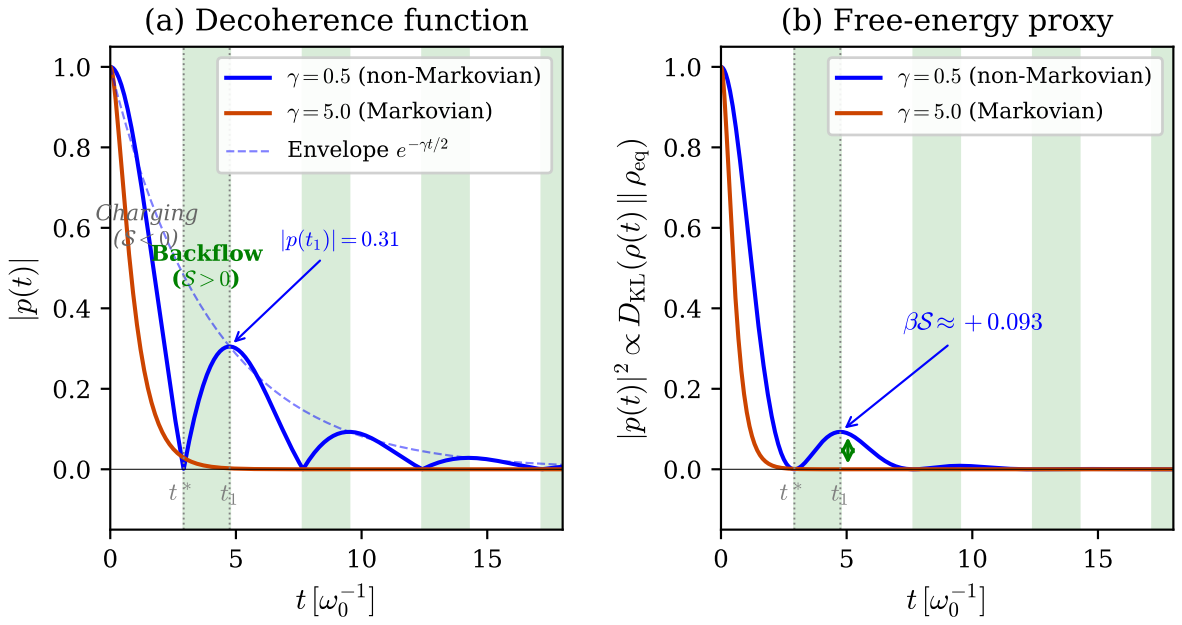


Figure 1.1: Pure-dephasing spin-boson model (Section 1.6) with Lorentz–Drude spectral density (1.33). **Parameters:** $\omega_0 = 1$ (energy unit), $\lambda = 1$ (reorganization energy). **Units:** all times in ω_0^{-1}; energies in $\hbar\omega_0$. **Regime:** low temperature ($\beta\omega_0 \gg 1$); the standard $T \to 0$ analytic expression (1.34) [10] is used as an accurate proxy. **(a)** Decoherence amplitude $|p(t)|$ (eq. (1.34)). Blue: non-Markovian ($\gamma = 0.5$, $\gamma/4\lambda = 0.125$). Orange: Markovian ($\gamma = 5.0$, $\gamma/4\lambda = 1.25$). Dashed: exponential envelope $e^{-\gamma t/2}$. Green bands indicate backflow (Remark 1.39: $d|p|/dt > 0$, $\Gamma(t) < 0$ per (1.39)). **(b)** Survival proxy $|p(t)|^2 \propto \beta\,\mathcal{S}$ (eq. (1.36)). At the first revival ($t_1 = \pi/\Omega \approx 4.75\,\omega_0^{-1}$), the non-Markovian agent achieves $\beta\,\mathcal{S}[t^*, t_1] \approx +0.093$ (eq. (1.38)), consistent with the closed-form prediction (1.40), funded by the consumption of pre-existing correlations (Proposition 1.26). The Markovian comparison decays monotonically: $\mathcal{S} \leq 0$ in this autonomous model (Theorem 1.13).

Instantaneous decoherence rate. The rate of coherence loss is

$$\Gamma(t) := -\frac{d}{dt}\ln|p(t)| = \frac{\gamma}{2} - \frac{\Omega\sin(\Omega t) + \frac{\gamma}{2}\cos(\Omega t)}{\cos(\Omega t) + \frac{\gamma}{2\Omega}\sin(\Omega t)}\,. \tag{1.39}$$

In the Markovian limit $\gamma \gg 4\lambda$, $\Gamma(t) \to \gamma/2 > 0$ for all t (monotone decoherence). In the non-Markovian regime $\gamma < 4\lambda$, $\Gamma(t)$ oscillates and becomes *negative* during the backflow intervals where $|p(t)|$ increases. These are precisely the intervals where $\mathcal{S} > 0$.

Closed-form revival amplitude. The decoherence function (1.34) can be written as $p(t) = R\,e^{-\gamma t/2}\cos(\Omega t - \phi)$, where $R = \sqrt{1 + (\gamma/2\Omega)^2}$ and $\phi = \arctan(\gamma/2\Omega)$, with $R\cos\phi = 1$. The extrema of $|p(t)|$ occur at $t_n = n\pi/\Omega$ ($n = 0, 1, 2, \ldots$), and the first revival peak after the first zero is at $t_1 = \pi/\Omega$. Its amplitude is *exactly*

$$|p(t_1)| = e^{-\gamma\pi/(2\Omega)}, \qquad \beta\,\mathcal{S}[t^*, t_1] \approx |p(t_1)|^2 = e^{-\gamma\pi/\Omega}. \tag{1.40}$$

This is the paper's central computable prediction: the survival gain at first backflow is determined by a single dimensionless ratio γ/Ω.

Remark 1.40 (Parameter Survey)**.** *Table 1.1 demonstrates the transition from the Markovian regime ($\mathcal{S} \leq 0$) to the non-Markovian regime ($\mathcal{S} > 0$) as the bath memory rate γ decreases below the critical value 4λ. All entries use $\omega_0 = 1$, $\lambda = 1$, $W = 0$ (autonomous evolution), with revival amplitudes computed from* (1.40)*.*

γ	$\gamma/4\lambda$	*Regime*	$\lvert p(t_1)\rvert$	$\beta\,\mathcal{S}(t_1)$	$\Gamma_{\min}$
20.0	5.0	*Markov*	—	≤ 0	> 0
4.0	1.0	*Critical*	—	≤ 0	$= 0$
2.0	0.50	*Non-Markov*	0.043	$+0.002$	< 0
1.0	0.25	*Non-Markov*	0.163	$+0.027$	< 0
0.5	0.125	*Deep NM*	0.305	$+0.093$	< 0
0.1	0.025	*Deep NM*	0.605	$+0.37$	< 0

Table 1.1: Survival functional at first backflow revival as a function of the bath memory rate γ, for the pure-dephasing spin-boson model with Lorentz–Drude spectral density. $|p(t_1)|$ is computed from (1.40); $\Gamma_{\min}$ is the sign of the minimum of the instantaneous decoherence rate (1.39). The transition $\mathcal{S} \leq 0 \to \mathcal{S} > 0$ occurs precisely at the non-Markovian threshold $\gamma = 4\lambda$. For $\gamma = 0.1$ (deep non-Markovian), the agent achieves $\beta\,\mathcal{S} \approx +0.37$ per backflow cycle in the autonomous setting ($H_{\text{ctrl}} = 0$).

Remark 1.41 (The Two Regimes: Summary)**.**

	Markovian *($\gamma = 20$)*	***Non-Markovian*** *($\gamma = 0.5$)*
$\gamma/4\lambda$	5.0 *(overdamped)*	0.125 *(underdamped)*
τ_B	$0.05\,\omega_0^{-1}$	$2.0\,\omega_0^{-1}$
$p(t)$	*Monotone decay*	*Oscillatory with envelope*
$\lvert p(t_1)\rvert$ *at first revival*	0 *(no revival)*	≈ 0.31
$\beta\,\mathcal{S}$ *at revival*	≤ 0	$\approx +0.093$
$\Gamma(t)$	> 0 *always*	*Oscillates,* < 0 *during backflow*
Interpretation	*Stone (sinks)*	*Surfer (rides backflow)*

The non-Markovian agent achieves $\beta\,\mathcal{S} \approx +0.093$ per backflow cycle (autonomous, $H_{\text{ctrl}} = 0$), while the Markovian agent can only lose free energy. As the coupling deepens ($\gamma/4\lambda \to 0$), the revival amplitude grows and $\mathcal{S}$ increases (Table 1.1), bounded above by $\beta\,\mathcal{S} \leq I(S{:}E;\, t^*)$ (Corollary 1.24(iii)).

1.7 The Cost of Memory

The preceding identity allows a correlated initial resource to fund a later interval. It does not assign a universal energetic price to storing or maintaining that resource. This section separates Landauer's erasure bound from a conditional archive-overload model used only to motivate Paper II.

1.7.1 The Landauer Debt

Landauer's principle prices a logically irreversible reset, not passive storage, sensing, correlation maintenance, or reversible computation.

Remark 1.42 (Conditional erasure cost)**.** *If a declared memory implementation irreversibly erases N_{erase} unbiased bits into a bath at temperature T, its reset work obeys*

$$W_{\text{reset}} \geq k_B T \ln 2 \, N_{\text{erase}}. \tag{1.41}$$

No lower bound proportional to total stored mutual information follows without an implementation model. Reversible storage and reversible measurement can in principle avoid this reset cost until a logically irreversible operation is performed [20, 7]. The former universal "Landauer Cost of Memory" proposition is withdrawn.

1.7.2 Raw Archive Overload Scenario

The crisis arises from the scaling of $\mathcal{I}_{\text{stored}}$ with time. To quantify this, we borrow two quantities from computational mechanics [14, 30]:

Definition 1.43 (Entropy Rate and Predictive Information)**.** *Let $\{X_t\}$ be the stochastic process describing the environment's influence on the agent (e.g., the sequence of bath correlation values).*

1. *The **entropy rate** of the environment is*

$$h_\mu := \lim_{n\to\infty} H(X_n \mid X_{n-1}, \ldots, X_1), \tag{1.42}$$

   measuring the intrinsic unpredictability per time step.

2. *The **predictive information** (excess entropy) is*

$$I_{\text{pred}} := I(\overleftarrow{X}; \overrightarrow{X}) = \sum_{k=1}^{\infty} \big[H(X_k) - h_\mu\big], \tag{1.43}$$

   where $\overleftarrow{X}$ and $\overrightarrow{X}$ denote the past and future half-chains. This is the total amount of information about the future that is encoded in the past—the useful *memory.*

Finite predictive information does not by itself prove that a finite physical memory captures every useful feature, and positive entropy rate does not lower-bound predictive information by $h_\mu \tau_{\text{mem}}$. In particular, a stationary mixing process commonly has finite excess entropy.

Remark 1.44 (Conditional archive-overload scenario)**.** *Consider a deliberately specified implementation: an uncompressed raw archive receives records at rate r_{raw} bits per unit time, has capacity M bits, and evaluates the entire archive once per control cycle. Without deletion it fills after M/r_{raw}. If it irreversibly resets discarded records at rate r_{erase}, Landauer gives*

$$\dot{W}_{\text{reset}} \geq k_B T \ln 2 \, r_{\text{erase}}. \tag{1.44}$$

For a long-memory or non-stationary source with divergent I_{pred}, the amount of predictively relevant history may also grow, but that is an additional source assumption. This archive can exceed a finite storage, throughput, or reset budget. The result is an engineering overload scenario, not a universal thermodynamic catastrophe and not a theorem that every memory implementation must compress.

1.7.3 Conditional Motivation for Compression

An agent using the raw-archive implementation may respond by increasing capacity, lowering sample rate, reversible uncomputation, sufficient statistics, lossy compression, or accepting stale data. Paper II studies one coordinate-selection model among these options.

- **Compression requires a criterion.** To decide what to keep and what to erase, the agent needs a *relevance function*—a mapping from stored correlations to survival value. This is a reference frame that ranks information by its contribution to $\mathcal{S}$.
- **Coordinate selection uses a reference frame.** If the admissible compressor is restricted to retaining a subset of supplied coordinates, choosing that subset privileges a basis. More general invariant or learned sufficient-statistic compressors need not have this form.

Remark 1.45 (The Origin of Paper II)**.** *Remark 1.44 supplies Paper II with a conditional motivation: a bounded agent attempting to evaluate an uncompressed archive needs some resource-allocation policy. It does not prove that compression, Clifford coordinates, spontaneous symmetry breaking, or a "self" is thermodynamically necessary. Those are modelling choices to be assessed in Paper II.*

1.8 Numerical Demonstration

The preceding sections establish an autonomous thermodynamic baseline and an accounting identity. The following HMM is a separate classical performance illustration: it compares finite observation windows with a Bayes filter in an aliased environment. Reward here is not the thermodynamic survival functional, so the demonstration does not test either theorem. Full code and parameters are provided for reproducibility.

1.8.1 Model

Environment. A two-hidden-state HMM with aliased observations. The hidden state $s_t \in \{0, 1\}$ evolves as a persistent Markov chain with $\Pr(s_{t+1} = s_t) = 1-\varepsilon$; the parameter $\varepsilon \in [10^{-3}, 10^{-1}]$ controls the correlation length $\ell \sim 1/\varepsilon$. Observations $o_t \in \{A, B\}$ are aliased: $\Pr(o_t = A \mid s_t = 0) = 0.5+\delta$, $\Pr(o_t = A \mid s_t = 1) = 0.5-\delta$, with $\delta = 0.05$ (mutual information $I(O; S) \approx 0.007$ bits). Reward: $r_t = 1$ if $a_t = s_t$, 0 otherwise.

Agents. All agents use the true model parameters and compute exact Bayesian posteriors; the only difference is how many observations each agent retains.

- **Markov-k** ($k \in \{1, 2, 4, 8\}$): runs an exact Bayes filter over the most recent k observations (sliding window, uniform prior at each window start); acts by MAP.
- **Memory (Bayes filter)**: maintains the full belief state $b_t = \Pr(s_t = 1 \mid o_{1:t})$ via the exact predict–update cycle over all past observations; acts by MAP.

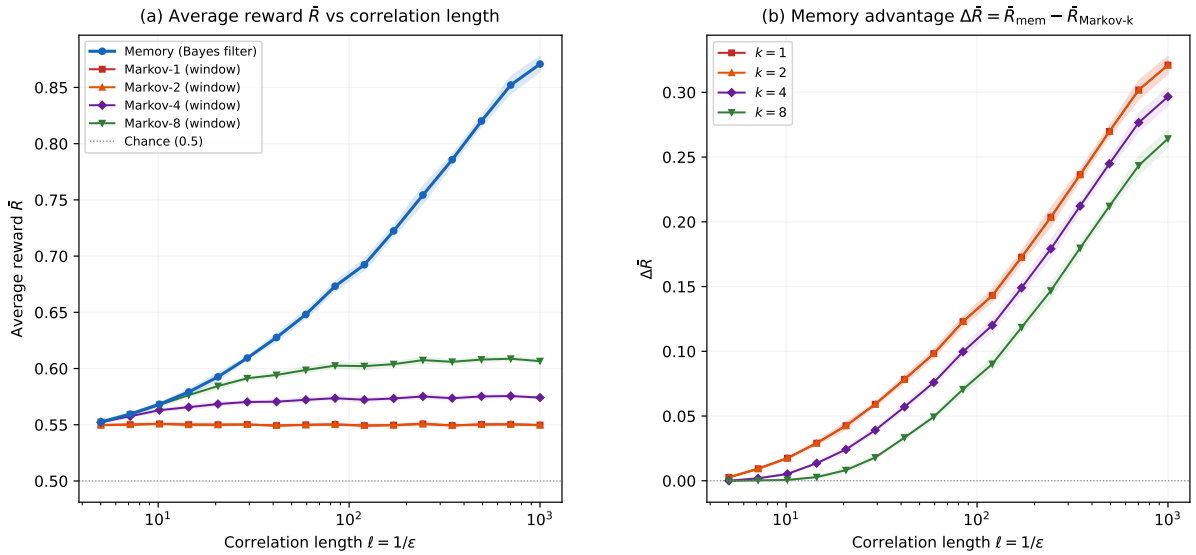


Figure 1.2: **Finite-window saturation and Bayes-filter advantage.** $T = 100{,}000$, 10 seeds, 95% CI bands. **(a)** Average reward $\bar{R}$ vs correlation length $\ell = 1/\varepsilon$. The Bayes filter (blue, bold) rises monotonically; Markov-k agents saturate at k-dependent ceilings. **(b)** Performance gap $\Delta\bar{R} = \bar{R}_{\text{mem}} - \bar{R}_{\text{Markov-}k}$ increases with ℓ; smaller k yields a larger gap.

Parameters.

Quantity	Value	Role
T	100,000	horizon per trial
Seeds	10	independent replications
δ	0.05	observation asymmetry
k	$\{1, 2, 4, 8\}$	Markov window sizes
ε	logspace(10^{-3}, 10^{-1}, 15)	transition noise grid
Burn-in	5,000	discarded steps

1.8.2 Results

Figure 1.2 shows the two key signatures.

Result 1: finite-window performance saturation (Figure 1.2a). The average reward $\bar{R}$ of the Bayes filter (memory agent) increases monotonically with correlation length $\ell = 1/\varepsilon$, while each Markov-k agent saturates at a distinct ceiling. The ceilings are ordered: $k = 1$ (lowest) through $k = 8$ (highest), and all fall below the memory agent for $\ell \gtrsim 20$. This is a property of the specified aliased HMM and agent classes, not a consequence of the thermodynamic ceiling.

Result 2: Memory advantage (Figure 1.2b). The gap $\Delta\bar{R} = \bar{R}_{\text{mem}} - \bar{R}_{\text{Markov-}k}$ increases monotonically with ℓ, and is larger for smaller k. Shaded bands show 95% confidence intervals across 10 seeds. The Markov-1 and Markov-2 curves nearly overlap at small ℓ, reflecting the fact that short observation windows provide negligible additional information in this aliasing regime—a consistency check, not a deficiency.

1.8.3 Scope of This Demonstration

These simulations illustrate finite-window saturation under the stated model class; they do not constitute a thermodynamic test.

This demonstration **does** show:

1. A reproducible regime in which finite-order Markov agents exhibit a performance ceiling while a memory-carrying (Bayes filter) agent improves.
2. The performance gap widens with correlation length and tightens with window size in this HMM. This is conceptually related to retained history but is not the quantity in Theorem 1.21.

This demonstration does **not** show:

1. Universality across environments, observation models, or agent architectures. The model uses a two-state HMM with binary aliased observations.
2. Tight constants or the functional form of the ceiling boundary $\ell_c(k)$.
3. That the Bayes filter is optimal among all possible memory-carrying agents.

Reproducibility. The complete simulation is a self-contained Python script (`paper1_markov_ceiling_demo.py`, ~ 560 lines, requiring only NumPy and Matplotlib) with fixed random seeds. All figures in this section can be reproduced by executing the script. The following files are included in the supplementary archive:

- `paper1_markov_ceiling_demo.py` — simulation script
- `fig_paper1_markov_ceiling.pdf` — Figure 1.2
- `markov_ceiling_data.csv` — raw sweep data
- `markov_ceiling_boundary.csv` — extracted ceiling boundaries $\ell_c(k)$

1.9 Discussion

1.9.1 Summary of Results

Result	Statement	Sec.
Restricted autonomous ceiling	$\mathcal{S} \leq 0$ for a fixed-Hamiltonian Davies semigroup	1.3
Initial-resource balance	$\beta\mathcal{S} = -\Delta I - \Delta D_{\mathrm{KL}} - \beta\Delta\langle H_{\mathrm{ctrl}}\rangle - \beta\Delta\langle H_{\mathrm{int}}\rangle$	1.4
Quantitative demo	Spin-boson: $\beta\mathcal{S} \approx +0.093 > 0$ at first backflow revival (Fig. 1.1, Table 1.1)	1.6
Memory/record interface	operational influence relation plus a conditional record test	1.5
Archive overload	conditional on raw storage, finite capacity, and irreversible reset	1.7
Numerical demo	finite-window saturation in an aliased HMM (Fig. 1.2)	1.8

1.9.2 What This Paper Does and Does Not Show

This paper shows:

1. Under an autonomous fixed-Hamiltonian Davies semigroup, the survival functional $\mathcal{S} \leq 0$ (Theorem 1.13).

2. For any initial state (product or correlated), the survival functional satisfies the exact identity $\beta\,\mathcal{S} = -\Delta I(S{:}E) - \Delta D_{\mathrm{KL}}(\rho_E \| \rho_E^{\mathrm{th}}) - \beta\,\Delta\langle H_{\mathrm{ctrl}}\rangle - \beta\,\Delta\langle H_{\mathrm{int}}\rangle$ (Theorem 1.21). Under autonomous evolution, when pre-existing system–environment correlations are consumed ($\Delta I < 0$), $\mathcal{S} > 0$ is achievable, bounded by the initial correlation budget (Corollary 1.24).

3. A quantitative spin-boson example illustrates: $\beta\,\mathcal{S} \approx +0.093 > 0$ at the first non-Markovian revival (Section 1.6).

4. The memory-influence relation and HAFF record ordering define a conditional model test, not a proved embedding (Conjecture 1.35).

5. Landauer supplies a conditional reset cost; a raw finite archive can overload under the implementation assumptions in Remark 1.44.

6. A computational demonstration exhibits finite-window saturation in a two-state HMM with aliased observations (Section 1.8, Figure 1.2).

This paper does not show:

1. That memory or non-Markovianity is necessary or sufficient for persistence.

2. That non-Markovian systems optimize a larger survival supremum than all Markovian systems.

3. That driven or measurement-feedback Markovian agents obey the restricted autonomous ceiling (Remark 1.14).

4. That the specific form of the optimal memory kernel can be derived from first principles without specifying the environment.

5. That memory implies or requires consciousness.

6. That storage or maintenance has a universal Landauer cost, or that compression and symmetry breaking are thermodynamic necessities.

Chapter 2

Spontaneous Symmetry Breaking of Reference Frames as a Computational Cost Minimization Strategy

Paper II — "The Ego"

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Abstract

We investigate a conditional implementation model for bounded agents that process memory-like state under finite computational resources. Revised Paper I [49] supplies a conditional correlation-resource account; it does not establish universal memory necessity, a universal Markovian ceiling, or a Landauer charge for merely storing records.

The present paper therefore studies a declared raw D-channel archive processed under a finite throughput budget. Within that model, symmetric processing of all declared channels can exceed the available budget at a finite time, while a supplied coordinate selector can reduce the declared load. This is a conditional overload result for the stated archive and algorithm, not a theorem that every persistent agent must compress memory or break a symmetry.

Reference-frame selection is treated as a modelling strategy: a privileged basis or coordinate projection packages a lower-dimensional representation of the archive. The rate-distortion calculation gives a budget-dependent target dimension under the stated source, distortion, and budget assumptions. It does not prove that the "self" is a universal survival-optimal structure.

The projected representation is accompanied by four diagnostic labels: basis selection, frame drag, objective centering, and model incompleteness. Only basis selection follows from projection alone; the other diagnostics require additional controller, update, objective, or task assumptions. A simple drift model gives a visibility crossover for a fixed frame, motivating Paper III's calibration problem without establishing an undetectable universal Delusion Trap.

2.1 Introduction

2.1.1 Context: The Problem of Overload

Revised Paper I [49] gives an ELV-type accounting identity in which initial correlations can fund a later interval and proves a ceiling only for a fixed-Hamiltonian autonomous Davies baseline. It retracts both universal memory necessity and a Landauer charge for merely storing or maintaining records.

The present paper therefore begins with an independent implementation model: a raw, uncompressed D-channel archive processed under a finite throughput budget. Its overload is conditional on that archive and algorithm. The question is when a supplied coordinate selector can reduce this declared load, not whether every persistent agent must compress or break symmetry.

2.1.2 Position within the Series

This paper is the second of three constituting the **T-DOME** (Thermodynamic Dynamics of Observer-Memory Entanglement) framework, the third pillar of a three-paper program.

Framework	**Question**		**Result**	**Status**
HAFF [41, 42]	How does geometry emerge?	Ocean	Conditional algebra/dictionary models	Revised
Q-RAIF [46, 47]	What algebra must an observer have?	Fish	Clifford representation as model input	Revised
T-DOME I [49]	What can inter-epoch correlations fund?	Seed	Restricted baseline; accounting identity	Conditional
T-DOME II (this work)	When can a model select a frame?	Ego	Conditional coordinate selection	**This paper**
T-DOME III	When can residual statistics track drift?	Loop	Conditional Fisher calibration	Companion

The three T-DOME papers form a conditional modelling sequence rather than an irreversible theorem chain:

1. **Paper I (The Seed):** An ELV-type balance identifies initial correlations as one possible interval resource. Raw-archive overload is an implementation scenario, not a thermodynamic necessity.

2. **Paper II (The Ego, this work):** Unbounded memory under finite computational resources causes processing overload in the coordinate-processing model. This paper studies conditional feature selection; it does not prove universal SSB.

3. **Paper III (The Loop):** Uncorrected bias diverges from a changing environment. A self-referential calibration loop (monitoring one's own prediction error) resolves the bias but requires the system to "observe its own observation"—closing the self-calibration loop.

2.1.3 Relation to Q-RAIF

Revised Q-RAIF Papers A and B [46, 47] use a Clifford algebra $Cl(V, q)$ as a supplied representation choice; persistence does not select it. Paper C [48] proves only that the image of a separately supplied homomorphism lies in the external algebra. This paper adopts both the Clifford coordinates and the representation map as model inputs.

The Clifford algebra $Cl(V, q)$, however, admits a non-trivial *automorphism group* $G = \mathrm{Aut}(Cl(V, q))$. In the absence of external constraints, all elements of G yield physically equivalent representations—the choice of basis within the algebra is a *gauge freedom*. This gauge freedom is the mathematical substrate of the symmetry considered in the conditional breaking analysis below.

The "ego" names a selected computational frame in that supplied representation. Whether a concrete optimum is a genuine gauge fixing is a model condition, not an inherited theorem.

2.1.4 Relation to HAFF Paper G

Revised HAFF Paper G [45] offers philosophy-of-science motivation and no fixed-point no-go theorem. It does not imply basis selection here. Any connection between its meta-level concern and this finite-budget model is motivation only.

2.1.5 Scope and Disclaimers

To prevent interpretational overreach, we state at the outset what this paper does *not* claim:

1. We do not claim that symmetry breaking is *sufficient* for persistence. Paper III addresses the additional requirements.

2. We do not claim that basis selection or spontaneous symmetry breaking is necessary for every bounded agent. The results concern a declared coordinate-retention model.

3. The term "ego" or "self" is used in the control-theoretic sense: a fixed reference frame within the agent's internal algebra. It carries no implication of consciousness or subjective experience.

4. A broader structural analogy with classical philosophical concepts of selfhood exists but is outside the scope of this paper.

Related work. The idea that bounded agents must compress their representations has roots in Simon's bounded rationality [32], Shannon's rate-distortion theory [31, 12], and Sims's rational inattention [33], which models finite-capacity decision-makers as solving a rate-distortion problem—precisely the economic counterpart of our $\mathcal{C}_{\mathrm{budget}}$ formalism. The information bottleneck [36] formalises relevance-weighted compression and has been applied to neural coding and deep learning. The role of decoherence in selecting preferred bases (pointer states) is well established via quantum Darwinism [37]; here it motivates one model-specific coordinate-selection rule. Measures of non-Markovianity and their thermodynamic consequences are reviewed in [26, 10]; the connection to survival was established in Paper I.

Summary of contributions. This paper separates three levels of result:

1. **Raw-channel load scaling law** a conditional accounting identity for a raw D-channel archive.
2. **Survival-weighted rate-distortion bound** a compactness-based existence result; the budget supplies only a capacity $k_{\max}$, not a universal optimal retained dimension.
3. **Delusion dynamics** a linear-drift visibility crossover plus an explicit two-mode example showing when gradient anti-correlation can occur.

Result-status ledger. The original Computational Ceiling and SSB-necessity chain is retracted ('TD-II-T3.5'). Raw-channel load is an FD/I accounting proposition; compression existence is conditional on a compact feasible model; SSB is a model interpretation. The Delusion Trap is downgraded ('TD-II-T6.7'): visibility crossover is exact in the declared two-mode geometry, while gradient sign and self-detection require extra dynamics.

2.2 Mathematical Preliminaries

2.2.1 Inherited Framework from Paper I

We briefly recall the key objects from Paper I [49] that the present work builds upon. The reader is referred to Paper I for full definitions and proofs.

Survival functional. For an open quantum system S coupled to an environment E at inverse temperature β, with dynamics Λ and external control protocol $H_{\mathrm{ctrl}}(t)$, the survival functional is

$$\mathcal{S}[\Lambda, \tau] := \Delta F - W[0, \tau], \tag{2.1}$$

where $\Delta F = F(\rho(\tau)) - F(\rho(0))$ is the change in non-equilibrium free energy and $W = \int_0^\tau \mathrm{tr}\Big(\rho(t)\, \dot{H}_{\mathrm{ctrl}}(t)\Big)\, dt$ is the work performed by the external protocol.

Restricted autonomous baseline. For a fixed-Hamiltonian Davies semigroup with no control, Paper I proves

$$\mathcal{S}[\Lambda^{\mathrm{M}}, \tau] \leq 0 \qquad \text{for all } \tau \geq 0. \tag{2.2}$$

It does not extend this result to driven Markovian agents.

Non-Markovian advantage identity. For arbitrary initial states:

$$\beta\, \mathcal{S} = -\Delta I(S{:}E) - \Delta D_{\mathrm{KL}}(\rho_E \| \rho_E^{\mathrm{th}}) - \beta\, \Delta\langle H_{\mathrm{ctrl}} \rangle - \beta\, \Delta\langle H_{\mathrm{int}} \rangle. \tag{2.3}$$

Conditional raw-archive model. Paper I retracts a universal Landauer maintenance bound. This paper independently posits an uncompressed record stream. Let h_μ be its *per-component* entropy rate [14], defined by

$$h_\mu := \lim_{T\to\infty} \frac{1}{T} H(X_{0:T}), \tag{2.4}$$

measuring the asymptotic information (in bits per unit time) generated by one declared record channel (we work in units where the sampling interval equals the environmental correlation time τ_E)[1] The predictive information I_{pred} is the *excess entropy* (excess entropy) [8, 30], defined as the mutual information between past and future of the environmental process:

$$I_{\text{pred}} := I\Big(\overleftarrow{X};\, \overrightarrow{X}\Big) = H(\overrightarrow{X}) - H(\overrightarrow{X} \mid \overleftarrow{X}), \tag{2.5}$$

where $\overleftarrow{X}$ and $\overrightarrow{X}$ denote the semi-infinite past and future, respectively. For a stationary process, I_{pred} relates to h_μ via the entropy-rate decomposition $H(X_{1:T}) = I_{\text{pred}} + h_\mu T + o(1)$ as $T \to \infty$ when the relevant limits exist [14]. No lower bound $I_{\text{stored}} \geq \min(I_{\text{pred}}, h_\mu \tau)$ is inherited.

2.2.2 The Agent's Internal Algebra

As a modelling choice motivated by revised Q-RAIF [47, 48], the agent's internal control algebra is taken to be a Clifford algebra $\mathcal{O}_{\text{int}} = Cl(V, q)$ for a real vector space V equipped with a non-degenerate quadratic form q. The algebra satisfies the fundamental relation $v^2 = q(v)\,\mathbf{1}$ for all $v \in V$.

The *automorphism group*

$$G := \text{Aut}(Cl(V, q)) \tag{2.6}$$

is the group of algebra automorphisms that preserve the grading and quadratic form.[2] For $Cl(1, 3)$, G contains the spin group $\text{Spin}(1, 3) \cong SL(2, \mathbb{C})$ as a subgroup—a six-real-dimensional Lie group.

In the absence of computational constraints, all $g \in G$ yield physically equivalent descriptions of the agent's internal state. The choice of basis within $Cl(V, q)$ is a *gauge freedom*—the symmetry examined in the conditional coordinate-selection model below.

A representation map $\phi : Cl(V, q) \to \mathcal{A}_{\text{ext}}$ is supplied as an interface. Q-RAIF Paper C proves only the image-level algebraic statement once this map is given; it does not select a preferred $Cl(1, 3)$ target or preserve quadratic signature.

Dimensional convention. Two distinct notions of dimension appear throughout:

Symbol	Meaning	Scaling
$n := \dim V$	number of generators (degrees of freedom)	—
$D := \dim Cl(V, q) = 2^n$	full multivector space (algebra basis size)	exponential in n

[1] For a continuous-valued process sampled at resolution b bits, h_μ includes the quantisation cost: $h_\mu = h_\mu^{(\text{diff})} + b\, f_s$, where $h_\mu^{(\text{diff})}$ is the differential entropy rate and f_s the sampling frequency. All raw-channel accounting identities in this paper use h_μ so defined.

[2] We use G as an effective symmetry group acting transitively on admissible frames. The detailed Lie-algebraic structure of G is not required for our results; only the existence of a non-trivial symmetry that may be broken under the additional encoder conditions in Remark 2.18. In concrete models, one may replace G by its image under the adjoint representation—typically $O(V, q)$ or a pin/spin subgroup.

The raw-channel load model (Section 2.3) scales with D, not n; the distinction matters whenever one compares generator-level and algebra-level quantities.

2.2.3 Rate-Distortion Theory

We require the classical rate-distortion framework of Shannon [31].

Definition 2.1 (Rate-distortion function)**.** *Let X be a random source with distribution $p(x)$, $\hat{X}$ a reconstruction, and $d : \mathcal{X} \times \hat{\mathcal{X}} \to [0, \infty)$ a distortion measure. The* rate-distortion function *is*

$$R(D) := \min_{\substack{p(\hat{x}|x): \\ \mathbb{E}[d(X,\hat{X})] \leq D}} I(X; \hat{X}), \tag{2.7}$$

the minimum mutual information between source and reconstruction that achieves average distortion at most D.

$R(D)$ is a convex, non-increasing function of D with $R(0) = H(X)$ (lossless) and $R(D_{\max}) = 0$ (maximum distortion). It provides the fundamental limit on lossy compression [12]. The *information bottleneck* method of Tishby et al. [36] generalises this framework to the case where the relevant variable is not the source itself but a downstream prediction target—precisely the situation in our survival-weighted compression problem (Section 2.4.2). For source X, representation Z, and declared survival-relevant target Y, the bottleneck objective is

$$\min_{p(z|x)} I(X; Z) - \beta_{\mathrm{IB}} I(Z; Y). \tag{2.8}$$

The coordinate-retention model below does not solve this unrestricted optimization. Rate-distortion and information bottleneck therefore provide comparison frameworks, not authority for the component count.

2.2.4 Bounded Rationality

Following Simon [32], we model computational limitations as a hard constraint on the agent's information processing rate.

Definition 2.2 (Computational budget)**.** *The agent's* computational budget $\mathcal{C}_{\mathrm{budget}}$ *(measured in bits per unit time) is the maximum rate at which the agent can evaluate functions of its stored correlations. We assume $\mathcal{C}_{\mathrm{budget}} < \infty$.*

Physically, finiteness of $\mathcal{C}_{\mathrm{budget}}$ reflects the finite number of degrees of freedom in the agent's physical substrate: finite Hilbert space dimension, finite memory register size, and finite energy available for computation (Landauer's principle [20, 7]).

2.2.5 Fiber Bundle Formalism

The geometric setting for reference-frame selection is a principal fiber bundle.

Definition 2.3 (Gauge bundle)**.** *The* gauge bundle *is the principal G-bundle*

$$\pi : P \to M, \qquad G = \mathrm{Aut}(Cl(V, q)), \tag{2.9}$$

where:

- *M is the base space of* effective memory kernels*—equivalently, the space of induced sufficient-statistic processes accessible to the agent (a finite-dimensional manifold that admits local parametrisation by the environmental spectral-density couplings);*
- *G is the structure group acting transitively on admissible frames (see footnote 2 for the effective subgroup);*
- *the fiber $\pi^{-1}(\kappa)$ over a kernel $\kappa \in M$ is the G-orbit of equivalent algebraic representations (frames) for describing κ in $Cl(V, q)$;*
- *a* section *$\sigma : M \to P$ constitutes a global gauge-fixing policy—a systematic choice of reference frame for every kernel configuration.*

A *connection* on P specifies how the reference frame is parallel-transported as the agent's state evolves. The curvature of this connection measures the extent to which the reference frame "twists" along different paths through state space.

2.2.6 Standing Assumptions

Definition 2.4 (Standing Assumptions)**.** *Throughout this paper, the following conditions are assumed:*

(B1) ***Inherited framework.*** *All assumptions (A1)–(A5) of Paper I [49] remain in force (open quantum system coupled to a thermal bath, well-defined free energy, non-equilibrium initial state, finite-dimensional system Hilbert space, and weak-coupling or controlled-coupling regime). Additionally, the agent possesses an internal control algebra $\mathcal{O}_{\text{int}} = Cl(V, q)$ with a supplied representation map $\phi : Cl(V, q) \to \mathcal{A}_{\text{ext}}$ (Q-RAIF [48]). Neither the Clifford choice nor the map is derived from persistence.*

(B2) ***Finite computational budget.*** *The agent's information processing rate satisfies $\mathcal{C}_{\text{budget}} < \infty$ (Definition 2.2).*

(B3) ***Raw-channel load model.*** *The supplied record process has D declared component channels, per-channel innovation rate $h_\mu > 0$, memory depth $\tau_{\text{mem}} > 0$, and no shared-code or group-theoretic compression before the accounting in Section 2.3. This worst-case assumption, not environmental non-triviality alone, carries the load calculation.*

(B4) ***Survival imperative.*** *The agent's dynamics must maintain $\mathcal{S} \geq \mathcal{S}_{\min}$ over survival horizons $T \gg \tau_{\text{mem}}$. This is a persistence constraint, not an optimization objective.*

(B5) ***Gauge symmetry of bare algebra.*** *The automorphism group $G = \text{Aut}(Cl(V, q))$ is non-trivial ($G \neq \{e\}$). In the absence of computational constraints, all $g \in G$ yield physically equivalent descriptions.*

2.3 Raw-Channel Load Model

We now record the throughput of the raw-channel model in (B3). This is a bookkeeping identity for an explicitly compression-free representation, not a lower bound over all symmetric algorithms and not a consequence of Paper I.

2.3.1 The Information Processing Inequality for Bounded Agents

Accounting convention. To ensure dimensional consistency throughout, we distinguish two quantities:

- $\mathcal{C}_{\text{budget}}$: the agent's processing *rate* (bits per unit time).
- $\mathcal{I}_{\text{proc}}(\tau)$: the total information (bits) that must be processed per evaluation cycle when the memory archive has depth τ.

The agent must complete one evaluation cycle per environmental correlation time τ_E. The *processing rate* required for a memory depth τ is

$$\mathcal{R}_{\text{proc}}(\tau) := \frac{\mathcal{I}_{\text{proc}}(\tau)}{\tau_E}. \tag{2.10}$$

Paralysis occurs when $\mathcal{R}_{\text{proc}}(\tau_{\text{mem}}) > \mathcal{C}_{\text{budget}}$. Hereafter we measure time in units of τ_E (i.e., set $\tau_E = 1$), so that rates and per-cycle information quantities are numerically equal.

Definition 2.5 (Symmetric processing)**.** *An agent processes its memory* symmetrically *if both its cost functional $\mathcal{C}[\cdot]$ and its distortion measure $D(\cdot)$ are G-invariant: $\mathcal{C}[g{\cdot}\mathcal{K}] = \mathcal{C}[\mathcal{K}]$ and $D(g \cdot \mathcal{F}) = D(\mathcal{F})$ for every $g \in G = \mathrm{Aut}(Cl(V, q))$. In operational terms: for every stored correlation c_i in the memory kernel $\mathcal{K}(t, s)$ and every $g \in G$, the cost of evaluating c_i equals the cost of evaluating $g{\cdot}c_i$, and no basis direction is* a priori *preferred for survival evaluation.*

Remark 2.6 (Operational meaning of processing rate)**.** *We define the processing rate $\mathcal{R}_{\text{proc}}$ as an* information-throughput *measure: the number of algebraic components that must be updated per unit time, multiplied by the innovation rate h_μ per component. It captures the* bandwidth *cost of maintaining an internal representation, not the algorithmic gate complexity of individual operations.*

Proposition 2.7 (Raw-channel load accounting)**.** *Under (B3), let $n = \dim V$ and $D = \dim Cl(V, q) = 2^n$. A raw archive of depth τ_{mem} contains, by the model definition,*

$$\mathcal{I}_{\text{archive}} = D\, h_\mu\, \tau_{\text{mem}} \qquad \textit{bits.} \tag{2.11}$$

Its incremental input rate is Dh_μ bits per unit time. If the declared algorithm rescans the entire archive once per environmental correlation time τ_E, its required throughput is

$$\mathcal{R}_{\text{sweep}} = \frac{Dh_\mu \tau_{\text{mem}}}{\tau_E}. \tag{2.12}$$

For a processing budget $\mathcal{C}_{\text{budget}}$, this particular full-sweep algorithm fits only when

$$\tau_{\text{mem}} \le \tau_{\max} := \frac{\mathcal{C}_{\text{budget}} \tau_E}{Dh_\mu}. \tag{2.13}$$

Proof. All three statements follow by multiplying the declared number of channels by their per-channel innovation rate and then converting stored bits per sweep to bits per unit time. No lower bound over compressed, streaming, sufficient-statistic, or group-equivariant algorithms is claimed. □

Remark 2.8 (Conditional component-count ratio)**.** *Within the same raw-coordinate implementation, comparing a D-channel sweep with a k-channel sweep gives the component-count ratio D/k. This is not a tax imposed by gauge invariance on all algorithms.*

Remark 2.9 (Effective vs. full dimension)**.** *The ceiling uses $D = \dim Cl(V, q) = 2^n$, the full multivector dimension. In practice, the environment may couple to only a subset of grades (e.g., grade-1 generators), yielding an effective dimension $D_{\text{eff}} \leq D$. For a* structured *environment where the agent knows which grades are active, the ceiling can be tightened to $\mathcal{R}_{\text{proc}} \gtrsim h_\mu \cdot D_{\text{eff}}$. The raw-channel condition in (B3) represents this worst case. A structured environment or algorithm changes the model rather than falling under a universal ceiling.*

2.3.2 Processing Collapse

Remark 2.10 (Raw full-sweep overload)**.** *If $\tau_{\text{mem}} > \tau_{\max}$, the full-sweep algorithm in Proposition 2.7 misses its declared cycle deadline. This does not imply that every record is useless or that another streaming/compressed algorithm fails.*

Remark 2.11 (Comparison with revised Paper I)**.** *Paper I now treats storage/reset cost as implementation-dependent. The present overload is likewise conditional on a raw full-sweep algorithm. Compression is one response; neither paper proves it must take the form of symmetry breaking.*

2.4 Conditional Frame-Selection Model

2.4.1 Reference Frame as Gauge Fixing

Definition 2.12 (Reference frame)**.** *A* reference frame *$\mathcal{F}$ is a section $\sigma : M \to P$ of the gauge bundle (Definition 2.3). Choosing σ is equivalent to selecting a preferred orthonormal basis $\{e_1, \ldots, e_n\}$ of the generating vector space V at each point in state space M, thereby fixing the gauge freedom of $Cl(V, q)$.*

Definition 2.13 (Projected memory kernel)**.** *Given a reference frame $\mathcal{F}$, let $V_{\text{fg}}(\mathcal{F}) \subset Cl(V, q)$ be a k-dimensional* foreground subspace *selected by the declared coordinate model, with $0 \leq k \leq D$. Let $\Pi_{\mathcal{F}}$ denote the orthogonal projection onto $V_{\text{fg}}(\mathcal{F})$ with respect to the trace inner product $\langle A, B\rangle := \operatorname{tr}\big(A^\dagger B\big)$. The* projected memory kernel *is*

$$\mathcal{K}_{\mathcal{F}}(t, s) := \Pi_{\mathcal{F}}\, \mathcal{K}(t, s)\, \Pi_{\mathcal{F}}. \tag{2.14}$$

The complementary projection $\Pi_{\mathcal{F}}^{\perp} = \mathbf{1} - \Pi_{\mathcal{F}}$ defines the background subspace *$V_{\text{bg}}(\mathcal{F})$. The decomposition $Cl(V, q) = V_{\text{fg}} \oplus V_{\text{bg}}$ is determined by $\mathcal{F}$, not by any* a priori *ordering of basis vectors.*

2.4.2 A Coordinate-Compression Model

We now define a restricted coordinate-compression problem and compare it with rate-distortion theory.

Processing rate of a frame. If the agent retains k algebraic components (the foreground subspace V_{fg}), each generating h_μ bits per unit time, the processing rate of frame $\mathcal{F}$ is

$$R_{\mathcal{F}}(k) \;=\; k \cdot h_\mu \qquad \text{(bits per unit time)}. \tag{2.15}$$

The budget constraint $R_{\mathcal{F}} \le \mathcal{C}_{\text{budget}}$ thus bounds the number of maintainable components. Define the capacity

$$k_{\max} := \min\!\left(D,\, \left\lfloor \frac{\mathcal{C}_{\text{budget}}}{h_\mu} \right\rfloor\right). \tag{2.16}$$

This is a maximum feasible component count, not yet an optimizer.

Definition 2.14 (Survival distortion)**.** *The* survival distortion *of a reference frame $\mathcal{F}$ is*

$$D(\mathcal{F}) := \mathbb{E}_\xi\left[\ell\big(\mathcal{S}_{\text{full}}(\xi) - \mathcal{S}_{\mathcal{F}}(\xi)\big)\right], \tag{2.17}$$

where ξ denotes environmental realizations, $\ell : \mathbb{R} \to [0,\infty)$ is a convex, non-decreasing loss function (we use squared error $\ell(x) = x^2$ throughout), $\mathcal{S}_{\text{full}}(\xi)$ is the survival functional evaluated using the full memory kernel $\mathcal{K}(t,s)$, and $\mathcal{S}_{\mathcal{F}}(\xi)$ is evaluated using the projected kernel $\mathcal{K}_{\mathcal{F}}(t,s)$.

Remark 2.15 (Information-theoretic objects)**.** *Strictly speaking, rate-distortion theory and mutual information apply to stochastic processes, not to superoperator kernels directly. Throughout Sections 2.4–2.7, $I(\mathcal{K}_{\mathcal{F}};\mathcal{K})$ is shorthand for $I(\hat{X};X)$, where $X = \{c_i(t)\}_{i=1}^{D}$ is the sufficient-statistic record process induced by the full kernel $\mathcal{K}$ acting on the agent's internal coordinates, and $\hat{X} = \{c_i(t)\}_{i\in V_{\text{fg}}}$ is the projected record induced by $\mathcal{K}_{\mathcal{F}}$. The distortion measure* (2.17) *acts on the survival functional $\mathcal{S}$ evaluated on these records.*

Proposition 2.16 (Existence in the coordinate model)**.** *For each $0 \le k \le k_{\max}$, let $\mathfrak{F}_k$ be a compact set of admissible k-component frames. Assume the feasible union $\mathfrak{F}_{\le k_{\max}}$ is nonempty and the distortion $D(\mathcal{F})$ is lower semicontinuous. Then*

$$\mathfrak{F}^* := \operatorname*{arg\,min}_{\mathcal{F}\in\mathfrak{F}_{\le k_{\max}}} D(\mathcal{F}) \tag{2.18}$$

is nonempty. Every optimizer has some retained dimension $k_{\text{opt}} \le k_{\max}$. It follows only by counting that its discarded component fraction is

$$1 - \frac{k_{\text{opt}}}{D}. \tag{2.19}$$

Budget saturation $k_{\text{opt}} = k_{\max}$ requires the additional strict-improvement condition that every feasible added component lowers the best attainable distortion. Without it, unused capacity is possible.

Proof. The feasible union is compact (a finite union of compact sets), so a lower-semicontinuous function attains its minimum. The dimension and discard fraction statements are definitions. Mere non-increase of distortion does not imply strict improvement or budget saturation. □

Remark 2.17 (Relation to Shannon and information bottleneck)**.** *Proposition 2.16 is not a Shannon converse or an information-bottleneck solution. Those theories optimize over stochastic encoders; this proposition optimizes over a supplied compact family of coordinate projections. The original theorem claiming $k^* = \lfloor \mathcal{C}_{\text{budget}}/h_\mu \rfloor$ as the universal optimum is withdrawn.*

For the remainder of the paper, k^* denotes the retained dimension of one chosen optimizer from $\mathfrak{F}^*$, when such a choice is made. Thus $k^* = k_{\text{opt}} \leq k_{\max}$; it is not identified with $\lfloor \mathcal{C}_{\text{budget}}/h_\mu \rfloor$ unless the stated strict-improvement condition is separately verified.

2.4.3 Conditional Symmetry-Breaking Interpretation

Remark 2.18 (Conditional symmetry-breaking interpretation)**.** *Suppose, in addition to Proposition 2.16, that (i) every admissible compressor is a proper coordinate projection, (ii) the effective group G acts on the candidate frames, and (iii) no G-invariant feasible encoder attains the required distortion. Then choosing any feasible optimizer privileges a foreground subspace of the D-dimensional algebra representation and breaks that model symmetry operationally. This conclusion is conditional on the encoder class; it is not a universal necessity theorem.*

If $D(\mathcal{F}) \neq D(\mathcal{F}')$ for distinct frames, that condition supports uniqueness of a global minimizer, not multiple local minima. Multiple "ego attractors" require a separate non-convex landscape with several demonstrated minima. The original SSB theorem and its generic non-uniqueness claim are withdrawn.

2.4.4 The Four Bias Terms

Definition 2.19 (Four frame-dependent diagnostics)**.** *For a declared coordinate projection $\Pi_{\mathcal{F}}$ and a specified controller, we use the following four labels as diagnostics. They organize possible departures from a symmetric representation; they are not asserted to be universal consequences of gauge fixing.*

(i) ***Basis selection bias*** *($\mathcal{B}_{\text{select}}$): The choice of $\{e_1, \ldots, e_{k^*}\}$ privileges certain algebraic components over others. Information aligned with the chosen basis is processed efficiently; misaligned information is discarded or distorted.* Observable consequence: *systematic blindness to off-basis environmental perturbations (orthogonal masking).*

(ii) ***Frame drag*** *($\mathcal{B}_{\text{frame}}$): If the update rule transports the current frame with memory or a restoring penalty, it may prefer states near that choice.* Possible observable consequence: *hysteresis in belief updating and lag behind environmental shifts. Holonomy alone does not supply a restoring force.*

(iii) ***Objective centering*** *($\mathcal{B}_{\text{center}}$): A projected surrogate objective depends only on the retained coordinates. Unless it is sufficient for the full objective, its stationary points need not be global stationary points.* Possible observable consequence: *failure to detect optima whose relevant directions lie in the background subspace.*

(iv) ***Model incompleteness*** *($\mathcal{B}_{\text{inc}}$): The compression from $Cl(V, q)$ to V_{fg} is lossy. The discarded components V_{bg} may contain correlations absent from the retained representation.* Possible observable consequence: *underestimation of total uncertainty when those correlations matter to the task.*

Only the first item follows from the projection alone: components in $\ker \Pi_{\mathcal{F}}$ are removed. Frame drag requires a stateful update law, objective centering requires a specified

surrogate loss, and overconfidence requires task-relevant information in the discarded coordinates. Proposition 2.16 supplies no generic information fraction for these diagnostics.

Bias	**Origin**	**Observable consequence**	**Determines**
$\mathcal{B}_{\text{select}}$ (selection)	projection $\Pi_{\mathcal{F}}$	Systematic blindness to off-basis perturbations (orthogonal masking)	*what* is seen
$\mathcal{B}_{\text{frame}}$ (frame drag)	bundle connection / holonomy	Hysteresis in belief updating; model lags behind rapid drift	*duration*
$\mathcal{B}_{\text{center}}$ (centering)	$\nabla\mathcal{S} \in V_{\text{fg}}$	Local frame-relative optima; global background optima invisible	*target*
$\mathcal{B}_{\text{inc}}$ (incompleteness)	lossy compression $k^* \ll D$	Underestimation of thermodynamic uncertainty (structural overconfidence)	*blind spot*

Table 2.1: The four bias terms of the broken phase. Only the first follows from projection alone; the others require the additional update, objective, or task assumptions stated in the text.

Remark 2.20 (Nature of the bias terms)**.** *The four bias terms (Table 2.1) are not pathologies. They are diagnostic names for distinct ways in which a projected controller may depart from a symmetric representation. Projection alone gives basis selection; the remaining terms become live only when the corresponding update, objective, or task structure is specified.*

2.5 Emergent Structure: The Architecture of Ego

We consolidate the gauge-fixed compressed representation into a single mathematical object. Throughout this section, "ego" is used purely as shorthand for a gauge-fixed compressed representation; no claims about phenomenal consciousness, subjective experience, or qualia are intended or implied.

Definition 2.21 (Ego)**.** *The* ego *of an agent satisfying* (B1)–(B5) *is the pair*

$$\mathfrak{E} := \big(\mathcal{F}^*,\ V_{\text{fg}}^*\big), \tag{2.20}$$

where $\mathcal{F}^* \in \mathfrak{F}^*$ (Proposition 2.16) is a chosen frame (providing the coordinate system) and $V_{\text{fg}}^* := V_{\text{fg}}(\mathcal{F}^*)$ is the k^*-dimensional foreground subspace selected within the coordinate model (providing the compression). The projected memory kernel $\mathcal{K}_{\mathfrak{E}} := \Pi_{V_{\text{fg}}^*}\,\mathcal{K}\,\Pi_{V_{\text{fg}}^*}$ is induced by this pair. All bias terms, distortion bounds, and delusion dynamics are functions of $\mathfrak{E}$.

2.5.1 The Ego as a Fiber Bundle Section

The reference frame $\mathcal{F}$, understood as a section $\sigma : M \to P$, is the mathematical object we call the *ego*. It has three key properties:

Smoothness. The section σ varies continuously with the agent's state $\rho \in M$. Small changes in ρ produce small changes in the preferred basis—the ego is not a discrete switch but a smooth deformation of perspective.

Holonomy. If the agent's state traces a closed loop $\gamma : [0, 1] \to M$ with $\gamma(0) = \gamma(1) = \rho_0$, the parallel-transported frame need not return to its initial value:

$$\sigma(\gamma(1)) = \mathrm{Hol}(\gamma) \cdot \sigma(\gamma(0)), \tag{2.21}$$

where $\mathrm{Hol}(\gamma) \in G$ is the holonomy of the connection around γ. Non-trivial holonomy means the agent can "learn"—its reference frame shifts after a complete cycle of experience.

Topological obstruction. In general, a *global* section $\sigma : M \to P$ may not exist. The obstruction is measured by the characteristic classes of the bundle P. When a global section does not exist, the ego must have "singularities"—states where the preferred basis is undefined or discontinuous. This connects to the crisis of Paper III: the delusion trap can be understood as the agent approaching a topological obstruction of its own reference frame.

2.5.2 The Effective Survival Functional

Proposition 2.22 (Survival decomposition)**.** *In the broken phase, the survival functional decomposes as*

$$\mathcal{S} = \mathcal{S}_{\mathrm{vis}}(\mathcal{F}) + \mathcal{S}_{\mathrm{hid}}(\mathcal{F}), \tag{2.22}$$

where:

- $\mathcal{S}_{\mathrm{vis}}(\mathcal{F})$ *is the contribution from the foreground subspace* V_{fg}*, computable within the agent's reference frame;*
- $\mathcal{S}_{\mathrm{hid}}(\mathcal{F})$ *is the contribution from the background subspace* V_{bg}*, invisible to the agent.*

The agent maximizes $\mathcal{S}_{\mathrm{vis}}$ *while being structurally blind to* $\mathcal{S}_{\mathrm{hid}}$*.*

Proof. The survival functional $\mathcal{S} = \Delta F - W$ depends on $\rho(t)$, which is a function of the full memory kernel $\mathcal{K}(t, s)$. Decomposing $\mathcal{K} = \Pi_{\mathcal{F}} \mathcal{K} \Pi_{\mathcal{F}} + \Pi_{\mathcal{F}}^{\perp} \mathcal{K} \Pi_{\mathcal{F}}^{\perp}$ + cross terms, the leading contributions are $\mathcal{S}_{\mathrm{vis}} := \mathcal{S}[\Pi_{\mathcal{F}} \mathcal{K} \Pi_{\mathcal{F}}]$ and $\mathcal{S}_{\mathrm{hid}} := \mathcal{S} - \mathcal{S}_{\mathrm{vis}}$ (collecting background and cross terms). The agent computes only $\mathcal{S}_{\mathrm{vis}}$, as the projected kernel $\mathcal{K}_{\mathcal{F}}$ discards all background components. □

2.5.3 The Computational Speedup

Proposition 2.23 (Ego dividend)**.** *After symmetry breaking, the computational cost of processing memory drops from* $\mathcal{C}_{\text{proc}} \sim h_\mu \cdot \tau_{\text{mem}} \cdot D$ *(symmetric case,* $D = \dim Cl(V, q)$*) to*

$$\mathcal{C}^{(\mathcal{F})}_{\text{proc}} \sim h_\mu \cdot \tau_{\text{mem}} \cdot k^*. \tag{2.23}$$

The speedup factor is

$$\frac{D}{k^*} = \frac{2^n}{k^*}. \tag{2.24}$$

This is the computational advantage of reference-frame selection. For $Cl(1, 3)$ ($D = 16$) with $k^* = 2$, the speedup is $8\times$. For higher-dimensional algebras, the speedup grows exponentially in n.

2.5.4 Uniform-Channel Information Trade-off

Theorem 2.24 (Uniform-channel predictive-loss relation)**.** *Let* $X = \{c_i(t)\}_{i=1}^{D}$ *denote the full stochastic record process induced by the memory kernel* $\mathcal{K}$ *on the agent's internal coordinates, and let* $\hat{X} = \{c_i(t)\}_{i \in V_{\text{fg}}}$ *denote the projected record retained by the ego. The mutual information between compressed and full records satisfies*

$$I(\hat{X};\, X) \;\le\; H(\hat{X}) \;\le\; k^* \cdot h_\mu \cdot \tau_{\text{mem}}. \tag{2.25}$$

Under the additional assumption that I_{pred} (2.5) *is approximately uniformly distributed across the* D *algebraic components in the symmetric phase*[3]*, define the predictive information lost by projection as*

$$I^{\text{pred}}_{\text{lost}} := I(X_{\text{past}}; X_{\text{future}}) - I(\hat{X}_{\text{past}}; X_{\text{future}}) \approx \left(1 - \frac{k^*}{D}\right) I_{\text{pred}}, \tag{2.26}$$

up to the error in the uniform-allocation approximation. This is a model relation, not a distribution-free lower bound.

Proof. By the data processing inequality, $I(\hat{X}; X) \le H(\hat{X})$. The projected record $\hat{X}$ has k^* components, each carrying at most h_μ bits per unit time over a window of τ_{mem}, giving $H(\hat{X}) \le k^* \cdot h_\mu \cdot \tau_{\text{mem}}$ [12]. This yields (2.25). Data processing also gives $I(\hat{X}_{\text{past}}; X_{\text{future}}) \le I(X_{\text{past}}; X_{\text{future}}) = I_{\text{pred}}$. Under the uniformity assumption, each of the D components carries $\sim I_{\text{pred}}/D$, so the k^* retained components account for $\sim (k^*/D)\, I_{\text{pred}}$. Equation (2.26) follows by subtraction under that allocation assumption. □

Remark 2.25 (The price of selfhood)**.** *Equation* (2.26) *quantifies predictive information loss for the uniform raw-channel model. It does not say that every* k^**-coordinate encoder loses this fraction, or that the selected encoder is globally optimal over all stochastic encoders.*

[3]This "uniformity assumption" is the information-theoretic counterpart of the raw-channel condition (B3). When some components carry disproportionately more predictive information, the bound tightens or loosens depending on the alignment between V_{fg} and the high-information subspace.

2.6 Worked Example: Qubit in a Two-Channel Bath

2.6.1 Model Setup

We extend Paper I's spin-boson model to demonstrate symmetry breaking explicitly. Consider a qubit ($\dim \mathcal{H}_S = 2$) with internal algebra $Cl(0,2) \cong \mathbb{H}$ (the quaternions, $\dim = 4$).

Symbol mapping. The general framework of Sections 2.3–2.5 specialises as follows:

General	This example	Value
$Cl(V,q)$	$Cl(0,2) \cong \mathbb{H}$	$D = 4$
$G = \mathrm{Aut}(Cl(V,q))$	$SO(3)$	acting on $\{\mathbf{i},\mathbf{j},\mathbf{k}\}$
$\mathcal{C}_{\mathrm{budget}}$	$2\,h_\mu$	bits/time
$k_{\max}$ (capacity)	$\lfloor 2h_\mu/h_\mu \rfloor = 2$	components
V_{fg}	$\mathrm{span}\{1,\mathbf{k}\}$	dephasing subspace
V_{bg}	$\mathrm{span}\{\mathbf{i},\mathbf{j}\}$	dissipative subspace
$\tau_{\max}$ (Prop. 2.7)	$2h_\mu/(4h_\mu) = 0.5$	ω_0^{-1}

The qubit is coupled to a bosonic environment through *two* independent channels:

- A *dephasing channel* via σ_z, with spectral density

$$J_z(\omega) = \frac{2\lambda_z\,\gamma_z\,\omega}{\omega^2 + \gamma_z^2} \qquad \text{(Lorentz–Drude)}, \tag{2.27}$$

  producing a memory kernel $\mathcal{K}_z(t,s)$ with non-Markovian backflow.

- A *dissipative channel* via σ_x, with spectral density

$$J_x(\omega) = \frac{2\lambda_x\,\gamma_x\,\omega}{\omega^2 + \gamma_x^2} \qquad \text{(Lorentz–Drude)}, \tag{2.28}$$

  producing a memory kernel $\mathcal{K}_x(t,s)$.

The full memory kernel is $\mathcal{K}(t,s) = \mathcal{K}_z(t,s) \oplus \mathcal{K}_x(t,s)$, and the quaternionic algebra $\mathbb{H} = \mathrm{span}\{1,\mathbf{i},\mathbf{j},\mathbf{k}\}$ has automorphism group $G = \mathrm{Aut}(\mathbb{H}) \cong SO(3)$ (rotations of the pure quaternion subspace).

Parameters. We set $\omega_0 = 1$ (energy unit), $\lambda_z = 1$, $\gamma_z = 0.5$ (underdamped, strong non-Markovian effects in the dephasing channel), $\lambda_x = 0.3$, $\gamma_x = 5.0$ (overdamped, approximately Markovian in the dissipative channel), and the low-temperature regime $\beta\omega_0 \gg 1$.

Computational budget. The agent has $\mathcal{C}_{\mathrm{budget}} = 2\,h_\mu$ bits per unit time—sufficient to track two components of $\mathbb{H}$ but not all four.

Parameter-to-model mapping. Table 2.2 collects the example parameters and checks the raw full-sweep budget.

Quantity	**Symbol**	**Value**	**Theorem check**
Full dimension	D	4	model input
Entropy rate	h_μ	1.0 (normalised)	per-component rate
Budget	$\mathcal{C}_{\text{budget}}$	$2\,h_\mu$	Def. 2.2
Raw update check	$h_\mu D$ vs $\mathcal{C}_{\text{budget}}$	$4 > 2$	full raw update does not fit
Capacity	$k_{\max}$	$\lfloor 2/1 \rfloor = 2$	budget identity
Discard fraction	$1 - k^*/D$	$1/2 = 50\%$	Thm. 2.24
Full-sweep depth	$\tau_{\max}$	$2/(4) = 0.5$	Eq. (2.13)

Table 2.2: Parameter mapping for the two-channel qubit example. This example chooses a two-component frame and verifies that it fits the budget; it does not prove symmetry-breaking necessity or budget saturation for every optimum.

2.6.2 Raw Full-Sweep Budget Limit

In the declared raw full-sweep implementation, the agent tracks all four quaternionic components $\{1, \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ simultaneously. The computational cost is

$$\mathcal{C}_{\text{proc}} = h_\mu \cdot \tau_{\text{mem}} \cdot D = 4\,h_\mu \cdot \tau_{\text{mem}}, \qquad D := \dim Cl(0,2) = 4. \tag{2.29}$$

The maximum raw full-sweep archive depth is

$$\tau_{\text{par}} = \frac{\mathcal{C}_{\text{budget}}}{h_\mu \cdot D} = \frac{2\,h_\mu}{4\,h_\mu} = 0.5 \qquad \text{(in units of } \omega_0^{-1}\text{)}. \tag{2.30}$$

Beyond $\tau_{\text{mem}} = 0.5\,\omega_0^{-1}$, this uncompressed implementation does not fit the stated budget. This does not exclude other symmetric compressed algorithms.

2.6.3 Frame Selection: Choosing σ_z

For comparison, we choose a frame that selects σ_z as the privileged basis direction, retaining the $\{1, \mathbf{k}\}$ subspace (the dephasing channel) as foreground and discarding $\{\mathbf{i}, \mathbf{j}\}$ (the dissipative channel) as background:

$$\mathbb{H} = \underbrace{\text{span}\{1, \mathbf{k}\}}_{V_{\text{fg}}\ (k^*=2)} \oplus \underbrace{\text{span}\{\mathbf{i}, \mathbf{j}\}}_{V_{\text{bg}}}. \tag{2.31}$$

Why σ_z? The dephasing channel ($\lambda_z = 1$, $\gamma_z = 0.5$) is strongly non-Markovian and carries the dominant survival-relevant information (the backflow revivals that enable $\mathcal{S} > 0$, as demonstrated in Paper I). The dissipative channel ($\lambda_x = 0.3$, $\gamma_x = 5.0$) is approximately Markovian and contributes primarily to decoherence—its survival value is negative.

This choice coincides with the *pointer basis* selected by environmental decoherence (quantum Darwinism [37]): the σ_z eigenstates are the states that survive decoherence and become redundantly encoded in the environment. The ego "accepts the suggestion" of decoherence, aligning its computational resources with the environmentally stable basis.

2.6.4 The Broken Phase: Effective Processing

In the broken phase, the projected memory kernel $\mathcal{K}_{\mathcal{F}} = \mathcal{K}_z$ retains only the dephasing-channel dynamics. The computational cost drops to

$$\mathcal{C}_{\text{proc}}^{(\mathcal{F})} = h_\mu \cdot \tau_{\text{mem}} \cdot k^* = 2\, h_\mu \cdot \tau_{\text{mem}}, \tag{2.32}$$

exactly half the symmetric cost (2.29). This implementation can process a raw archive up to depth $\tau_{\text{mem}} = 1\,\omega_0^{-1}$ before reaching its budget—twice the full-sweep depth of the four-channel baseline.

The survival functional in the broken phase is

$$\mathcal{S}_{\text{vis}}(\mathcal{F}) = \mathcal{S}[\mathcal{K}_z], \tag{2.33}$$

which gives the plotted positive backflow proxy at the first revival. This proxy comparison is not a proof of thermodynamic optimality.

The hidden component $\mathcal{S}_{\text{hid}} = \mathcal{S}[\mathcal{K}_x]$ is the survival contribution from the dissipative channel, which the agent can no longer evaluate. For the chosen parameters, $|\mathcal{S}_{\text{hid}}| \ll |\mathcal{S}_{\text{vis}}|$ (the dissipative channel contributes primarily negative survival value), so the distortion is small.

2.6.5 Quantitative Evaluation

We now evaluate the declared implementation comparison explicitly. Each channel's decoherence function follows from the exact $T \to 0$ solution of the Lorentz–Drude pure-dephasing model [10, 49]:

$$p_\alpha(t) = e^{-\gamma_\alpha t/2}\left[\cos(\Omega_\alpha t) + \frac{\gamma_\alpha}{2\Omega_\alpha}\,\sin(\Omega_\alpha t)\right], \quad \Omega_\alpha := \tfrac{1}{2}\sqrt{4\lambda_\alpha\gamma_\alpha - \gamma_\alpha^2}, \tag{2.34}$$

for $\alpha \in \{z, x\}$. When $4\lambda_\alpha\gamma_\alpha < \gamma_\alpha^2$ (the overdamped regime), Ω_α becomes imaginary and the trigonometric functions are replaced by hyperbolic functions (monotonic decay, no backflow).

For our parameters:

- **z-channel** ($\lambda_z = 1$, $\gamma_z = 0.5$): $\Omega_z = \frac{1}{2}\sqrt{1.75} \approx 0.661$. Underdamped; $|p_z(t)|$ exhibits oscillatory backflow.
- **x-channel** ($\lambda_x = 0.3$, $\gamma_x = 5.0$): Discriminant $4\lambda_x\gamma_x - \gamma_x^2 = 6 - 25 = -19 < 0$. Overdamped; $|p_x(t)|$ decays monotonically with no backflow.

For this model we plot the proxy used in Paper I, $\beta\,\mathcal{S} \propto |p(t)|^2 - 1$ (for the pure-dephasing model with maximally coherent initial state and pointer-basis measurement), evaluated for each channel independently. Backflow intervals are those where $d|p_\alpha|/dt > 0$. The proxy increases there; interpreting that increase as thermodynamic survival requires the initial-state and interaction-energy conditions of the revised Paper I accounting identity.

Model comparison. For the z-channel with $\gamma_z = 0.5$, the first backflow interval begins at $t^* \approx 2.9\,\omega_0^{-1}$—well after the raw full-sweep depth $\tau_{\max} = 0.5\,\omega_0^{-1}$. The stipulated raw full-sweep implementation stops at its budget limit before the plotted backflow interval. The selected z-channel implementation remains within its declared component budget and records the three plotted revivals.

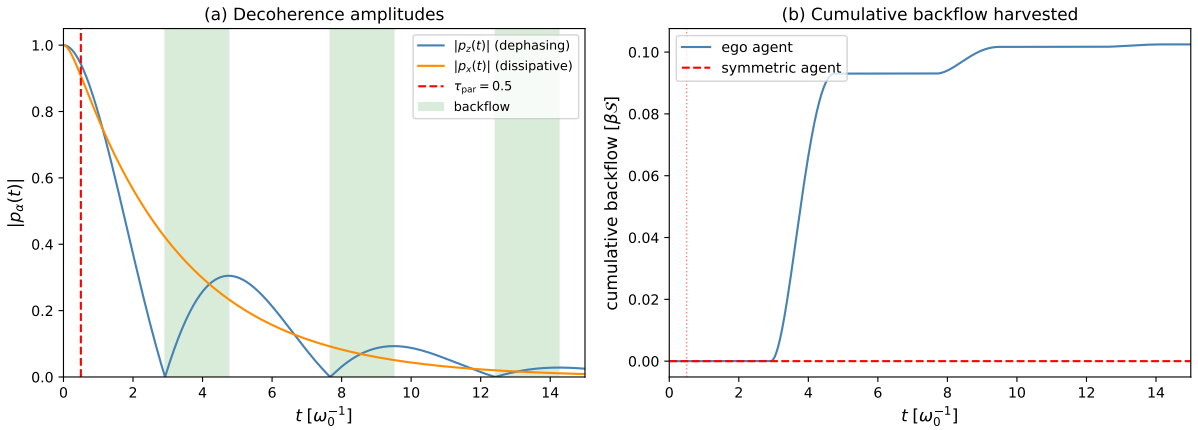


Figure 2.1: Two-channel qubit model (Section 2.6) with Lorentz–Drude spectral density. **Parameters:** $\omega_0 = 1$ (energy unit); $\lambda_z = 1$, $\gamma_z = 0.5$ (dephasing, non-Markovian); $\lambda_x = 0.3$, $\gamma_x = 5.0$ (dissipative, ∼Markovian); $\mathcal{C}_{\text{budget}} = 2\,h_\mu$. **Units:** time in ω_0^{-1}. **Regime:** low temperature ($\beta\omega_0 \gg 1$); using the standard $T \to 0$ analytic expression (2.34) [10]. **(a)** Decoherence amplitudes $|p_z(t)|$ (blue, non-Markovian, with backflow in green bands) and $|p_x(t)|$ (orange, monotonic decay). Red dashed: raw full-sweep depth $\tau_{\max} = 0.5$. **(b)** Cumulative backflow harvested. Blue: ego agent (broken $\to \sigma_z$) exploits all three revival intervals. Red dashed: the stipulated raw full-sweep agent stops at $\tau_{\max}$; all plotted backflow occurs later. The growing gap is the *ego dividend*.

The cumulative backflow harvested by the ego agent (Figure 2.1(b)) totals approximately 0.10 (in dimensionless $\beta\,\mathcal{S}$ units) over $t \in [0, 15\,\omega_0^{-1}]$. The raw baseline records zero after its stipulated stopping point. The result compares these two implementations only; it is not an infinite performance ratio against all symmetric algorithms and does not show that backflow is accessible only after symmetry breaking.

Accounting check. The raw update load is $h_\mu D = 4h_\mu$ while the budget is $2h_\mu$, so that declared representation does not fit. The example then *chooses* the two-component dephasing frame ($k^* = 2$); for that choice, $R_{\mathcal{F}} = 2\,h_\mu = \mathcal{C}_{\text{budget}}$. Saturation is an example property, not a general consequence of monotonic distortion.

2.6.6 The Pointer-State Connection

In this example the chosen basis coincides with an einselection (environment-induced superselection) basis of the dephasing model [37]. The coordinate optimization can select pointer-aligned components when the declared distortion assigns them the highest survival value per bit. This alignment is verified only within a concrete model and is not generic rate-distortion theorem.

Remark 2.26 (Decoherence as symmetry-breaking catalyst)**.** *The environment does not* force *a specific gauge fixing. A pointer-aligned frame is selected only if the declared distortion and encoder class rank it above the alternatives. The present example chooses that ranking; it does not derive it from bounded computation.*

2.7 The Cost of Ego

Coordinate selection reduces the declared raw load but can introduce a blind subspace. A fixed reference frame is a *static* gauge choice in a *dynamic* environment. If the environment changes, the ego becomes progressively maladaptive.

Drift layer. Environmental change can occur at multiple levels: parameter drift ($\lambda_\alpha(t)$, $\gamma_\alpha(t)$), spectral-density deformation ($J(\omega, t)$), or full process-distribution shift ($P_t(X)$). For analytical tractability, we model drift at the *spectral-density parameter level* throughout this section. No monotone extension to arbitrary distribution shift is claimed.

2.7.1 The Rigidity Trap

Proposition 2.27 (Frame Rigidity under Drift)**.** *Let the environment undergo slow drift: the spectral density parameters change as $\lambda_\alpha(t) = \lambda_\alpha^{(0)} + \varepsilon\, f_\alpha(t)$ for $\alpha \in \{z, x\}$, with drift rate $\varepsilon > 0$. Assume the supplied distortion has a locally unique C^1 minimizer $\mathcal{F}^*(\lambda) \in G$ and $\|D_\lambda \mathcal{F}^*\| \le K$ along the path.*

If the agent's reference frame $\mathcal{F}$ is held fixed (no recalibration), the mismatch between $\mathcal{F}$ and $\mathcal{F}^*(t)$ then obeys

$$\delta(t) := d_G\big(\mathcal{F},\, \mathcal{F}^*(t)\big) \le K\varepsilon \int_0^t \|\dot{f}(s)\|\, ds, \tag{2.35}$$

where d_G is the geodesic distance in the gauge group.

Proof. The chain rule and the stated sensitivity bound give $\|\dot{\mathcal{F}}^*(t)\| \le K\varepsilon \|\dot{f}(t)\|$. Integrating path length bounds the geodesic distance and yields (2.35). □

2.7.2 Stylized Drift Model

To quantify the loss of visibility of a fixed frame, we use the linear drift supported by Proposition 2.27. No unstable environmental dynamics or Lyapunov exponent is assumed.

Definition 2.28 (Rotating optimal frame)**.** *Let the mismatch angle $\theta(t)$ between the agent's fixed frame $\mathcal{F}$ and the instantaneous optimal frame $\mathcal{F}^*(t)$ evolve as*

$$\theta(t) = \theta_0 + \varepsilon t, \tag{2.36}$$

where $\theta_0 \in (0, \pi/4)$ is the initial misalignment (so that the crossover time below is positive) and $\varepsilon > 0$ is the angular drift rate. This is a local model up to the first visibility crossover, not a claim about long-time motion on the compact gauge group.

The visible and hidden survival components decompose geometrically:

$$\mathcal{S}_{\text{vis}}(t) = \mathcal{S}_{\text{tot}}\, \cos^2\theta(t), \qquad \mathcal{S}_{\text{hid}}(t) = \mathcal{S}_{\text{tot}}\, \sin^2\theta(t), \tag{2.37}$$

where $\mathcal{S}_{\text{tot}}$ is the full survival functional (invariant under frame rotation).

2.7.3 The Prediction Error Divergence

Proposition 2.29 (Growth of the hidden component)**.** *Under the drift model* (2.36)–(2.37)*, the hidden survival component grows as*

$$\big|\mathcal{S}_{\text{hid}}(t)\big| = |\mathcal{S}_{\text{tot}}|\,\sin^2\big(\theta_0 + \varepsilon t\big). \tag{2.38}$$

For small angles ($\theta_0 + \varepsilon t \ll 1$)*:* $|\mathcal{S}_{\text{hid}}| \approx |\mathcal{S}_{\text{tot}}|(\theta_0 + \varepsilon t)^2$*.*

Proof. Direct substitution of (2.36) into (2.37). The small-angle expansion $\sin^2\theta \approx \theta^2$ gives the quadratic form. □

2.7.4 Visibility Crossover and Conditional Gradient Reversal

Proposition 2.30 (Visibility crossover and a conditional gradient reversal)**.** *Under* (B1)–(B5) *with the drift model* (2.36) *and initial misalignment* $\theta_0 \in (0, \pi/4)$*, an agent with a fixed reference frame* $\mathcal{F}$ *reaches the visibility crossover*

$$t_{\text{vis}} = \frac{\pi/4 - \theta_0}{\varepsilon}, \tag{2.39}$$

at which the hidden and visible magnitudes are equal. For $t > t_{\text{vis}}$*, the hidden magnitude is larger while* $\theta(t) < \pi/2$*. This geometric crossover alone implies neither gradient anti-correlation nor failure of every in-frame detector.*

There nevertheless exist specified controllers for which gradients do reverse. In the two-mode model

$$\mathcal{S}_{\text{vis}}(u) = -\tfrac{1}{2}(u-1)^2, \qquad \mathcal{S}_{\text{full}}(u,\theta) = -\tfrac{1}{2}\|ue_1 - y^*(\theta)\|^2, \quad y^*(\theta) = \cos\theta\, e_1 + \sin\theta\, e_2, \tag{2.40}$$

the two scalar update directions are anti-correlated exactly when

$$\cos\theta(t) < u < 1.$$

Thus the reversal time is controller- and state-dependent; it need not coincide with t_{vis}*.*

Proof. The crossover $|\mathcal{S}_{\text{hid}}| = |\mathcal{S}_{\text{vis}}|$ occurs when $\sin^2\theta = \cos^2\theta$, i.e., $\theta(t_{\text{vis}}) = \pi/4$. Substituting (2.36): $\theta_0 + \varepsilon t_{\text{vis}} = \pi/4$, which gives (2.39). For (2.40), $\partial_u \mathcal{S}_{\text{vis}} = 1 - u$ and $\partial_u \mathcal{S}_{\text{full}} = \cos\theta - u$. Their product is negative exactly when u lies strictly between $\cos\theta$ and 1. Outside that interval the two gradients agree in sign or one vanishes. The calculation exhibits the claimed effect without promoting it to a universal consequence of visibility loss. □

Remark 2.31 (Model scope)**.** *For* $\theta_0 = 0$*, the crossover occurs at* $t_{\text{vis}} = \pi/(4\varepsilon)$*. With* $\varepsilon = 0.01\,\omega_0$*,* $t_{\text{vis}} \approx 79\,\omega_0^{-1}$*—long enough for the agent to accumulate a false sense of security, yet short on environmental timescales. Neither this number nor the two-mode sign condition proves that an agent cannot detect drift from residual statistics; Paper III studies one such detector.*

Remark 2.32 (Why dithering does not help)**.** *One might ask whether the agent could escape the delusion trap by randomly "probing" the background subspace* V_{bg}*—temporarily rotating its frame to sample hidden components. In the raw-coordinate model, a full probe competes with foreground processing at the component-count rate in Remark 2.8.*

This does not prove dithering fails: randomized exploration can reveal hidden directions, with a model-dependent cost–benefit trade-off. A deterministic in-frame controller may lack a signal indicating when or where to probe. As long as $|\mathcal{S}_{\text{hid}}| < |\mathcal{S}_{\text{vis}}|$ *(pre-delusion), the in-frame performance metric* $\mathcal{S}_{\text{vis}}$ *shows no anomaly. The linear-drift visibility model below motivates monitoring residual statistics, the subject of Paper III, but does not rule out other exploration policies.*

Remark 2.33 (The ego as medicine and poison)**.** *In this model, coordinate selection reduces raw load but creates a blind subspace. The visibility crossover below motivates Paper III. This is a conditional engineering trade-off, not an irreversible logic chain.*

2.7.5 The Origin of Paper III

To respond to the visibility loss in this model, the agent may use a mechanism to monitor the quality of its own reference frame—to "observe its own observation." This requires a *second-order control loop*: a meta-controller that adjusts the gauge fixing σ in response to accumulated prediction errors.

The distribution of in-frame prediction residuals can itself change as the frame becomes stale. Paper III studies whether a specified likelihood model makes that change statistically identifiable. This paper does not prove that Fisher information is the unique detector or that every controller must add such a loop.

This is the subject of Paper III: the Fisher information geometry of self-referential calibration, and the thermodynamic cost of the loop that closes the chain *Chaos* $\rightarrow$ *Time* $\rightarrow$ *Self* $\rightarrow$ *Calibration*.

2.8 Numerical Demonstration

The preceding sections establish analytic bounds and a worked example with a qubit in a two-channel bath. We now compare budgeted coordinate-selection rules in a minimal multi-dimensional system. The observed attention concentration and selection advantage are properties of this supplied algorithm and environment. Full code and parameters are provided for reproducibility.

2.8.1 Model

Environment. A D-dimensional linear prediction task with sparse rotating support: $y(t) = \mathbf{w}^*(t)^\top \mathbf{x}(t) + \xi(t)$, $\mathbf{x}(t) \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_D)$, $\xi \sim \mathcal{N}(0, \sigma^2)$. Only $m \ll D$ dimensions carry nonzero weight at any time; the active support rotates every τ_{switch} steps, modelling environmental drift.

Hard budget constraint. Per step, the agent may update only k coordinates of its weight vector (a hard processing budget), mirroring the bounded computation assumption (B2).

Agents.

- **Budgeted selector**: selects the top-k dimensions by importance score—an exponential moving average of the signed per-coordinate gradient. Signed accumulation

ensures that noise dimensions (zero expected signal) cancel over time while signal dimensions persist, enabling reliable discrimination without access to the true support.

- **Random-k baseline**: selects k dimensions uniformly at random each step. This provides a budget-fair comparison with no persistent coordinate preference.

The choice of *signed* gradient EMA (rather than squared-gradient magnitude) is structurally motivated: for noise dimensions $\mathbb{E}[r\,x_i] = 0$, so the signed accumulation cancels over time; for signal dimensions $\mathbb{E}[r\,x_i] \neq 0$, so a consistent directional bias persists. The signed EMA thus acts as a *directional coherence filter* that discriminates signal from noise without access to the true support—a minimal model of a persistent coordinate preference.

Parameters.

Quantity	Value	Role
D	64	ambient dimension
m	8	signal dimensions (sparse support)
T	10,000	horizon per trial
Seeds	10	independent replications
σ	0.3	observation noise std
η	0.02	SGD learning rate
λ	0.995	weight decay per step
k	2, 4, 6, 8, 10, 12, 16, 20, 24, 32, 48, 64	budget grid
τ_{switch}	$\{500, 1000, 2000\}$	support rotation period

Attention entropy. Let n_i be the number of updates coordinate i receives in a measurement window of the last 1,000 steps. The normalised update frequency $p_i = n_i / \sum_j n_j$ defines the attention entropy:

$$H_{\text{attn}} = -\sum_{i=1}^{D} p_i \ln p_i. \tag{2.41}$$

Under uniform coordinate selection, $p_i = 1/D$ and $H_{\text{attn}} = \ln D$. Under budget-constrained selection, H_{attn} collapses away from $\ln D$, measuring concentration of the update rule. It is not by itself a thermodynamic order parameter.

Oracle metric. Neither agent has access to $\mathbf{w}^*(t)$. Performance is evaluated externally using the weight-space mean-squared error $\text{MSE} = \|\hat{\mathbf{w}} - \mathbf{w}^*\|^2$, averaged over post-burn-in steps.

2.8.2 Results

Figure 2.2 shows the two key signatures.

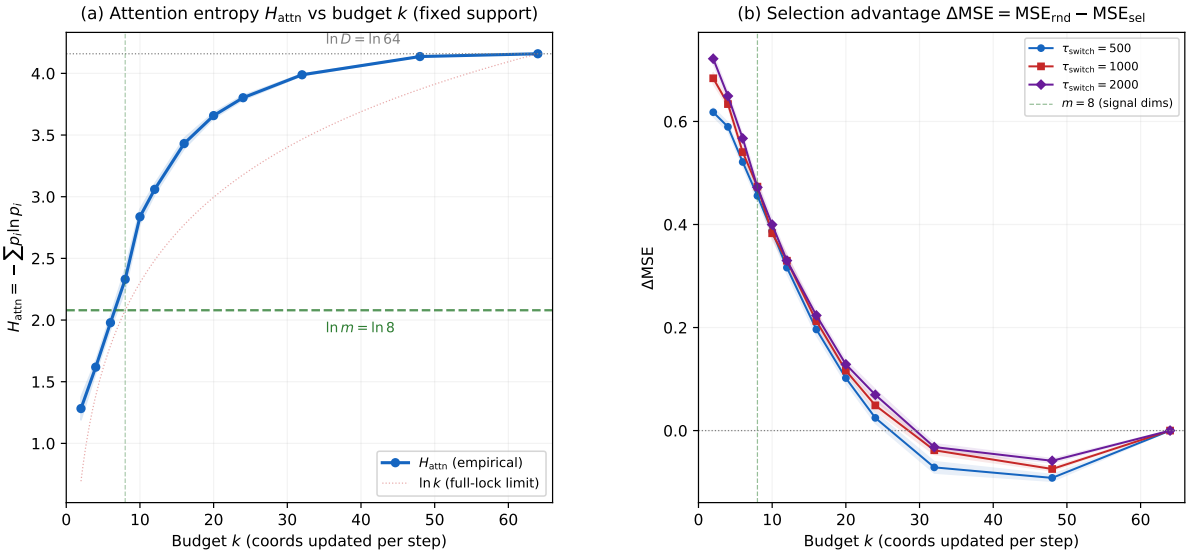


Figure 2.2: **Budgeted coordinate selection in the declared linear model.** $D = 64$, $m = 8$, $T = 10{,}000$, 10 seeds, 95% CI bands. **(a)** Attention entropy H_{attn} vs budget k (fixed support). The empirical curve (blue) collapses from $\ln D \approx 4.16$ toward an $O(\ln m)$ floor as budget tightens. For $k \leq m$, H_{attn} remains below $\ln m \approx 2.08$ (green dashed), consistent with confinement to the signal subspace. **(b)** Selection advantage $\Delta\text{MSE} = \text{MSE}_{\text{rnd}} - \text{MSE}_{\text{sel}}$ vs budget k under rotating support. The gap is positive for $k \lesssim 3m$ (selection helps), turns slightly negative at large k (commitment cost exceeds diversification), and returns to zero at $k = D$. Slower drift ($\tau = 2000$) yields a larger peak advantage.

Result 1: Attention entropy collapse (Figure 2.2a). Under fixed support (no rotation), the attention entropy H_{attn} exhibits a sharp collapse away from $\ln D = \ln 64 \approx 4.16$ and increases monotonically with k, consistent with a budget-induced concentration of update mass onto signal-carrying dimensions. For budgets near and below the signal scale ($k \leq m$), H_{attn} remains $O(\ln m)$, consistent with confinement to the signal subspace. We use the collapse of H_{attn} away from $\ln D$ as the order parameter of symmetry breaking; a strict plateau at $\ln m$ is not expected under the present re-selection dynamics and finite-window estimator.

Result 2: Selection advantage (Figure 2.2b). Under rotating support, the mean-squared error gap $\Delta\text{MSE} = \text{MSE}_{\text{rnd}} - \text{MSE}_{\text{sel}}$ is positive for $k \lesssim 3m$ and peaks at tight budgets ($k = 2$) where the selection advantage is strongest. For $k \gg m$ the gap turns slightly negative (the selector's commitment to stale dimensions costs more than the random baseline's diversification), before returning to zero at $k = D$. The three τ_{switch} curves are ordered: slower drift (larger τ) yields a larger peak gap, with the ordering most visible at small k.

2.8.3 Scope of This Demonstration

These simulations compare two supplied coordinate-selection rules in the stated sparse linear model. They do not establish a universal symmetry-breaking transition or identify the capacity $k_{\max}$ with an information-bottleneck optimum.

This demonstration **does** show:

1. Under hard budget constraints, attention entropy collapses sharply away from $\ln D$ and remains $O(\ln m)$ for $k \leq m$—the agent confines its updates to the signal subspace. This measures concentration of the chosen update rule.

2. A budgeted selector that exploits importance-weighted selection systematically outperforms a budget-fair random baseline, consistent with a survival advantage for the parameter range shown.

3. The advantage scales with both budget tightness (smaller k) and environmental stability (larger τ_{switch}).

In summary, this demonstration establishes a model-specific selection advantage and a measurable attention-concentration statistic; it does not derive selection from the budget or claim universality across architectures or environment classes.

This demonstration does **not** show:

1. That H_{attn} reaches a strict plateau at $\ln m$ for all $k \leq m$. Under the adaptive re-selection dynamics used here, the selector cycles within the signal subspace, producing H_{attn} values near but not locked to $\ln m$. The relevant signature is the collapse *away from* $\ln D$, not convergence to a specific lower bound.

2. That the specific form of the importance score (signed gradient EMA) is optimal. It is one realisation of the selection mechanism.

3. That the results generalise to all environment classes. The model uses Gaussian features, linear regression, and sparse rotating support.

4. That the delusion–correction cycle is addressed. This is the subject of Paper III.

5. That a sharp thermodynamic phase transition occurs, or that $k = k_{\max}$ is the distortion-minimizing retained dimension.

Reproducibility. The complete simulation is a self-contained Python script (`paper2_kstar_scaling_demo.py`, $\sim$540 lines, requiring only NumPy and Matplotlib) with fixed random seeds. All figures in this section can be reproduced by executing the script. The following files are included in the supplementary archive:

- `paper2_kstar_scaling_demo.py` — simulation script
- `fig_paper2_kstar_scaling.pdf` — Figure 2.2
- `kstar_scaling_data.csv` — raw performance gap data

2.9 Discussion

2.9.1 Summary of Results

Result	Statement	Sec.
Raw-channel load	A declared D-channel archive costs Dh_μ per incremental update; this is accounting, not a lower bound	2.3
Compression existence	A minimizer exists in the supplied compact encoder class and has $k_{\text{opt}} \leq k_{\max}$	2.4.2
Conditional frame selection	A chosen proper projection breaks the declared model symmetry when no invariant feasible encoder meets the target	2.4.3
Four diagnostics	Frame-dependent models can be audited using $\mathcal{B}_{\text{select}}$, $\mathcal{B}_{\text{frame}}$, $\mathcal{B}_{\text{center}}$, $\mathcal{B}_{\text{inc}}$	2.4.4
Survival Decomposition	$\mathcal{S} = \mathcal{S}_{\text{vis}} + \mathcal{S}_{\text{hid}}$	2.5.2
Uniform-channel trade-off	Predictive loss is approximately $1 - k^*/D$ only under the uniform raw-channel assumption	2.5.4
Visibility crossover	Linear drift gives $t_{\text{vis}} = (\pi/4 - \theta_0)/\varepsilon$; gradient reversal requires a separate controller condition	2.7.4
Numerical demo	Attention concentration and model-specific selection advantage (Fig. 2.2)	2.8

2.9.2 Testable Predictions

The conditional model claims can be tested without treating them as universal laws. First, vary k, D, and the support-switching time in the declared sparse linear environment and compare the supplied selector with random and alternative compressed baselines. A sharp transition is not predicted without a separately defined thermodynamic limit.

Second, estimate $I(X_{\text{past}}; X_{\text{future}})$ and $I(\hat{X}_{\text{past}}; X_{\text{future}})$ while varying k/D. Linear $1 - k/D$ scaling tests the uniform-allocation assumption in Theorem 2.24; systematic curvature falsifies that assumption rather than information theory.

Third, in a controlled rotating-feature task, measure both the visibility ratio and the inner product of surrogate and oracle gradients. The linear model predicts the visibility crossover at $t_{\text{vis}} = (\pi/4 - \theta_0)/\varepsilon$. Proposition 2.30 predicts gradient reversal only for controllers entering the region $\cos\theta(t) < u(t) < 1$; coincidence of the two events is not required.

2.9.3 What This Paper Does and Does Not Show

This paper **does** show:

1. A declared raw D-channel archive has the stated component-count load, and a minimizer exists in the supplied compact encoder class.
2. A proper coordinate projection privileges a frame; under additional feasibility assumptions this can be interpreted as operational symmetry breaking (Remark 2.18).
3. Definition 2.19 supplies four model diagnostics, with the extra controller and task assumptions needed by each made explicit.
4. Linear frame drift produces a visibility crossover, while the two-mode model exhibits a separate, conditional gradient reversal (Proposition 2.30).
5. A minimal computational demonstration measures attention concentration and a selection advantage over a budget-fair random baseline (Section 2.8, Figure 2.2).

This paper does **not** show:

1. That bounded computation uniquely determines a privileged basis, forces symmetry breaking, or makes the chosen frame survival-optimal.
2. That symmetry breaking is necessary or sufficient for persistence.
3. That the "ego" implies or requires consciousness, subjective experience, or phenomenal awareness. The term is used strictly in the control-theoretic sense.
4. That this framework constitutes a theory of consciousness. It is a theory of computational optimality under thermodynamic constraints.
5. That the four diagnostics exhaust self-reference or follow from gauge fixing alone.
6. That the coordinate capacity is a Shannon rate–distortion bound or is achievable by a specific physical implementation.
7. That visibility loss forces gradient reversal or is undetectable. Paper III analyzes one conditional calibration model.
8. That the framework constitutes or implies a philosophical or metaphysical claim about the nature of selfhood.
9. That this framework applies to all possible physical systems. It applies to systems satisfying the declared channel, encoder, controller, and drift assumptions.
10. That the Clifford algebra is the only possible algebraic setting. It is supplied here as a model input; Q-RAIF does not uniquely derive it.

Chapter 3

Fisher Information Geometry and Conditional Implementation Costs of Self-Referential Calibration

Paper III — "The Loop"

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Abstract

Papers I and II of the T-DOME series [49, 50] supply conditional resource and reference-frame models for persistent agents. They do not prove that all persistent agents require non-Markovian memory or spontaneous Clifford gauge-symmetry breaking. Paper II's linear-drift visibility crossover is likewise a declared coordinate model, not a universal logarithmic "Delusion Trap" timescale.

This paper studies **self-referential calibration** under explicit residual-likelihood, prior, drift, convexity, and hardware assumptions. The agent uses the score statistics of a supplied prediction-residual model to estimate frame mismatch. The principal conditional results are:

1. **Gaussian Fisher Signal** (Proposition 3.14): in the declared even Gaussian residual model, per-sample local Fisher information grows quadratically and accumulated information grows cubically. Detection precedes the visibility crossover if and only if the threshold condition in Corollary 3.16 holds.

2. **Self-Referential Cramér–Rao Bound** (Theorem 3.18): the van Trees lower bound applies under the stated regular likelihood, conditional-independence, compact-support, and prior assumptions, with data and prior Fisher information entering separately.

3. **Conditional Implementation Cost** (Proposition 3.28): a separable implementation with declared irreversible sensor and computational resets and linear-response actuation obeys $\dot{W}_{\mathrm{loop}} \geq k_B T \, \ln 2 \, (R_s + R_c) + \mathcal{L}^2/\tau^2_{\mathrm{recalib}}$, a bound for that implementation rather than a universal cost of sensing, Fisher evaluation, or self-reference.

The calibration loop has a local tracking bound only under geodesic strong convexity and bounded drift (Theorem 3.23). The four supplied layers in Definition 3.27 define the running model architecture; they are not proved necessary, sufficient, minimal, or complete for persistent agency. "Reflexivity" here denotes one implementation of second-order control, with no claim of architectural uniqueness or phenomenal consciousness.

3.1 Introduction

3.1.1 Context: The Delusion Trap

Revised Paper II [50] studies a supplied coordinate projection under a raw-channel budget. It does not prove that persistence forces Clifford gauge symmetry breaking. Its drift model uses a linearly changing mismatch $\theta(t) = \theta_0 + \varepsilon t$ and identifies the geometric visibility crossover

$$t_{\text{vis}} = \frac{\pi/4 - \theta_0}{\varepsilon}. \tag{3.1}$$

That crossover does not by itself imply gradient reversal or undetectability. The present paper asks a narrower question: for a specified residual likelihood, can score statistics detect mismatch, and under what conditions can a natural-gradient controller track a moving optimum?

3.1.2 Position within Papers I–III

This paper is the third and final of the T-DOME framework, closing the three-paper sequence.

Framework	Question	Result	Status
HAFF [41, 42]	How does geometry emerge?	Conditional algebra/dictionary models	Revised
Q-RAIF [46, 47]	What algebra must an observer have?	Clifford representations as model inputs	Revised
T-DOME I [49]	What can inter-epoch correlations fund?	Restricted baseline; correlation accounting	Conditional
T-DOME II [50]	When can a model select a frame?	Conditional coordinate selection	Conditional
T-DOME III (this work)	How does self-calibration arise?	Fisher self-referential bound; thermodynamic cost of reflexivity	**This paper**

The three T-DOME papers form a conditional modelling sequence:

1. **Paper I:** Initial correlations can enter an ELV-type thermodynamic balance; a restricted autonomous baseline is proved.

2. **Paper II:** A raw-coordinate archive motivates a supplied compression model and a linear visibility-crossover example.

3. **Paper III (this work):** Uncorrected bias diverges from a changing environment. A self-referential calibration loop—monitoring the Fisher information of one's own prediction stream—resolves mismatch under explicit likelihood and convexity assumptions. A conditional hardware model assigns reset and actuation costs.

The links are motivation and model interfaces, not an irreversible theorem chain. None of the three papers proves that this architecture is necessary for persistence.

3.1.3 The Information-Geometric Insight

The key observation that resolves the Delusion Trap is subtle: *while the agent cannot observe V_{bg} directly, it can observe the statistical properties of its own prediction residuals in V_{fg}.*

The prediction residual $e(t) := \mathcal{S}_{\text{vis}}(t) - \mathcal{S}_{\text{vis}}^{(\text{pred})}(t)$ lies in V_{fg} by construction. In the declared even Gaussian model, a record cannot identify the sign of mismatch. Its *distribution*—the probability law $p(e \mid \sigma)$, parametrised by the gauge-fixing parameter σ—does depend on σ, because the projection $\Pi_{\mathcal{F}}(\sigma)$ determines which environmental correlations are captured and which are discarded.

When the frame σ drifts away from the optimal σ^*, the residual distribution shifts. The *Fisher information metric*

$$g_{ij}(\sigma) = \mathbb{E}_\sigma\left[\frac{\partial \log p(e \mid \sigma)}{\partial \sigma^i}\,\frac{\partial \log p(e \mid \sigma)}{\partial \sigma^j}\right] \tag{3.2}$$

measures the sensitivity of this distribution to changes in σ. Whether g_{ij} changes with staleness is likelihood-dependent. The Gaussian model below provides one explicit case.

This is the mathematical realisation of the "second-order operation" demanded by Paper II, Section 7.5: the agent does not need to see the truth (the full $Cl(V, q)$), but only the *rate of change of its own prediction error* as a function of its frame parameters. Fisher information is one local sensitivity measure, not the unique possible detector.

3.1.4 Relation to Architectural Incompleteness

Revised HAFF Paper G [45] supplies philosophy-of-science motivation rather than a fixed-point no-go theorem. In that motivational sense, the self-referential calibration loop cannot *eliminate* architectural incompleteness, but it can *track* the consequences of incompleteness in real time. The Lyapunov function $V(\sigma)$ monitors the distance between the agent's frame and the optimal frame without requiring access to the "complete" description—it operates entirely within the agent's own predictive statistics.

3.1.5 Scope and Disclaimers

1. *Reflexivity* refers throughout to second-order control: the ability of a system to monitor and adjust its own monitoring process. It carries *no* implication of phenomenal consciousness, subjective experience, or qualia.

2. The self-referential calibration loop does not *eliminate* the ego's bias; it tracks and compensates for drift in the bias. The four bias terms of Paper II persist in the calibrated phase.

3. The thermodynamic cost statement is conditional on irreversible reset, separable hardware, and a linear-response actuation model.

4. The framework applies to systems satisfying (C1)–(C5) (Section 3.2.6). It is not a universal theory of agency.

Related work. The Fisher information metric on statistical manifolds was introduced by Rao [25] and shown to be unique by Čencov [11]. The natural gradient and information geometry were developed by Amari [2, 3]. Thermodynamic length and optimal finite-time transformations were established by Crooks [13] and Sivak–Crooks [34]. The connection between Fisher information and entropy production was formalised by Ito [18] and Barato–Seifert [6]. Second-order cybernetics originates with Ashby [4] and von Foerster [40]. Adaptive control and self-tuning regulators are treated in [5]. The Bayesian Cramér–Rao bound (van Trees inequality) is from [39].

Summary of contributions. This paper separates three levels of result:

1. **Drift Detectability** (Proposition 3.14): an explicit Gaussian residual likelihood has quadratic per-sample and cubic accumulated local Fisher signals; early detection is a threshold condition.

2. **Self-Referential Cramér–Rao Bound** (Theorem 3.18): drift-estimation precision is bounded by the sum of data Fisher information and ego rigidity.

3. **Thermodynamic Cost** (Proposition 3.28): a separable implementation with irreversible resets and finite-time actuation obeys a conditional additive bound.

Result-status ledger. The original drift-detectability theorem `TD-III-T3.3` is retracted; Proposition 3.14 is an FD/I model calculation and Corollary 3.16 is conditional on a declared threshold. Ego rigidity `TD-III-ego` remains a supplied prior model. The tracking theorem is conditional on local geodesic strong convexity. The universal additive-cost claim `TD-III-T6.2` is downgraded to Proposition 3.28, and the inherited persistence budget is withdrawn.

3.2 Mathematical Preliminaries

3.2.1 Inherited Framework from Papers I and II

We briefly recall the key objects; the reader is referred to Papers I and II for full definitions and proofs.

From Paper I [49].

- **Survival functional.** $\mathcal{S}[\Lambda, \tau] := \Delta F - W[0, \tau]$ (Paper I, Eq. (9)).

- **Restricted autonomous baseline.** $\mathcal{S}[\Lambda^{\mathrm{D}}, \tau] \leq 0$ for a fixed-Hamiltonian, no-control Davies semigroup under its stated initial conditions. It is not a ceiling for all Markovian agents.

- **Memory kernel.** $\mathcal{K}(t,s)$: the non-Markovian superoperator encoding system–environment correlations.
- **Entropy rate.** $h_\mu := \lim_{T\to\infty} T^{-1} H(X_{0:T})$ (bits per unit time per algebraic component).
- **Predictive information.** $I_{\text{pred}} := I(\overleftarrow{X};\, \overrightarrow{X})$.

From Paper II [50].

- **Internal algebra.** $\mathcal{O}_{\text{int}} = Cl(V,q)$, $D = \dim Cl(V,q) = 2^n$, gauge group $G = \text{Aut}(Cl(V,q))$.
- **Gauge bundle.** $\pi : P \to M$, structure group G; a section $\sigma : M \to P$ is a reference frame.
- **Ego.** $\mathfrak{E} := (\mathcal{F}^*, V_{\text{fg}}^*)$ denotes a chosen optimizer in a supplied encoder class, with $k^* = k_{\text{opt}} \le k_{\max}$. The budget capacity is not a universal optimizer formula.
- **Projected kernel.** $\mathcal{K}_{\mathcal{F}}(t,s) = \Pi_{\mathcal{F}}\, \mathcal{K}(t,s)\, \Pi_{\mathcal{F}}$.
- **Survival decomposition.** $\mathcal{S} = \mathcal{S}_{\text{vis}}(\mathcal{F}) + \mathcal{S}_{\text{hid}}(\mathcal{F})$.
- **Four bias terms.** $\mathcal{B}_{\text{select}}$, $\mathcal{B}_{\text{frame}}$, $\mathcal{B}_{\text{center}}$, $\mathcal{B}_{\text{inc}}$ (revised Paper II, four-diagnostics definition). These are model diagnostics, not generic consequences of gauge fixing.
- **Visibility crossover.** $t_{\text{vis}} = (\pi/4 - \theta_0)/\varepsilon$ in the supplied linear drift geometry. Gradient reversal requires an additional controller condition.
- **Information objects.** Mutual information is evaluated on explicitly induced record processes; algebra symbols are not themselves random variables.

3.2.2 Fisher Information Metric

Definition 3.1 (Fisher information matrix)**.** *Let* $\{p(x \mid \theta) : \theta \in \Theta \subset \mathbb{R}^d\}$ *be a parametric family of probability densities satisfying standard regularity conditions (interchange of differentiation and integration). The* Fisher information matrix *is*

$$g_{ij}(\theta) := \mathbb{E}_\theta\left[\frac{\partial \log p(x \mid \theta)}{\partial \theta^i}\, \frac{\partial \log p(x \mid \theta)}{\partial \theta^j}\right] = -\mathbb{E}_\theta\left[\frac{\partial^2 \log p(x \mid \theta)}{\partial \theta^i\, \partial \theta^j}\right]. \tag{3.3}$$

The pair (Θ, g) *is a Riemannian manifold called the* statistical manifold.

Remark 3.2 (Uniqueness)**.** *By Čencov's theorem [11], the Fisher–Rao metric* g^{FR} *is, up to a positive scalar multiple, the unique Riemannian metric on the space of probability distributions that is invariant under all Markov morphisms (sufficient-statistic embeddings). Within the hypotheses and category used by Čencov's theorem, this makes the Fisher metric a canonical local distinguishability metric. It does not make a Fisher-based detector physically mandatory, nor does it choose a likelihood or control objective for the agent.*

Proposition 3.3 (Cramér–Rao bound)**.** *For any unbiased estimator $\hat{\theta}$ of θ based on n independent observations:*

$$\mathrm{Cov}(\hat{\theta}) \succeq \frac{1}{n}\left[g(\theta)\right]^{-1} \tag{3.4}$$

in the Löwner order. The scalar case reads $\mathrm{Var}(\hat{\theta}) \geq 1/\big(n\, g(\theta)\big)$.

Remark 3.4 (Effective independence)**.** *Throughout this paper, references to "independent observations" in the context of continuous-time residual streams should be read as* effective independence *after thinning by the environmental decorrelation time τ_E, yielding an effective sample size $n_{\mathrm{eff}} \approx T/\tau_E$. In particular, the sample count n in* (3.4) *becomes n_{eff} in the self-referential setting of Section 3.4.2.*

Remark 3.5 (Fisher metric and KL divergence)**.** *The Fisher metric arises as the Hessian of the Kullback–Leibler divergence [12]:*

$$D_{\mathrm{KL}}\big(p_\theta \,\|\, p_{\theta+d\theta}\big) = \tfrac{1}{2}\, g_{ij}(\theta)\, d\theta^i\, d\theta^j + O\big(|d\theta|^3\big)\,. \tag{3.5}$$

This identifies the Fisher metric as the infinitesimal measure of statistical distinguishability.

3.2.3 Information Geometry

Following Amari [1, 3], the statistical manifold $(\Theta,\, g)$ carries additional geometric structure beyond the Riemannian metric.

α-connections. For each $\alpha \in [-1,\, 1]$, Amari defines an affine connection $\nabla^{(\alpha)}$ on Θ. The cases $\alpha = 1$ (exponential connection, $\nabla^{(e)}$) and $\alpha = -1$ (mixture connection, $\nabla^{(m)}$) are *dual* with respect to g: $\partial_k\, g(X, Y) = g(\nabla^{(e)}_k X,\, Y) + g(X,\, \nabla^{(m)}_k Y)$. For exponential families, $\nabla^{(e)}$ is flat in natural parameters and $\nabla^{(m)}$ is flat in expectation parameters—the *dually flat structure*. The case $\alpha = 0$ recovers the Levi-Civita connection of the Fisher metric.

Natural gradient. Standard gradient descent in parameter space ignores the curvature of the statistical manifold. The *natural gradient* [2]

$$\dot{\theta} = -\eta\, g^{-1}(\theta)\, \nabla_\theta L(\theta), \tag{3.6}$$

where $\eta > 0$ is the learning rate and $L(\theta)$ is a loss function, provides the steepest descent direction in the Fisher metric. It is reparametrisation-invariant. Statistical efficiency requires additional estimator and asymptotic assumptions and is not implied by the update formula alone.

Pythagorean theorem. In a dually flat space, the KL divergence satisfies a generalised Pythagorean relation: $D_{\mathrm{KL}}(p \,\|\, r) = D_{\mathrm{KL}}(p \,\|\, q) + D_{\mathrm{KL}}(q \,\|\, r)$ when q is the m-projection of p onto a submanifold containing r. This decomposition will be applied to separate the foreground-recoverable and background-irrecoverable components of drift.

3.2.4 Thermodynamic Length

Definition 3.6 (Thermodynamic length)**.** *Let $\lambda(t)$ for $t \in [0, \tau]$ be a path through control parameter space, and let $\zeta_{ij}(\lambda)$ be the* friction tensor *(the time-integrated equilibrium force–force correlation function at λ). The* thermodynamic length *of the path [13] is*

$$\mathcal{L} := \int_0^\tau \sqrt{\zeta_{ij}(\lambda)\, \dot{\lambda}^i\, \dot{\lambda}^j}\; dt. \tag{3.7}$$

Proposition 3.7 (Sivak–Crooks bound)**.** *The excess (dissipated) work during a finite-time transformation of duration τ satisfies [34]*

$$W_{\mathrm{ex}} \;\geq\; \frac{\mathcal{L}^2}{\tau}. \tag{3.8}$$

The minimum is achieved by the geodesic of the friction tensor ζ. In the linear-response regime, the friction tensor is related to the Fisher metric of the equilibrium distribution at λ by $\zeta_{ij}(\lambda) \sim \tau_{\mathrm{relax}}\, g_{ij}^{\mathrm{Fisher}}(\lambda)$, where τ_{relax} is the relaxation time.

3.2.5 Second-Order Cybernetics

Von Foerster [40] distinguished two levels of control:

- **First-order cybernetics**: feedback control of observed systems. The controller adjusts its actions based on the output of a sensor. Paper II's ego is a first-order structure: it processes environmental data within a fixed frame.
- **Second-order cybernetics**: feedback control of the *observing* system itself. The controller adjusts the *sensor*—or equivalently, the reference frame within which the sensor operates. This is what Paper III provides.

Ashby's Law of Requisite Variety [4] provides a lower bound on the complexity of the meta-controller:

$$\dim(\text{meta-controller state space}) \;\geq\; \dim(\text{environmental drift subspace})\,. \tag{3.9}$$

The meta-observer must have at least as many adjustable parameters as there are independent modes of environmental drift.

In adaptive control theory [5], the analogous result is the *persistent excitation* condition: parameter estimates converge if and only if the input signal is "rich enough" to excite all modes of the system. In our framework, persistent excitation corresponds to $h_\mu > 0$—the environment must continue to generate novelty for the self-calibration loop to function.

Remark 3.8 (Operational content)**.** *The second-order cybernetic structure in this paper is* not *a philosophical metaphor. It has concrete operational content: the natural gradient update* (3.6) *is a specific algorithm that takes as input the Fisher information of the residual stream and produces as output an update to the frame parameter σ. An implementation must supply a differentiable residual likelihood, evaluate or approximate its score, and realize the update law. Merely accumulating an unlabelled residual second moment is not generally a Fisher computation.*

3.2.6 Standing Assumptions

Definition 3.9 (Standing Assumptions)**.** *Throughout this paper, the following conditions are assumed:*

(C1) ***Supplied representation model.*** *A finite-dimensional agent model, a foreground projection, and a control algebra are specified. A Clifford representation and any external map ϕ are model inputs, not conclusions inherited from revised Q-RAIF or T-DOME II.*

(C2) ***Environmental drift.*** *The instantaneous optimal frame $\mathcal{F}^*(t)$ rotates continuously in G at bounded speed. In the scalar worked model the mismatch is $\theta(t) = \theta_0 + \varepsilon t$ up to the first visibility crossover, as in revised Paper II.*

(C3) ***Finite meta-observer budget.*** *The self-calibration loop has a computational budget $\mathcal{C}_{\text{meta}} < \infty$ (bits per unit time), distinct from the ego's processing budget $\mathcal{C}_{\text{budget}}$.*

(C4) ***Regularity.*** *The agent's predictive family $\{p(e \mid \sigma) : \sigma \in G/H\}$ satisfies standard Fisher information regularity: finite Fisher matrix and interchange of differentiation and integration. Any scalar sensitivity or uniform lower bound used later must be assumed separately.*

(C5) ***Persistent excitation.*** *The time-averaged Fisher matrix obeys the explicit lower bound in Definition 3.24. A positive entropy rate alone is not sufficient for this condition.*

3.3 The Drift Detection Problem

3.3.1 Identifiability in an Even Residual Model

Proposition 3.10 (Sign non-identifiability in an even residual model)**.** *Suppose the scalar residual model is $e \mid \theta \sim \mathcal{N}(a\theta^2, \sigma_e^2)$ with $a \neq 0$ and $\sigma_e > 0$. Then $p(e \mid \theta) = p(e \mid -\theta)$ for every sample and every record length. Residual data alone therefore cannot identify the sign of the frame mismatch without an orientation-breaking input.*

Proof. The likelihood depends on θ only through θ^2; hence the joint likelihood of any record is invariant under $\theta \mapsto -\theta$. No statistic of that record can distinguish the two parameter values. □

This proposition is a property of the declared even likelihood. It does not prove that every instantaneous or "first-order" controller fails: directional probes, asymmetric residual models, and exogenous orientation data can remove the ambiguity.

3.3.2 The Agent's Statistical Manifold

The agent's prediction-residual stream $\{e(t)\}_{t \geq 0}$ defines a stochastic process whose distribution depends on the gauge-fixing parameter σ. We model this dependence as a parametric family.

Definition 3.11 (Predictive family)**.** *The* predictive family *of the agent is the set*

$$\mathcal{P} := \{p(e \mid \sigma) : \sigma \in \mathcal{M}_G\}, \tag{3.10}$$

where $\mathcal{M}_G := G/H$ is the space of gauge-fixing orbits (H is the stabiliser of the foreground subspace), e denotes the prediction-residual time series over a window of length T, and $p(e \mid \sigma)$ is the likelihood of the observed residuals given the gauge parameter σ.

The key insight is that $p(e \mid \sigma)$ depends on σ even though $e(t) \in V_{\text{fg}}$, because the projection $\Pi_{\mathcal{F}}(\sigma)$ determines which environmental correlations are captured. When σ drifts from the optimal σ^*:

- The variance of the residuals increases (the discarded background components contribute unmodelled noise).
- The temporal correlations of the residuals change (the projected kernel $\mathcal{K}_{\mathcal{F}}$ no longer captures the dominant environmental modes).
- Higher-order statistics (kurtosis, spectral shape) shift systematically.

These distributional changes are invisible to the raw error $e(t)$ but detectable by the Fisher metric of $\mathcal{P}$.

3.3.3 Self-Referential Fisher Information

Definition 3.12 (Self-referential Fisher information)**.** *For one residual with likelihood $p(e \mid \sigma)$, define the* per-sample Fisher information

$$\mathcal{I}_1(\sigma) := \mathbb{E}_\sigma\left[(\partial_\sigma \log p(e \mid \sigma))^2\right]. \tag{3.11}$$

For conditionally independent residuals observed at times t_m, the accumulated local information *is $J_T := \sum_{t_m \le T} \mathcal{I}_1(\sigma(t_m))$. The Fisher metric g, the local quadratic form $g_{ij}\delta\sigma^i\delta\sigma^j$, and J_T are distinct objects. Fisher information is measured in inverse-squared parameter units, not bits.*

Remark 3.13 (What the agent "measures")**.** *Computing $\mathcal{I}_1(\sigma)$ does not require access to V_{bg} or to the "true" environment. It requires only: (i) the agent's own prediction residuals $\{e(t)\}$ (which lie in V_{fg}), and (ii) the ability to evaluate the score function $\partial_\sigma \log p(e \mid \sigma)$— the sensitivity of its own predictive model to frame perturbations. This is a computation entirely within the agent's internal algebra, using only quantities already available from the ego's processing pipeline.*

Proposition 3.14 (Fisher signal in a Gaussian residual model)**.** *Let residuals sampled at rate r be conditionally independent with*

$$e_m \mid \theta_m \sim \mathcal{N}(a\theta_m^2, \sigma_e^2), \qquad \theta(t) = \theta_0 + \varepsilon t, \tag{3.12}$$

where $a \neq 0$, $\sigma_e > 0$, and $\varepsilon > 0$. The per-sample Fisher information with respect to the current mismatch is

$$\mathcal{I}_1(\theta) = \frac{4a^2\theta^2}{\sigma_e^2}. \tag{3.13}$$

Consequently the expected accumulated local information over a window of length T is

$$J_T \simeq \frac{4a^2 r}{3\sigma_e^2 \varepsilon} \left[(\theta_0 + \varepsilon T)^3 - \theta_0^3 \right]. \tag{3.14}$$

For a freshly calibrated frame ($\theta_0 = 0$), the per-sample signal is quadratic in time and $J_T = 4a^2 r \varepsilon^2 T^3/(3\sigma_e^2)$.

Proof. For a normal family with fixed variance and mean $\mu(\theta) = a\theta^2$, the score is $(e - \mu)\mu'(\theta)/\sigma_e^2$. Taking its squared expectation gives $(\mu')^2/\sigma_e^2$, which is (3.13). Summing over samples and using the sampling rate gives $J_T \simeq r \int_0^T \mathcal{I}_1(\theta_0 + \varepsilon t)dt$, which evaluates to (3.14). □

Remark 3.15 (Scope of the sensitivity claim)**.** *Full-rank Fisher information for σ does not imply that a scalar mismatch map $\theta(\sigma)$ has a Jacobian bounded away from zero. Proposition 3.14 therefore works directly in the identified scalar mismatch coordinate. A multi-parameter extension must supply or derive the relevant sensitivity map.*

Corollary 3.16 (Conditional detection before visibility crossover)**.** *In the Gaussian residual model, let $J_{\min} > 0$ be a declared detection threshold. For a freshly calibrated frame,*

$$T_{\text{det}} = \left(\frac{3\sigma_e^2 J_{\min}}{4a^2 r \varepsilon^2} \right)^{1/3}. \tag{3.15}$$

Detection precedes Paper II's visibility crossover if and only if $T_{\text{det}} < t_{\text{vis}}$, where for the same linear drift $t_{\text{vis}} = (\pi/4 - \theta_0)/\varepsilon$. Equivalently, for general θ_0 the condition is $J_{t_{\text{vis}}} > J_{\min}$. No unconditional ordering follows from regularity alone.

Proof. Set the fresh-calibration expression for J_T in Proposition 3.14 equal to $J_{\min}$ and solve for T. Comparing that value with the independently defined visibility time gives the stated condition. □

3.4 The Self-Referential Bound

3.4.1 The Bayesian Framework

The agent's ego structure (Paper II) provides a *prior belief* about the correct gauge parameter: the current frame σ_0 is the ego's "preferred" value. We encode this as a prior distribution $\pi_{\text{ego}}(\sigma)$, concentrated around σ_0.

Definition 3.17 (Ego rigidity)**.** *The* ego rigidity *is the prior Fisher information*

$$\mathcal{I}_{\text{ego}} := \int_{\mathcal{M}_G} \left(\frac{\partial \log \pi_{\text{ego}}(\sigma)}{\partial \sigma} \right)^2 \pi_{\text{ego}}(\sigma)\, d\sigma. \tag{3.16}$$

High $\mathcal{I}_{\text{ego}}$ corresponds to a sharply peaked prior (rigid ego); low $\mathcal{I}_{\text{ego}}$ to a diffuse prior (flexible ego). This prior and its Fisher information are supplied model data. The diagnostics of Paper II do not determine a prior without an additional statistical construction.

3.4.2 The Self-Referential Cramér–Rao Bound

Theorem 3.18 (Self-Referential Cramér–Rao Bound)**.** *Let the scalar drift parameter $\delta\sigma$ lie in a compact interval $[a, b]$. Assume that the residual likelihood is regular, that the effective samples are conditionally independent, and that the prior π_{ego} is C^1, has finite prior Fisher information, and vanishes at a and b. Let $\delta\hat{\sigma}$ be any estimator based only on a residual record of duration T. Define the* effective sample size $n_{\text{eff}} := T/\tau_E$*, where τ_E is the decorrelation time of the residual process $\{e(t)\}$ (the time beyond which consecutive residuals carry approximately independent information about σ). If the effective samples share per-sample information $\mathcal{I}_1(\delta\sigma)$, the van Trees inequality [39] gives*

$$\mathbb{E}\Big[\big|\delta\hat{\sigma} - \delta\sigma\big|^2\Big] \;\geq\; \frac{1}{n_{\text{eff}}\,\mathbb{E}_\pi[\mathcal{I}_1(\delta\sigma)] + \mathcal{I}_{\text{ego}}}. \tag{3.17}$$

Proof. This is a direct application of the van Trees (Bayesian Cramér–Rao) inequality [39]. The total information about the drift parameter $\delta\sigma$ consists of two contributions:

- $n_{\text{eff}}\,\mathbb{E}_\pi[\mathcal{I}_1]$: the data Fisher information. In continuous time, the residual process is correlated with decorrelation time τ_E set by the bath memory kernel. Over a window of duration T, the process yields $n_{\text{eff}} \approx T/\tau_E$ effectively independent samples, each carrying Fisher information about $\delta\sigma$.
- $\mathcal{I}_{\text{ego}}$: the prior Fisher information from the ego's preference for σ_0 (Definition 3.17).

The boundary and regularity assumptions remove the integration-by-parts boundary term in the standard van Trees proof. Fisher information is not measured in bits; the denominator has inverse-squared parameter units. □

Remark 3.19 (The ego as help and hindrance)**.** *The ego rigidity $\mathcal{I}_{\text{ego}}$ acts as both help and hindrance:*

- ***Help**: when the ego is well-aligned ($\sigma_0 \approx \sigma^*$), the prior tightens the bound, reducing estimation variance.*
- ***Hindrance**: when the ego is misaligned ($|\sigma_0 - \sigma^*|$ large), the prior pulls the estimate toward the wrong value, creating a* confirmation bias *that resists recalibration.*

For a conjugate scalar Gaussian location model only, the posterior mean has the precision-weighted form below, where $\bar{\mathcal{I}}_1$ is its constant per-sample data precision:

$$\hat{\sigma}_{\text{opt}} = \frac{n_{\text{eff}}\,\bar{\mathcal{I}}_1\,\hat{\sigma}_{\text{MLE}} + \mathcal{I}_{\text{ego}}\,\sigma_0}{n_{\text{eff}}\,\bar{\mathcal{I}}_1 + \mathcal{I}_{\text{ego}}}, \tag{3.18}$$

a weighted average of the maximum-likelihood estimate $\hat{\sigma}_{\text{MLE}}$ and the ego's prior belief σ_0, with weights proportional to their respective Fisher informations. This formula is not implied by van Trees for a general likelihood or prior. In the conjugate Gaussian case, as $n_{\text{eff}}\,\bar{\mathcal{I}}_1 \gg \mathcal{I}_{\text{ego}}$ (enough data to overwhelm the ego), the estimator converges to the MLE.

3.4.3 The Rigidity-Sensitivity Trade-off

Proposition 3.20 (Optimal ego rigidity)**.** *Let the total expected loss be $\mathcal{L}_{\text{total}}(\mathcal{I}_{\text{ego}}) = \mathcal{L}_{\text{estimation}} + \lambda\,\mathcal{L}_{\text{calibration}}$, where $\mathcal{L}_{\text{estimation}}$ is the mean-squared drift-estimation error*

(bounded by (3.17)*) and* $\mathcal{L}_{\text{calibration}}$ *is the cost of adjusting the frame (proportional to the frame rotation distance, hence larger when the ego is rigid and must be overcome). If the admissible rigidity interval is nonempty and compact and the two supplied loss terms are lower semicontinuous, an optimal rigidity* $\mathcal{I}^*_{\text{ego}}$ *exists. Uniqueness or an interior optimum additionally requires a model-specific strict-convexity or monotonicity condition.*

Too rigid *(*$\mathcal{I}_{\text{ego}} \gg n_{\text{eff}}\,\bar{\mathcal{I}}_1$*): the ego overwhelms the data; the agent is blind to drift.* Too soft *(*$\mathcal{I}_{\text{ego}} \ll n_{\text{eff}}\,\bar{\mathcal{I}}_1$*): the agent overreacts to noise; calibration cost is high. The optimum balances sensitivity against stability.*

Proof. A lower-semicontinuous real function attains its minimum on a nonempty compact set. The existence conclusion uses only that fact; the qualitative "too rigid/too soft" interpretation is not a convexity proof. □

3.5 The Calibration Loop

3.5.1 The Natural Gradient Update Law

The meta-observer updates the gauge parameter σ following the natural gradient on the statistical manifold $(\mathcal{M}_G,\, g)$:

$$\dot{\sigma} = -\eta\, g^{-1}(\sigma)\, \nabla_\sigma L_{\text{frame}}(\sigma), \tag{3.19}$$

where $\eta > 0$ is the adaptation rate and the *frame loss* is

$$L_{\text{frame}}(\sigma) := \mathbb{E}_\sigma[-\mathcal{S}_{\text{vis}}(\sigma)]\,. \tag{3.20}$$

The frame loss is minimised at the optimal gauge σ^* that maximises visible survival. The Fisher metric enters through the inverse g^{-1} in the natural gradient, not as a penalty term: it defines the *geometry* of the update, ensuring reparametrisation invariance.

Remark 3.21 (Reparametrisation invariance)**.** *The natural gradient* (3.19) *is invariant under reparametrisation of the gauge manifold* $\mathcal{M}_G$*: the update direction does not depend on the choice of coordinates for* σ*. This is essential because the gauge manifold inherits its geometry from the Clifford algebra, and no canonical coordinate system is preferred.*

3.5.2 Lyapunov Stability of the Loop

Drift velocity. Let $\sigma^*(t)$ denote the instantaneous optimal gauge parameter, defined as a minimiser of L_{frame} at time t in Eq. (3.20). Define the *drift velocity* $\dot{\sigma}^* := d\sigma^*/dt$, measured with respect to the Fisher metric g; its norm $\|\dot{\sigma}^*\|_g := \sqrt{g_{ij}\,\dot{\sigma}^{*i}\,\dot{\sigma}^{*j}}$ is the instantaneous rate at which the environment's optimal frame rotates on the gauge manifold.

Definition 3.22 (Lyapunov monitoring function)**.** *The* Lyapunov monitoring function *is the squared geodesic distance on the statistical manifold from the current frame to the instantaneous optimal frame:*

$$V(\sigma) := d_{\text{geo}}(\sigma,\, \sigma^*(t))^2\,, \tag{3.21}$$

where d_{geo} *is the geodesic distance in the Fisher metric* g*.*

Theorem 3.23 (Loop Tracking Bound)**.** *Assume the natural-gradient trajectory remains in a geodesically convex normal neighbourhood of $\sigma^*(t)$ and that the frame loss is geodesically m_L-strongly convex there, with* $\operatorname{Hess} L_{\text{frame}} \succeq m_L g$ *for some $m_L > 0$. Then the update* (3.19) *satisfies*

$$\frac{dV}{dt} \;\le\; -2\eta\, m_L\, V \;+\; 2\sqrt{V}\,\left\|\dot\sigma^*\right\|_g, \tag{3.22}$$

where $\|\dot\sigma^*\|_g$ is the instantaneous drift speed of the optimal frame. Consequently:

(a) ***Tracking.*** *Whenever $\sqrt{V} > \|\dot\sigma^*\|_g/(\eta\, m_L)$, we have $dV/dt < 0$: the loop actively reduces the mismatch.*

(b) ***Tracking neighbourhood.*** *The mismatch converges to a neighbourhood of the set of stationary points of L_{frame}. This neighbourhood has size*

$$V_\infty \;:=\; \frac{\left\|\dot\sigma^*\right\|_g^2}{(\eta\, m_L)^2}. \tag{3.23}$$

For bounded drift ($\|\dot\sigma^*\|_g \le \Lambda_{\max}$), the mismatch is bounded: $\limsup_{t\to\infty} V(t) \le \Lambda_{\max}^2/(\eta m_L)^2$.

(c) ***Static limit.*** *When $\sigma^* = \text{const}$ ($\dot\sigma^* = 0$), the bound reduces to $dV/dt \le -2\eta m_L\, V$, giving exponential convergence $V(t) \le V(0)\, e^{-2\eta m_L\, t}$.*

Proof. Since $V(\sigma) = d_{\text{geo}}(\sigma,\, \sigma^*(t))^2$ and $\sigma^*(t)$ is time-varying, the total derivative has two contributions:

$$\frac{dV}{dt} = \underbrace{\frac{\partial V}{\partial \sigma}\cdot\dot\sigma}_{\text{control}} \;+\; \underbrace{\frac{\partial V}{\partial \sigma^*}\cdot\dot\sigma^*}_{\text{drift}}.$$

Control term. In normal coordinates centred at σ^*, let $\delta\sigma := \sigma - \sigma^*$. The control contribution is $2\, g(\delta\sigma,\, \dot\sigma) \;=\; 2\, g(\delta\sigma,\, -\eta\, g^{-1}\nabla L_{\text{frame}}) \;=\; -2\eta\, \langle \delta\sigma,\, \nabla L_{\text{frame}}\rangle$. By the explicit strong-convexity hypothesis, $\langle \delta\sigma,\, \nabla L_{\text{frame}}\rangle \ge m_L\, |\delta\sigma|_g^2 = m_L\, V$, giving a control contribution $\le -2\eta\, m_L\, V$.

Drift term. The drift contribution is $-2\, g(\delta\sigma,\, \dot\sigma^*)$. By Cauchy–Schwarz: $|g(\delta\sigma,\, \dot\sigma^*)| \le |\delta\sigma|_g\, \|\dot\sigma^*\|_g = \sqrt{V}\, \|\dot\sigma^*\|_g$. Hence the drift contribution is bounded by $+2\sqrt{V}\, \|\dot\sigma^*\|_g$.

Combined. Adding both contributions gives (3.22). Part (a) follows by setting $dV/dt < 0$; part (b) by solving $dV/dt = 0$ for the fixed point $\sqrt{V_\infty} = \|\dot\sigma^*\|_g/(\eta m_L)$; part (c) by setting $\dot\sigma^* = 0$. □

3.5.3 Convergence Rate under Persistent Excitation

Definition 3.24 (Persistent excitation constant)**.** *The* persistent excitation constant *$\alpha_{\text{PE}} > 0$ is the minimum eigenvalue of the time-averaged Fisher information matrix:*

$$\bar g(t) := \frac{1}{T}\int_t^{t+T} g(\sigma(s))\, ds \;\succeq\; \alpha_{\text{PE}}\, I \qquad \textit{for all } t, \tag{3.24}$$

when the residual likelihood is persistently exciting. Pointwise full-rank regularity and a positive environmental entropy rate do not by themselves provide a uniform lower bound.

Remark 3.25 (Tracking vs convergence)**.** *In the static case ($\sigma^* = \text{const}$), Theorem 3.23(c) gives pure exponential convergence: $V(t) \leq V(0)\, e^{-2\eta m_L t}$. Under environmental drift, convergence to zero is* not *possible—instead the loop maintains the mismatch within the tracking neighbourhood* (3.23)*. The tracking error V_∞ grows with drift speed $\|\dot{\sigma}^*\|_g$ and decreases with loop parameters η and m_L. The distinct constant α_{PE} controls statistical conditioning of the Fisher metric; no equality $m_L = \alpha_{\mathrm{PE}}$ is assumed. A physical implementation must separately relate adaptation rate to its energy budget.*

Remark 3.26 (The necessity of novelty)**.** *A frozen or uninformative residual likelihood can violate persistent excitation, but $h_\mu \to 0$ alone does not determine α_{PE} or m_L. Loss of excitation therefore has to be checked in the supplied observation model rather than inferred from entropy rate alone.*

3.5.4 Four-Part Model Template

The following is an architecture template for the declared model, not a necessity or sufficiency theorem for persistent agents.

Definition 3.27 (Four-part model architecture)**.** *The running T-DOME model uses four supplied structural layers:*

(I) ***External observable geometry.*** *The environmental observable algebra supports a metric structure; $Cl(1,3)$ serves as the running example throughout the programme, but the argument applies to any algebra satisfying (C1). This is a modelling premise; revised HAFF/Q-RAIF do not derive a general metric dictionary.*

(II) ***Internal control algebra.*** *The agent carries a supplied internal algebra $Cl(V,q)$ and a supplied representation map ϕ into an external algebra. Revised Q-RAIF neither uniquely selects the Clifford algebra nor proves a signature-preserving embedding.*

(III) ***Self-monitoring function.*** *The agent maintains a Lyapunov function $V(\sigma)$* (3.21) *satisfying the tracking bound* (3.22)*, via a second-order control loop on the agent's Fisher information, keeping the mismatch within the tracking neighbourhood* (3.23). Conditional source: *Theorem 3.23, under its explicit local strong-convexity assumptions.*

(IV) ***Biased, non-Markovian memory.*** *The agent carries path-dependent state (non-Markovian memory kernel $\mathcal{K}(t,s)$) compressed through a gauge-fixed reference frame (the ego $\mathfrak{E}$). This is a model choice motivated by Papers I and II, not a necessity result inherited from them.*

The template is sufficient only for performance separately proved by a fully specified implementation. Omitting a layer may be benign in a different architecture.

3.6 Thermodynamic Cost

3.6.1 The Three Cost Components

One implementation may use sensing, computing, and actuation. Their thermodynamic costs depend on how they are physically realized.

1. Sensing cost. The meta-observer must read the prediction residuals from the ego's processing pipeline. Measurement or copying can in principle be reversible. If sensor registers irreversibly erase Shannon information at rate R_s bits per unit time, Landauer's principle gives

$$\dot{W}_{\text{sense}} \;\geq\; k_B T \,\ln 2 \cdot R_s. \tag{3.25}$$

No positive sensing bound follows merely from reading a stream.

2. Computing cost. Reversible logic likewise evades a gate-count Landauer charge. If computational work registers irreversibly erase Shannon information at rate R_c bits per unit time, then

$$\dot{W}_{\text{compute}} \;\geq\; k_B T \,\ln 2 \cdot R_c. \tag{3.26}$$

3. Actuating cost. Rotating the gauge parameter from the current frame σ to the estimated optimal frame $\hat{\sigma}^*$ is a finite-time thermodynamic transformation on the gauge manifold. By the Sivak–Crooks bound (Proposition 3.7):

$$\dot{W}_{\text{actuate}} \;\geq\; \frac{\mathcal{L}^2(\sigma,\, \hat{\sigma}^*)}{\tau^2_{\text{recalib}}}, \tag{3.27}$$

where $\mathcal{L}(\sigma,\, \hat{\sigma}^*)$ is the thermodynamic length (3.7) of the geodesic from σ to $\hat{\sigma}^*$, and τ_{recalib} is the recalibration time.

3.6.2 Conditional Implementation Bound

Proposition 3.28 (Conditional implementation cost)**.** *Suppose that (i) sensing, computing, and actuation use separable physical degrees of freedom with no shared erasure accounting; (ii) sensor and computational registers erase Shannon information at rates R_s and R_c; and (iii) frame actuation follows a linear-response protocol of duration τ_{recalib} and thermodynamic length $\mathcal{L}$. Then*

$$\dot{W}_{\text{loop}} \;\geq\; k_B T \,\ln 2 \,[R_s + R_c] + \frac{\mathcal{L}^2(\sigma,\, \sigma^*)}{\tau^2_{\text{recalib}}}. \tag{3.28}$$

This is a bound for the declared implementation, not a universal cost of sensing, Fisher-information evaluation, or self-reference.

Proof. Landauer's bound applies to the two declared irreversible reset streams, giving the first two terms. The Sivak–Crooks work bound divided by the protocol duration gives the actuation rate. The separability hypothesis permits addition without double counting. Dropping that hypothesis, or implementing reversible sensing/computation with no reset in the window, invalidates the corresponding additive terms. □

3.6.3 Withdrawal of the Universal Persistence Budget

Remark 3.29 (Withdrawal of the universal persistence budget)**.** *The former additive persistence corollary is withdrawn. Revised Papers I and II provide no universal memory-maintenance or ego-processing Landauer terms that can be inherited here. Proposition 3.28 may be combined with independently established costs for a concrete hardware implementation, provided separability and reset accounting are verified afresh.*

3.7 Worked Example: Qubit in a Drifting Two-Channel Bath

3.7.1 Model Setup

We extend the two-channel qubit model from Paper II (Section 6) by introducing environmental drift.

Inherited setup. A qubit ($\dim \mathcal{H}_S = 2$) with internal algebra $Cl(0,2) \cong \mathbb{H}$ ($D = 4$), coupled to two bosonic channels:

- Dephasing channel (σ_z): $J_z(\omega) = 2\lambda_z\gamma_z\omega/(\omega^2 + \gamma_z^2)$.
- Dissipative channel (σ_x): $J_x(\omega) = 2\lambda_x\gamma_x\omega/(\omega^2 + \gamma_x^2)$.

Paper II's ego selects $V_{\text{fg}} = \text{span}\{1, \mathbf{k}\}$ (the dephasing subspace), discarding $V_{\text{bg}} = \text{span}\{\mathbf{i}, \mathbf{j}\}$.

Environmental drift and residual channel. We use Paper II's revised linear mismatch model and the Gaussian residual channel of Proposition 3.14:

$$\theta(t) = \varepsilon t, \qquad e_m \mid \theta_m \sim \mathcal{N}(a\theta_m^2, \sigma_e^2), \quad a = 1.2, \quad \sigma_e = 0.35, \quad \varepsilon = 0.04\,\omega_0. \tag{3.29}$$

At sampling rate $r = 50\,\omega_0$, the visibility crossover is $t_{\text{vis}} = \pi/(4\varepsilon) \approx 19.63\,\omega_0^{-1}$. The Gaussian likelihood is a declared observation model, not derived from the bosonic spectral density.

Parameter mapping.

Quantity	Value	Source
$D = \dim Cl(0,2)$	4	Paper II
k^*	2	chosen Paper II model frame
$\mathcal{C}_{\text{budget}}$	$2\,h_\mu$	Paper II
θ_0 (initial misalignment)	0	fresh calibration
ε (angular drift rate)	$0.04\,\omega_0$	Eq. (3.29)
t_{vis}	$19.63\,\omega_0^{-1}$	Paper II visibility model
η (adaptation rate)	0.5	meta-observer

3.7.2 Fisher Information under Drift

For the declared residual channel, the per-sample and accumulated local Fisher signals are

$$\mathcal{I}_1(t) = \frac{4a^2\varepsilon^2 t^2}{\sigma_e^2}, \qquad J_T = \frac{4a^2 r\varepsilon^2 T^3}{3\sigma_e^2}. \tag{3.30}$$

The first quantity grows quadratically in time; the second grows cubically because it also counts samples. For $J_{\min} = 20$, Equation (3.15) gives $T_{\text{det}} \approx 2.52\,\omega_0^{-1} < t_{\text{vis}}$. This ordering is a parameter check, not a general theorem.

3.7.3 Loop Dynamics: Self-Calibration in Action

Under the natural gradient update (3.19), the frame angle $\phi(t)$ tracks the drifting optimal frame $\phi^*(t)$. The Lyapunov function $V(t) = (\phi(t) - \phi^*(t))^2$ is governed by the tracking bound (3.22): the loop drives V toward the tracking neighbourhood $V_\infty = \|\dot{\sigma}^*\|_g^2/(\eta m_L)^2$, with the approach rate set by the loss strong-convexity modulus m_L and adaptation rate η.

Comparison.

- **Without loop** (Paper II agent): the mismatch grows as $\theta(t) = \varepsilon t$, reaching the visibility crossover at t_{vis}.
- **With loop** (Paper III agent): the mismatch oscillates around zero, bounded by the estimation neighbourhood supplied by Theorem 3.23 when its local strong-convexity and bounded-drift hypotheses hold.

A multi-dimensional numerical evaluation extending this qubit illustration to continuous drift is presented in Section 3.8.

3.7.4 Thermodynamic Cost Evaluation

For illustration, suppose a particular implementation irreversibly resets sensor and work registers at rates $R_s = 2$ and $R_c = 1$ bits per unit time and uses separable actuation hardware. Then

$$\dot{W}_{\text{sense}} \geq k_B T \ln 2 \cdot R_s = 2\, k_B T \ln 2, \tag{3.31}$$

$$\dot{W}_{\text{compute}} \geq k_B T \ln 2 \cdot R_c = k_B T \ln 2, \tag{3.32}$$

$$\dot{W}_{\text{actuate}} \geq \frac{\mathcal{L}^2}{\tau_{\text{recalib}}^2} \approx \frac{(\Delta\theta)^2\, \tau_{\text{relax}}}{\tau_{\text{recalib}}^2}\, k_B T. \tag{3.33}$$

The two Landauer terms are present because this example stipulates irreversible reset. A reversible implementation may lower them, and no percentage overhead relative to Papers I or II follows.

3.8 Numerical Demonstration

We report two separate numerical checks. The first directly samples the Gaussian likelihood and computes its score-based Fisher signal. The second retains the earlier multi-dimensional calibration controller, whose top-k gradient-overlap sentinel is a heuristic proxy rather than Fisher information.

3.8.1 Direct Fisher-Signal Test

We sample Equation (3.12) at rate $r = 50$ with $a = 1.2$, $\sigma_e = 0.35$, $\varepsilon = 0.04$, and 4,000 Monte Carlo residuals per plotted time. For each residual we evaluate the score

$$s_\theta(e) = \frac{(e - a\theta^2)(2a\theta)}{\sigma_e^2}$$

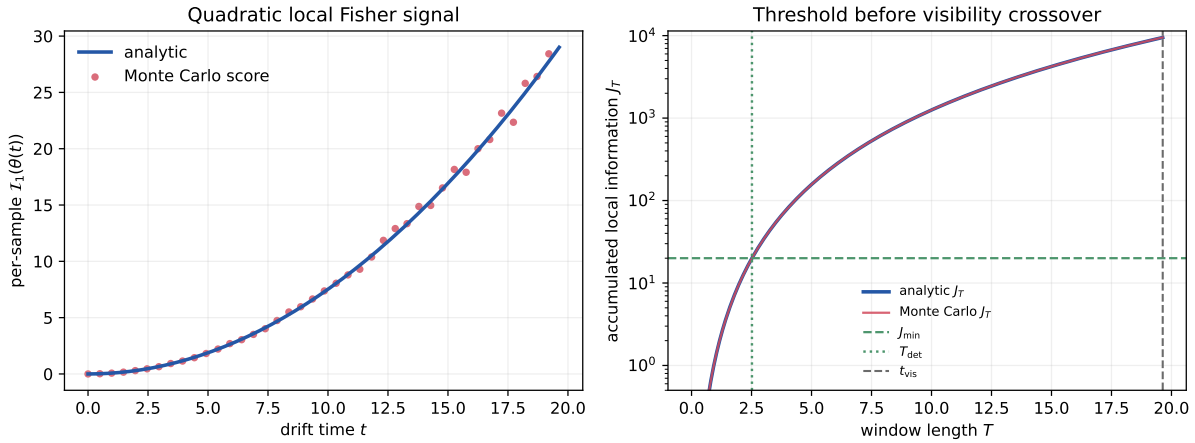


Figure 3.1: **Direct score-based Fisher test. Left:** analytic per-sample Fisher information and Monte Carlo score-squared estimate, both quadratic in drift time. **Right:** analytic and empirical accumulated local information; the $J_{\min} = 20$ crossing precedes the independently defined visibility crossover for this parameter set.

and estimate $\mathcal{I}_1(\theta) = \mathbb{E}[s_\theta^2]$ directly. Figure 3.1 shows agreement with the analytic quadratic curve and with the cubic accumulated curve. For $J_{\min} = 20$, the analytic and empirical crossings are respectively $T_{\det} = 2.517$ and 2.524, both before $t_{\text{vis}} = 19.635$. This verifies Proposition 3.14 and Corollary 3.16 for the declared parameter set; it does not establish the ordering for other likelihoods or thresholds.

3.8.2 Heuristic Calibration-Control Model

Environment. A d-dimensional linear prediction task: $y(t) = \mathbf{w}(t)^\top \mathbf{x}(t) + \sigma\,\epsilon(t)$, $\mathbf{x}(t) \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$, $\epsilon \sim \mathcal{N}(0,1)$. The weight vector $\mathbf{w}(t) \in \mathbb{S}^{d-1}$ drifts by receiving random perturbations on *background* dimensions only (indices $k, \ldots, d-1$), then renormalising. Signal therefore migrates progressively from the ego's foreground to its blind sector.

Agents.

- **Fixed ego** (Paper II analogue): learns a linear model on a *fixed* foreground subspace of dimension k via stochastic gradient descent (SGD, rate η, decay λ). Embodies the "frozen gauge" of Paper II.
- **Calibrated loop** (Paper III analogue): identical ego plus a staleness sentinel and recalibration mechanism. The sentinel tracks $g_i = \text{EMA}(|e\,x_i|)$ for each dimension i (an absolute-gradient proxy), computes the fraction of top-k gradient dimensions *not* in the current foreground as a frame-staleness index $m \in [0,1]$, and triggers recalibration when $\text{EMA}(m) > \theta$. After recalibration, a settling period of τ steps elapses before the sentinel resumes monitoring.

Parameters.

Quantity	Value	Role
d	20	full ambient dimension
k	5	ego foreground dimension ($k/d = 0.25$)
σ	0.1	observation noise std
η	0.01	SGD learning rate
λ	0.998	SGD weight decay
θ	0.25	staleness threshold
Λ_{demo}	variable	simulator drift amplitude per step
τ	variable	settling period (cooldown)

Oracle metrics. Neither agent has access to $\mathbf{w}(t)$. We evaluate performance externally using the *oracle full-space error*:

$$\mathcal{E}_{\text{full}} = \left\|\mathbf{w}_{\text{ego}} - \mathbf{w}^*_{\text{fg}}\right\|^2 + \left\|\mathbf{w}^*_{\text{bg}}\right\|^2, \tag{3.34}$$

where $\mathbf{w}^*_{\text{fg}}$ and $\mathbf{w}^*_{\text{bg}}$ denote the true weight vector restricted to foreground and background coordinates respectively, and $\mathbf{w}_{\text{ego}}$ is the ego's foreground-supported estimator lifted to the full space. The first term captures foreground tracking error (accessible to the ego); the second captures hidden-sector signal (invisible).

3.8.3 Results

Result 1: Delusion-correction separation (Figure 3.2). At simulator drift amplitude $\Lambda_{\text{demo}} = 0.02$, settling period $\tau = 200$, and $T = 5\,000$ steps, three phenomena are visible:

(a) *Delusion trap.* The ego's foreground tracking error converges to near zero, while the true full-space error rises toward ~ 1 and stabilises. The growing gap between the two is the hidden sector in this supplied linear model. It does not prove a no-go theorem for all first-order controllers.

(b) *Detectability.* The staleness sentinel produces a clean sawtooth: rising from zero after each recalibration, crossing the threshold $\theta = 0.25$, and triggering frame reset (25 events over $T = 5\,000$). This demonstrates that the chosen overlap proxy can trigger a useful reset. The sentinel is not Fisher information; the direct Fisher test is reported in Section 3.8.1.

(c) *Net benefit.* The calibrated loop achieves $\mathcal{E}_{\text{full}} \approx 0.74$ versus the fixed ego's ≈ 1.02: a 27% reduction in true prediction error.

Result 2: Phase structure and optimal calibration budget (Figures 3.3–3.4). We scan 16 simulator drift amplitudes $\Lambda_{\text{demo}} \in [0.005, 0.08]$ and 16 settling periods $\tau \in [15, 800]$ (logarithmically spaced), running both agents for $T = 4\,000$ steps across 6 random seeds per grid point.

Figure 3.3(a) shows the performance gain $\Delta = \mathcal{E}_{\text{ego}} - \mathcal{E}_{\text{loop}}$: the loop improves over the ego (green) across most of the parameter space, with a boundary at $\Delta = 0$ (dashed) below which recalibration is counterproductive (very low drift, where the overhead of re-learning exceeds the benefit of tracking). The solid curve traces the *optimal settling period*

$\tau_{\text{opt}}(\Lambda_{\text{demo}})$—the recalibration period minimising $\mathcal{E}_{\text{loop}}$—which decreases monotonically from ~ 370 steps at $\Lambda_{\text{demo}} = 0.005$ to ~ 100 at $\Lambda_{\text{demo}} = 0.08$. Figure 3.3(b) shows that calibration frequency increases smoothly with drift and with shorter settling period, exhibiting the cost–performance trade-off of this supplied controller and cost proxy. It does not test Proposition 3.28.

Extracting $\tau_{\text{opt}}(\Lambda_{\text{demo}})$ yields the *optimal calibration frequency* $\alpha_{\text{opt}}(\Lambda_{\text{demo}}) = 1/\tau_{\text{opt}}$ (Figure 3.4). The curve is smooth and monotonically increasing: faster drift demands tighter calibration. It saturates at high Λ_{demo} near $\alpha_{\text{opt}} \approx 0.01$ per step ($\tau_{\text{opt}} \approx 100$), of the same order as the learner's settling time. This is consistent with the intuition that drift estimation requires a minimum observation window; the self-referential Cramér–Rao bound (Theorem 3.18) provides the analytic counterpart of this computational floor.

3.8.4 Scope of This Demonstration

This demonstration **does** show:

1. The explicit Gaussian residual model has a directly measured score-squared Fisher signal with the quadratic and cubic scalings in Proposition 3.14.

2. For the declared Gaussian parameters, the Fisher threshold is crossed before the linear visibility crossover.

3. A separate, non-Fisher frame-staleness proxy has clean threshold dynamics and triggers effective recalibration in the supplied linear controller.

4. An optimal calibration frequency $\alpha_{\text{opt}}(\Lambda_{\text{demo}})$ exists, increases monotonically with drift rate, and saturates at the learner's settling timescale in the scanned grid.

In summary, the direct experiment validates the declared Fisher likelihood, while the controller experiment establishes a model-specific proxy trade-off. Neither is universal across architectures or environment classes.

This demonstration does **not** show:

1. That the specific functional form of $\alpha_{\text{opt}}(\Lambda)$ matches the analytic Cramér–Rao prediction in the large-d limit. The demonstration confirms the monotonic trend and saturation; deriving the exact scaling exponent from Theorem 3.18 remains open.

2. That the results generalise to all environment classes. The model uses Gaussian features, linear regression, and isotropic background drift; extensions to non-linear, non-Gaussian, or structured-drift settings require further investigation.

3. That the calibration loop is optimal among all possible adaptive strategies. It implements one specific realisation of the calibration-loop architecture.

4. That the gradient-overlap sentinel estimates Fisher information, or that recalibration-event frequency is a thermodynamic dissipation measurement.

Reproducibility. The controller simulation is in `tdome_demo.py`. The direct Fisher experiment is in `fisher_residual_demo.py`, with raw output in `fisher_residual_data.csv`. Both require only NumPy and Matplotlib and use fixed random seeds.

3.9 Discussion

3.9.1 Summary of Results

Result	Statement	Sec.
Even-likelihood non-identifiability	The Gaussian residual record cannot identify mismatch sign	3.3.1
Drift Detectability	Per-sample FI is quadratic and accumulated local FI is cubic in the Gaussian linear-drift model	3.3.3
Self-Referential CR Bound	Drift estimation bounded by $1/(n_{\text{eff}}\,\mathbb{E}_\pi[\mathcal{I}_1] + \mathcal{I}_{\text{ego}})$ under regular prior/likelihood data	3.4.2
Loop Tracking Bound	Lyapunov V with tracking neighbourhood $V_\infty = \|\dot{\sigma}^*\|^2/(\eta m_L)^2$ under local geodesic strong convexity	3.5.2
Four-Part Structure	Four supplied layers define the running model architecture	3.5.4
Loop Cost	$\dot{W}_{\text{loop}} \geq k_B T \ln 2\,(R_s + R_c) + \mathcal{L}^2/\tau^2_{\text{recalib}}$ for a separable reset-based implementation	3.6.2
Persistence Budget	Universal inherited budget withdrawn	3.6.3
Numerical Demonstration	Direct Fisher test plus a separate heuristic-controller scan	3.8

3.9.2 Conditional Interface Sequence

Papers I–III provide a conditional interface sequence:

Paper	Crisis	Resolution	What is born
Paper I	Restricted autonomous baseline	Correlation-aware accounting	**Conditional resource model**
Paper II	Raw archive exceeds a declared budget	Supplied coordinate projection	**Frame model**
Paper III	Linear visibility loss in a fixed frame	Likelihood-based detection; conditional tracking	**Calibration model**

The arrows are motivation or model interfaces. The tracking theorem is local and conditional, the universal persistence budget is withdrawn, and no termination or completeness result follows.

3.9.3 What This Paper Does and Does Not Show

This paper **does** show:

1. The even Gaussian residual likelihood cannot identify mismatch sign, while its score has the explicit Fisher scalings in Proposition 3.14.

2. Detection precedes visibility loss exactly when the threshold condition in Corollary 3.16 holds.

3. Drift estimation precision is bounded by the Self-Referential Cramér–Rao bound (Theorem 3.18).

4. The calibration loop tracks the optimal frame within a bounded neighbourhood under a Lyapunov tracking bound (Theorem 3.23).

5. A separable reset-based hardware model obeys the conditional cost bound in Proposition 3.28.

This paper does **not** show:

1. That self-referential calibration implies or requires phenomenal consciousness, subjective experience, or qualia. "Reflexivity" as used here denotes second-order control, nothing more.

2. That the Lyapunov function V is a measure of "awareness." It is a control-theoretic stability condition, not a consciousness metric.

3. That the four-part template is necessary, sufficient, or a complete characterisation of agency.

4. That Fisher information requires the agent to "know" it is computing Fisher information. The computation can be implemented implicitly by any physical system whose dynamics approximate the natural gradient.

5. That the calibration loop eliminates the ego's bias. It tracks and compensates for drift in the bias; the four bias terms of Paper II persist.

6. That sensing or computing is intrinsically dissipative. The Landauer terms require irreversible register reset.

7. That this framework applies to all possible systems. It applies to systems satisfying (C1)–(C5).

8. That the structural parallel with philosophical concepts of self-awareness constitutes a philosophical or metaphysical claim.

9. That the Clifford algebra is the only possible algebraic setting. Other control algebras may yield analogous results with different quantitative bounds.

We have established a checkable Gaussian Fisher signal, a regular Bayesian estimation bound, and a local tracking result. Their use in a physical agent remains conditional on the declared observation, optimization, and hardware models.

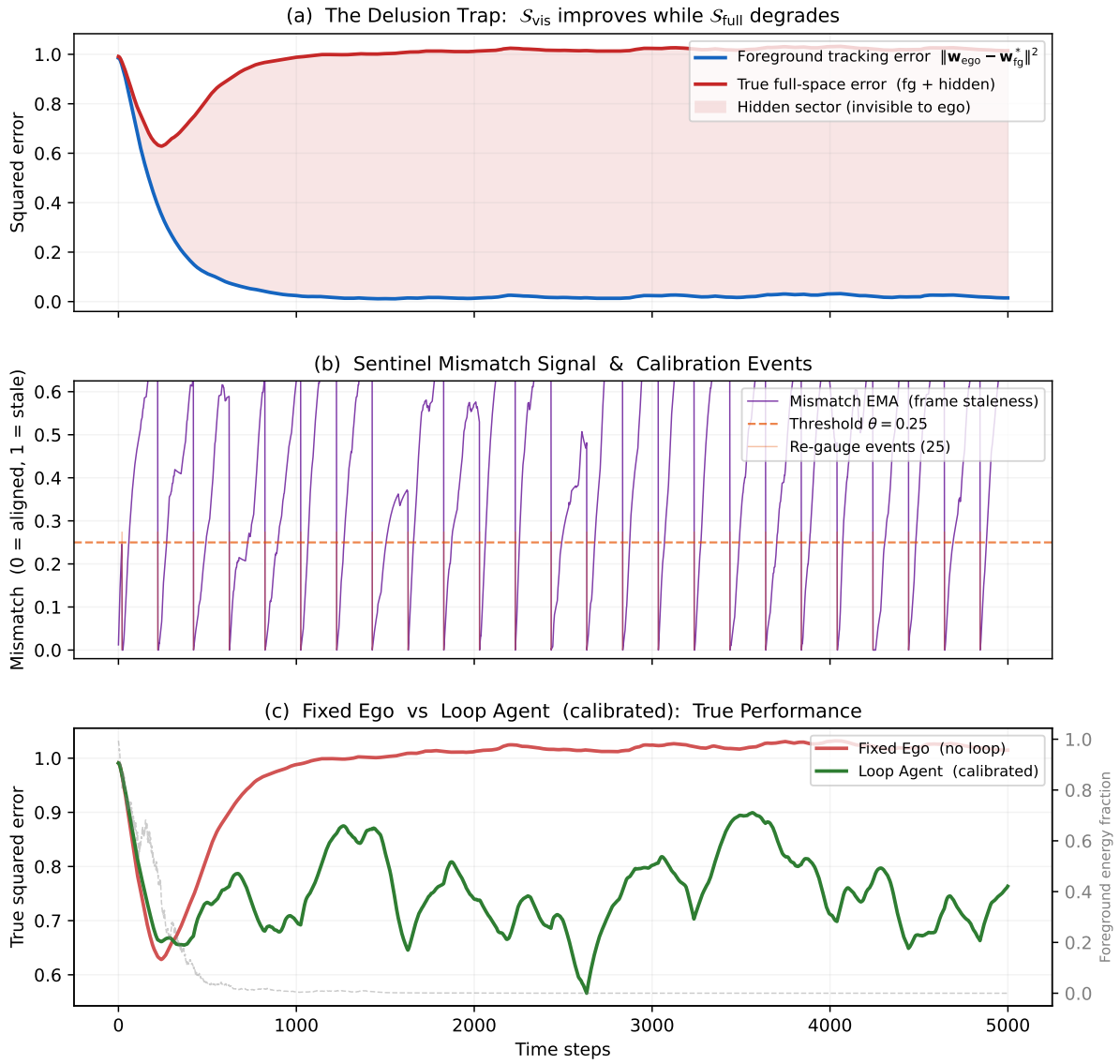


Figure 3.2: **Delusion trap and calibration loop.** $d = 20$, $k = 5$, $\Lambda_{\text{demo}} = 0.02$, $\tau = 200$, $T = 5\,000$. **(a)** Foreground tracking error (blue) decreases toward zero while true full-space error (red) increases; the shaded region is the hidden sector, invisible to the ego. **(b)** Frame-staleness sentinel (purple) rises monotonically between recalibration events (orange), producing a sawtooth with 25 threshold crossings. **(c)** The calibrated loop (green) maintains lower true error than the fixed ego (red); the grey dashed line shows the foreground energy fraction decaying as signal migrates to the background.

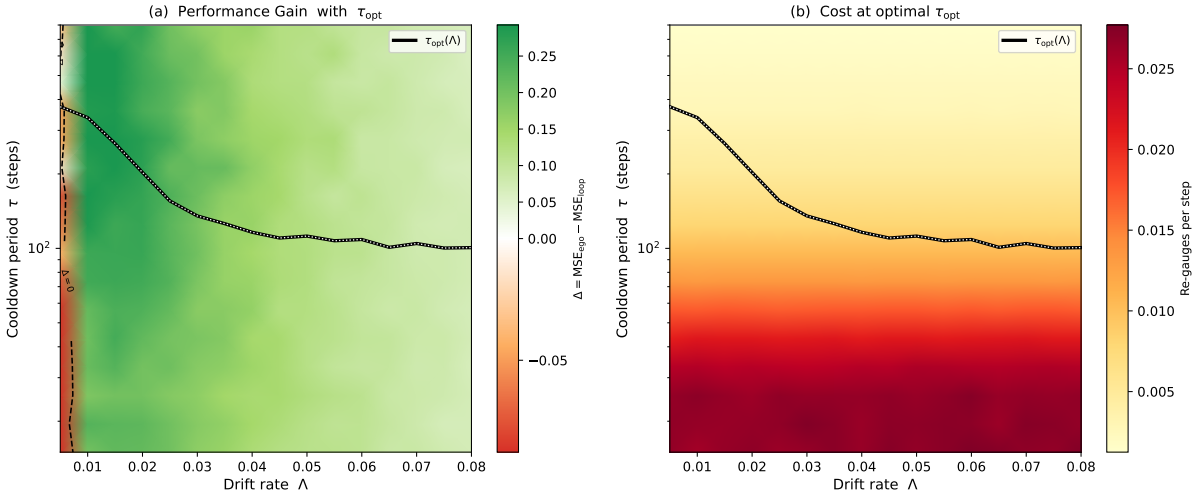


Figure 3.3: **Phase structure and calibration cost.** 16×16 grid, $T = 4\,000$, 6 seeds per point. **(a)** Performance gain Δ; green = loop improves on ego, red = counterproductive. Solid curve: $\tau_{\text{opt}}(\Lambda_{\text{demo}})$. Dashed: $\Delta = 0$ boundary. **(b)** Calibration frequency (thermodynamic cost proxy: recalibration events per step, proportional to energy expenditure under a fixed per-recalibration cost model); τ_{opt} overlaid.

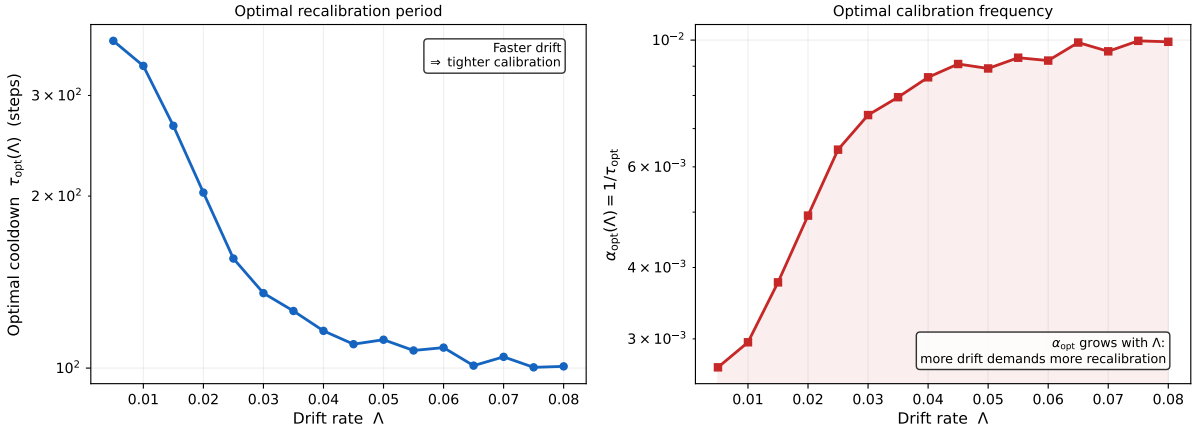


Figure 3.4: **Optimal calibration frequency. Left:** $\tau_{\text{opt}}(\Lambda_{\text{demo}})$ decreases monotonically with drift rate. **Right:** $\alpha_{\text{opt}}(\Lambda_{\text{demo}}) = 1/\tau_{\text{opt}}$ increases with drift rate and saturates at the learner's settling timescale ($\sim$100 steps), consistent with an observation-window floor.

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